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Masthead



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Editor's Note

Dear Reader,

I am pleased to present the second edition of the *Stevens Journal of Law and Technology*. Since our inaugural volume, SJOLT has grown into a working institution capable of soliciting, reviewing, and publishing meaningful scholarship. The questions this Journal exists to address have not slowed down. Laws and policies are still catching up to technology on nearly every front, and the need for early-career voices in those conversations remains as great as ever.

This volume reflects how far our Journal has come. Our new “author assistance” system has allowed editors to work one-on-one with interested authors, helping them rigorously develop and communicate their perspectives. This growth, in both reach and capability, is what I most hope SJOLT continues to build upon.

Each edition should be more capable than the last of fulfilling the promise made in our inaugural note: giving early-career voices real purchase on conversations that will otherwise be shaped without them. This edition is evidence that promise is being kept.

Per aspera ad astra.

Nicholas Russo

Founder and Editor-in-Chief

Mission Statement

The *Stevens Journal of Law and Technology* (SJOLT) is the first and only undergraduate-led academic publication dedicated to advancing scholarship at the intersection of law and technology. We publish original long-form pieces by undergraduate and graduate students worldwide, providing a platform for encouraging and distributing interdisciplinary inquiry and argumentation. Through this work, SJOLT seeks to elevate emerging voices, expand access to legal and technological scholarship, and advance rigorous discourse on the complex relationships between law, emerging technologies, and society.

Disclaimer

The views and opinions presented in this journal are solely those of the respective authors and do not necessarily reflect the official policy or position of the Stevens Journal of Law and Technology, its editors, or the Stevens Institute of Technology.

A Criticism of Congress’ TAKE IT DOWN Act: Why Removal is not Remedy

Jane Starr

Stevens Institute of Technology

**“FEDERAL LAW IS ENACTED, CONTENT GETS
REMOVED, THE PERPETRATOR MAY GET
CHARGED, AND THE VICTIM REMAINS
SCARRED WITH NO FEDERAL CIVIL REMEDY.”**

Abstract

The rapid proliferation of artificial intelligence has intensified concerns about online safety for adults and minors, particularly regarding non-consensual intimate imagery (NCII) and digitally fabricated pornographic content. This paper evaluates recent federal efforts to address these harms, including the enacted TAKE IT DOWN Act (TIDA) and the stalled DEFIANCE Act. Central to this analysis is the imbalance created by TIDA's notice-and-removal framework, which obligates platforms to take down NCII but provides no corresponding private right of action for victims of deepfake abuse. This gap exposes a structural weakness in the current federal response to AI-enabled sexual exploitation, as the DEFIANCE Act, which would create a private right of action and has sat idly on the House floor for over 7 months. This must be incorporated into TIDA to ensure both platform obligations and meaningful remedies for victims.

Introduction

In January of 2024, 14 year old Francesca Mani of Westfield, New Jersey, rose before millions of viewers, attesting at a Press Conference calling for Congress to pass the "Preventing Deepfakes of Intimate Images Act" sponsored by Representative Joe Morelle (D-NY). That past summer, Mani had unknowingly been a victim of abuse by her male peers, through the means of which they weaponized artificial intelligence to use existing images of her face to portray her as nude and committing sexual activities and behaviors. These violating depictions of her were then shared around various group chats and stories via Snapchat. She was not the only victim of this abuse at her school, however she quickly became the face of this case after her and her mother brought it to their school district's attention. [1]

Mani's desperate efforts to retrieve some of her dignity and protect herself against her classmates was met with silence and avoidance from her peers. Only when she contacted her district's congressmen and local news outlets did her story, and this growing problem, begin to gain traction. In fact, the number of non-consensual pornographic deep-fake videos available online has increased nine-fold since 2019. [2]

After months of toiling and relentless self-advocacy, Mani's sufferance finally paved the path for bi-partisan legislation signed into law by New Jersey Governor Phil Murphy in April 2025 that criminalizes deepfake imagery of minors or non-consenting adults, while also allowing victims to sue for up to \$1,000 per image. [3] There are still 9 states in the United States that lack NCII deepfake laws of their own, evincing a significant problem where victims inhabiting those states are left without just remediation. [4]

Statutory Architecture

First introduced in June 2024, the TAKE IT DOWN Act (Tools to Address Known Exploitation by Immobilizing Technological Deepfakes on Websites and Networks) operates as a means of criminalizing the willful creation and publication of Non-Consensual Intimate Imagery (NCII).

This legislation obligates platforms and companies to comply with federal statutes and take down any NCII within 48 hours of notice, criminalizes the act of “knowingly publishing intimate visual depictions or digital forgeries of an identifiable individual,” and requires that digital platforms provide a clear process in which victims can request the removal of sensitive content shared without consent. [5] The Federal Trade Commission (FTC) is the primary enforcer of TIDA, imposing civil penalties of \$53,088 per violation collected by the U.S. Treasury, not the victim. [6] TIDA also criminalizes publication as an indictable offense, carrying prison sentences of up to 2 years for adult targets and up to 3 years if the victim is a minor. [7]

Federal law is enacted, content gets removed, the perpetrator may get charged, and the victim remains scarred with no federal civil remedy. Not only does the Take it Down Act conspicuously omit a private right of action for victims, but it forces reliance on a convoluted patchwork of state NCII statutes, the unsettled Violence Against Women Act, or common-law theories which have notoriously misaligned with the complexities of AI and digital forgeries. [8]

The DEFIANCE (Disrupt Explicit Forged Images and Non-Consensual Edits) Act, which passed the Senate unanimously in January 2026 and remains held at the House desk, would close this gap by amending § 6851 to add a private right of action covering "intimate digital forgeries," with \$150,000 in liquidated damages and a ten-year statute of limitations. [9] Congress's decision to split the regulatory response across two bills produces a structurally incoherent enforcement system that prioritizes platform compliance over restoration of justice for victims, thus the Take It Down Act should be amended to incorporate the DEFIANCE Act's civil remedy rather than waiting for pending separate House action.

Public Enforcement Gaps

The primary problem for public enforcement is rooted in agencies' misunderstandings, underpreparedness, and negligence. According to an empirical analysis in *Cyberstalking: A Growing Challenge for the U.S.*

Legal System by the Office of Justice Programs, the number of cyberstalking filings in the U.S. has risen significantly between 2010 and 2014, while actual convictions remain low. [10] The legal system is short of the knowledge, technology, and associated resources needed to address such cyberstalking cases, similarly as they are with the newly arising dilemma of NCII deepfake abuse. [11]

Matthew C. Stephenson, in his book *Public Regulation of Private Enforcement*, posits that private enforcement in any form, including a private right of action for victims, can foment innovation in the form of new legal theories, novel techniques of investigation, and alternative approaches to dispute settlements. [12] Stephenson further suggests that private litigation in the U.S. has paved the way towards many precedents in antidiscrimination and antitrust law, showing how lawyers forge social change as a concomitant of legal challenges. [13] Private litigation fills the gaps that agency enforcement cannot mend, evincing the argument that Congress's failure to amalgamate TIDA's compliancy requirement with DEFIANCE's private right of action fails to serve rightful justice to American citizens.

One of the most evident defeats of the Take it Down Act is revealed in the understanding that the legislation was conceptually intended to fix discrepancies in NCII state statutes, yet it forces reliance on uncertain state interpretations that vary depending on jurisdiction. [14] Experts have noted that deepfakes in the legal forum are typically treated as “isolated problems,” falling variously under pornography law, privacy statutes, or sexual abuse laws depending on the state. [15] The Take it Down Act does little to coordinate state efforts towards recovering damages for victims, leaving them both unfulfilled and lost at deciphering whichever state’s law happens to apply.

Public Enforcement Gaps

A concept most worthy of consideration is the associated emotional and reputational damages that victims incur after NCII abuse. Danielle Citron and Mary Anne Franks in *Criminalizing Revenge Porn* discuss emotional and financial harms imposed on victims, including loss of sexual privacy and autonomy, humiliation, shame, and detriments to

self-expression. [16] Financial harms include costs of “reputation repair,” legal assistance fees, takedown efforts, and reputation-management services that can bring about job loss and greater economic hardship. [17] The Take it Down Act fails to account for any of these associated costs, showcasing a negligent governing effort to support victims of deepfake abuse.

Legislative Splits

To understand why Congress may have split these pieces of legislation, it is important to explore different hypotheses. The first hypothesis is embedded in anxieties of Section 230, claiming that a private right of action against platforms would collide with the logic of § 230, which rids websites and platforms of liability if they host user-created content. The Take It Down Act’s FTC-only platform enforcement evades any potential collision with § 230 because it does not shield platforms from federal criminal prosecution, as asserted through case law in *FTC v. Accusearch*. [18]

However, a private right of action under DEFIANCE against platforms can potentially hold platforms accountable if they promote or solicit illegal content, deeming them as “creators” and exempting them from § 230 immunity. A response to this argument alleges that the DEFIANCE Act’s private right of action is enforceable against creators and distributors, not platforms. The *Doe v. Internet Brands* decision extends this idea further, framing the concept that simply complying with federal law to remove content under TIDA does not constitute a platform acting as a publisher or creator, therefore evading confusion regarding immunity under Section 230. [19]

Further elucidating Congress’ rationale, fears over constitutional complications, First Amendment erosion, and political viability may have warranted the separation of TIDA and DEFIANCE. The cost of such precaution, however, is most evident within the gaps of the fragmented single-issue bills. A unified bill with civil remedy provisions delineated in the DEFIANCE Act serving as amendments to TIDA would harmonize definitions of AI-generated content, provide victims with a single pathway for both a right of action and content removal, and ensure

consistent regulation. Efforts to harmonize civil remedies with federal statutes have proven successful in the past, such as the Telephone Consumer Protection Act (TCPA) featuring both FCC regulatory authority and private rights of action. [20]

Policy Proposal

Amending the TAKE IT DOWN ACT to incorporate the DEFIANCE Act's civil remedy as a tangential cause of action would prove feasible, most importantly by merging the damages provision into the platform-obligation framework of TIDA. Section 230 would remain intact by maintaining immunity for platforms complying with notice-and-removal requirements while declining immunity for cases where platforms enable such heinous content to be shared. This would incorporate the DEFIANCE Act's ten-year statute of limitations stipulation, considering that victims often discover they have been targeted long after content is initially generated. Victims could recover up to \$150,000 in liquidated damages, while the mens rea requirements—including lack of consent, clear identifiability, and a reckless-disregard standard—would be upheld under a single amendment to TIDA.

Conclusion

For survivors and advocates of NCII deepfake abuse, it is a long and formidable legal battle to restore their dignity. This criticism of the Take it Down Act is not to underscore the efforts of Congress in combating this unfamiliar societal dilemma, but rather to enlighten politicians and citizens alike of the dire need for strict governance of the online realm. Not only does the government have the responsibility to hold platforms and perpetrators accountable, but by establishing a private right of action they must further provide vulnerable victims a chance to reclaim themselves from the hands of the perpetrator. As targeted NCII deepfake abuse proliferates and puts the livelihood of adults and minors in peril, the two-track problem between TIDA and DEFIANCE evinces the notion that removal is not remedy, and significant governmental action must be taken to repair the gap between takedown and recovering damages.

Notes

[1] Kaylee Williams, “Minors Are on the Frontlines of the Sexual Deepfake Epidemic — Here’s Why That’s a Problem,” Tech Policy Press, October 10, 2024, <https://www.techpolicy.press/minors-are-on-the-frontlines-of-the-sexual-deepfake-epidemic-heres-why-thats-a-problem/>.

[2] United States Senate Committee on the Judiciary, “Durbin Successfully Passes Bill to Combat Nonconsensual, Sexually-Explicit Deepfake Images,” January 13, 2026, <https://www.judiciary.senate.gov/press/dem/releases/durbin-successfully-passes-bill-to-combat-nonconsensual-sexually-explicit-deepfake-images/>.

[3] Williams, “Minors Are on the Frontlines.”

[4] Recording Law, “Deepfake & AI Voice Cloning Laws by State (2026),” July 9, 2026, <https://www.recordinglaw.com/us-laws/deepfake-laws/>.

[5] Congressional Research Service, “The TAKE IT DOWN Act: A Federal Law Prohibiting the Nonconsensual Publication of Intimate Images,” May 20, 2025, <https://www.everycrsreport.com/reports/LSB11314.html>.

[6] Federal Trade Commission, “Complying with the Take It Down Act,” May 8, 2026, <https://www.ftc.gov/business-guidance/resources/complying-take-it-down-act>.

[7] Eric Rosen, “Navigating the TAKE IT DOWN Act in Litigation,” Dynamis LLP, June 16, 2025, <https://www.dynamisllp.com/knowledge/navigating-take-it-down-act-in-litigation>.

[8] U.S. Department of Justice, “Violence Against Women Act,” June 10, 2026, <https://www.justice.gov/ovw/violence-against-women-act>.

[9] GovTrack.us, “Summary of S. 1837: DEFIANCE Act of 2025,” n.d., <https://www.govtrack.us/congress/bills/119/s1837/summary>.

[10] Office of Justice Programs, “Cyberstalking: A Growing Challenge for the U.S. Legal System,” n.d., <https://www.ojp.gov/library/publications/cyberstalking-growing-challenge-us-legal-system>.

[11] *Id.*

[12] Matthew C. Stephenson, “Public Regulation of Private Enforcement: The Case for Expanding the Role of Administrative Agencies,” *Virginia Law Review* 91 (2005), <https://virginialawreview.org/wp-content/uploads/2020/12/93.pdf>.

[13] *Id.*

[14] U.S. Representative Debbie Dingell, “Dingell, Colleagues Reintroduce Bipartisan TAKE IT DOWN Act to Protect Victims of Non-Consensual Intimate Imagery,” January 22, 2025, <https://debbiedingell.house.gov/news/documentsingle.aspx?DocumentID=5397>.

[15] Blog Admin, “The Take It Down Act Shows the Fragmentation of Policies Addressing Deepfakes and Generative AI,” LSE United States Politics and Policy (blog), September 23, 2025, <https://blogs.lse.ac.uk/usappblog/2025/07/10/the-take-it-down-act-shows-the-fragmentation-of-policies-addressing-deepfakes-and-generative-ai/>.

[16] Danielle Keats Citron and Mary Anne Franks, “Criminalizing Revenge Porn,” *Digital Commons at University of Maryland Carey Law* 49, no. 2 (January 1, 2014): 345, https://digitalcommons.law.umaryland.edu/fac_pubs/1420.

[17] *Id.* at 345.

[18] “Internet Law — Communications Decency Act — Federal District Court Denies § 230 Immunity to Website That Solicits Illicit Content,” *Harvard Law Review* 121 (2008), https://harvardlawreview.org/wp-content/uploads/2008/05/FTC_v_accusearch.pdf.

[19] HLR, “*Doe v. Internet Brands, Inc.*,” *Harvard Law Review*, March 24, 2023, <https://harvardlawreview.org/print/vol-130/doe-v-internet-brands-inc/>.

[20] Recording Law, “TCPA Explained: Robocall, Text, and Telemarketing Law (2026),” July 11, 2026, <https://www.recordinglaw.com/us-laws/tcpa/>.

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Algorithmic Borders: Human Rights Implications of AI-Driven Visa Adjudication Systems

Ye Min Hong

Columbia University in the City of New York

**“THE TECHNOLOGY IS ALREADY HERE.
THE ACCOUNTABILITY IS NOT.”**

Abstract

The United States has deployed, as of 2026, a suite of artificial intelligence tools across its immigration adjudication system—evidence classifiers, fraud-detection systems, risk-scoring platforms, and translation engines—that screen, classify, and flag visa applications before any human officer reaches a decision. This paper argues that the central failure of this system is not that its automated tools are proven to discriminate, but that they operate without any legally enforceable duty of transparency, audit, or contestability. As a result, algorithmic error and bias can be neither detected nor challenged by the people they affect, and the costs of that opacity fall on the population least equipped to bear them: non-citizens whose only safeguards are administrative, and therefore revocable at a change of administration.

Through a comparative legal analysis of three jurisdictions—the United States, the European Union, and South Korea—the paper shows that the jurisdiction with the most aggressive AI deployment offers the weakest legal protection to those subject to it. It identifies three categories of rights harm implicating the International Covenant on Civil and Political Rights, and argues that the remedy is legislative rather than judicial: the same plenary power invoked to justify the accountability gap is precisely what authorizes Congress to close it. The paper proposes five statutory reforms—mandatory disclosure, explainability, independent auditing, a right to human review, and data-standard requirements—that would bring U.S. practice into alignment with the rights-based frameworks its peers already require.

I. Introduction

On April 8, 2025, I received an email from my university's International Students and Scholars Office informing me that my SEVIS record had been auto-terminated. The stated reason was: "Other — Individual identified in criminal records check and/or has had their visa revoked." The notice did not say how long I had to leave — only that I should depart as soon as possible. No one, including my university's own advisers, could tell me what that meant in practice.

I was not alone. By mid-April 2025, over 1,840 international students and recent graduates from more than 280 colleges and universities had reported SEVIS record shifts — each assigned the same vague "Other" designation, with no citation to a specific provision of the Immigration and Nationality Act, and no explanation of what any of us had done to trigger the action. [1] Universities often discovered that their students' records had been terminated only after checking the SEVIS database themselves. Students, meanwhile, were making consequential decisions — whether to attend class, whether to move out of housing, whether to purchase flights home — with no reliable information about what was actually happening or who had the authority to explain it. That scale of simultaneous, undifferentiated action is what distinguishes this episode from an ordinary administrative error: it is consistent with automated, undifferentiated processing operating at machine speed, without individual review. Whether an automated system in fact drove the terminations has never been disclosed — and that very indistinguishability, from the outside, between algorithmic and human action is itself the first instance of the accountability gap this paper examines: where the two cannot be told apart, the affected person cannot know which occurred, much less contest it.

The case was ultimately resolved through litigation. A temporary restraining order entered on April 18, 2025 by Judge Victoria M. Calvert of the United States District Court for the Northern District of Georgia ordered the government to reinstate affected students' SEVIS records retroactive to March 31, 2025. [2] The government's acknowledgment of error, however, arrived only after the material consequences —

disrupted studies, forfeited employment, and compounded financial costs — had already been absorbed by the students the system was supposed to govern. That sequence — automated action first, legal accountability only later, through costly litigation — is not an anomaly of the 2025 crisis. As Section II demonstrates, it is the operating logic of AI-assisted immigration decision-making more broadly.

What the litigation could not address — and what drew me to the research underlying this paper — was a prior question: how did a system designed to enforce immigration law manage to act on nearly two thousand students simultaneously, without individual review, without legally adequate notice, and without any mechanism for those students to understand, in real time, what had happened to them or what they could do about it? The answer, I came to believe, was not primarily a question of legal authority. It was a question of administrative design. And the design flaw at its center was not unique to the 2025 termination wave — it was structural, systemic, and increasingly technological in character.

As of 2026, USCIS has deployed roughly twenty-nine distinct AI and machine learning systems in its adjudicative workflows, according to the Department of Homeland Security’s publicly available AI Use Case Inventory. [3] The ELIS Evidence Classifier automatically categorizes documents before any human officer reviews a file; USCIS has credited it with saving an estimated twenty-four million page scrolls of reviewer effort over an eight-month period and with nearly doubling the share of cases adjudicated within thirty days, from about 30 to 58 percent. [4] ImmigrationOS — built by Palantir Technologies under a thirty-million-dollar contract with ICE — aggregates data from IRS records, Social Security Administration databases, DMV registrations, passport systems, and license plate readers to construct profiles of immigration status holders and determine enforcement priorities. [5] Fraud detection systems cross-reference submissions against large datasets in ways that practitioners report are generating requests for evidence for documents that were, in fact, submitted. [6] EB-2 National Interest Waiver denial rates rose sharply across FY2025 — reaching 46

percent of adjudicated petitions by the third quarter, up from the mid-30s earlier in the year — though the government has not disclosed what role, if any, automated screening played in that increase. [7]

What remains unpublished — because DHS has not released it — is whether those outcomes distribute evenly across national origin, race, and religion, or whether the algorithms producing them replicate the discriminatory patterns embedded in the historical data on which they were trained. As this paper documents below, that absence of disclosure is not a bureaucratic oversight — it is the central legal and human rights problem this paper addresses.

This paper argues that the central failure of U.S. immigration adjudication is not that its automated systems are proven to discriminate, but that they operate without any legally enforceable duty of transparency, audit, or contestability — so that algorithmic error and bias can be neither detected nor challenged by the people they affect. Because the only safeguards that have ever applied to these systems are administrative, and therefore revocable at a change of administration, the remedy this paper defends is not judicial but legislative: the same plenary power invoked to justify the gap is precisely what authorizes Congress to close it, through the transparency, audit, and human-review obligations that the European Union and South Korea already require.

The argument proceeds in four sections. Section II maps the operational landscape of AI-driven visa adjudication, examining the key systems currently deployed, their scale, and their documented effects on outcomes. Section III conducts a comparative legal analysis across three jurisdictions — the United States, the European Union, and South Korea — assessing the adequacy of each framework to protect individuals subjected to AI-assisted adjudication. Section IV identifies three specific categories of rights harm: the unauditable risk of algorithmic bias under ICCPR Article 26, the transparency deficit and its relationship to due process, and the erosion of meaningful procedural

rights through the “decision support” framing. Section V proposes five legislative reforms that would bring U.S. practice into alignment with the rights-based frameworks developed by peer jurisdictions.

The claim of this paper is not that AI has no place in immigration administration. The scale of the U.S. system — over eleven million pending cases as of mid-2025 [8] — makes some degree of automated assistance operationally defensible. The claim, rather, is that the deployment of such tools without transparency, accountability, or meaningful recourse for those they affect is not an administrative oversight. It is a structural choice that transfers the full cost of algorithmic error onto the population least equipped to absorb it: the visa applicants, students, and immigrants whose legal status depends on systems they cannot see, cannot challenge, and are never told exist. I know what that cost looks like from the inside. This paper is an attempt to trace what it looks like from the outside — and to argue that the law should do better.

II. The Operational Landscape of AI-Driven Visa Adjudication

A. From Bureaucratic Processing to Algorithmic Triage

The United States Citizenship and Immigration Services has undergone a fundamental operational shift. What was once a paper-intensive, officer-driven review process has increasingly become an algorithmically mediated one, in which artificial intelligence tools screen, categorize, and flag cases — in a growing share of the workflow, before any human adjudicator reaches a substantive decision. The Department of Homeland Security publishes an AI Use Case Inventory listing every artificial intelligence and machine learning system deployed across DHS agencies; the January 28, 2026 update includes roughly twenty-nine distinct USCIS AI use cases — a figure drawn directly from the government’s own public disclosure, not from practitioner estimates. [9] These are not pilot programs or future-facing initiatives — they are embedded in daily adjudicative workflows. This

paper centers on that adjudicative layer — the systems that screen, classify, and score benefit applications submitted by people seeking or holding lawful status. It treats the enforcement tools described below, such as ImmigrationOS and ICE’s risk-scoring systems, not as its primary subject but as the more extreme expression of the same accountability gap: the legal posture of a person already in removal proceedings differs from that of a visa applicant, yet both are governed by automated processing that operates without any duty to disclose, explain, or justify what it has done.

A note on the evidentiary basis of what follows is warranted, because the same opacity this paper criticizes also constrains the evidence available to study it. Government disclosures establish that these systems exist, at what scale, and — in the case of the ELIS Evidence Classifier — with what efficiency gains. What those disclosures do not provide is per-case performance data: error rates, the weight automated outputs carry in individual adjudications, or their distributional effects across national origin, race, and religion. For the operational effects described below, this paper therefore relies in part on practitioner reporting — accounts by immigration attorneys and legal-services organizations of what they observe in their cases — rather than on primary agency data. Such reporting is probative but not conclusive; it is flagged as practitioner observation throughout, and the resulting account of how these systems affect applicants should be read as indicative of documented mechanisms rather than as a quantified measure of their aggregate impact. That evidentiary gap is not incidental to the argument. It is a direct consequence of the disclosure vacuum this paper identifies as the central problem, and it is precisely why the reforms proposed in Section V center on the auditing and transparency obligations that would make a more definitive account possible.

DHS maintains that human adjudicators retain final decision-making authority, and that AI systems do not grant or deny immigration benefits. [10] That formal position, however, obscures a more complex reality: automated tools shape which evidence surfaces for review,

which inconsistencies get flagged, and which cases arrive at a human officer's desk already marked for heightened scrutiny. Final authority and sole influence are not the same thing.

B. Key AI Systems Currently Deployed

1. ELIS Evidence Classifier

The Electronic Information System (ELIS) Evidence Classifier is a machine learning tool that automatically categorizes and indexes documents submitted with immigration petitions. Over the period from September 2021 to May 2022, USCIS credited the tool with saving roughly twenty-four million page scrolls of reviewer effort — on the order of 13,000 hours — and with nearly doubling the share of cases adjudicated within thirty days, from about 30 to 58 percent. These are efficiency gains, not measures of accuracy: USCIS has not published any error-rate data, and practitioners report requests for evidence for documents that were in fact submitted, consistent with classifier mis-tagging. [11]

Per the 2024 USCIS Privacy Impact Assessment, the Evidence Classifier categorizes and surfaces documents for human review; it does not make adjudication decisions. Practitioners describe a related downstream effect: when a document bundle is mis-scanned, out-of-order, or illegible, the classifier's confidence in its categorization drops, and the adjudicating officer is correspondingly more likely to request a secondary evidence submission. [12] This means a purely technical failure in the AI layer can generate a procedural burden on the applicant that mirrors the effect of a substantive evidentiary deficiency.

2. ImmigrationOS

ImmigrationOS represents a qualitatively different category of AI intervention. In April 2025, ICE awarded Palantir a thirty-million-dollar contract to build ImmigrationOS, a unified case-management and analytics platform that aggregates IRS tax records, Social Security Administration data, DMV registrations, passport and visa data, and

license plate reader data. [13] Federal officials have said the platform draws these disparate records together to prioritize enforcement targets and to track individuals as they move through the immigration system. [14]

Prior visa applications, travel histories, and business registrations may be cross-referenced in ways that were not previously routine. Errors that once went unnoticed may now trigger delays, requests for evidence, or referrals for further review. [15]

3. StateChat and the Document Translation Service

StateChat, a generative AI chatbot the State Department launched in late 2024, is used by consular and departmental staff to interpret internal guidance, draft communications, and coordinate across posts — functions that increasingly touch the visa-issuance process. [16]

Separately, USCIS officers can now upload foreign-language documents and receive near-instant AI-generated translations via Microsoft Azure AI Translator. Whether that translation is preserved in the administrative record, and in what form, remains an open question — one that matters most in proceedings where credibility may hinge on precise wording. [17]

4. Fraud Detection Systems (FDNS) and Babel X

USCIS and its Fraud Detection and National Security Directorate rely on AI systems to flag anomalies and cross-reference filings, while Customs and Border Protection uses Babel X, a tool that mines social media and other publicly available data to build profiles of travelers and screen them at ports of entry. [18]

C. Measurable Consequences for Applicants

The cumulative effect of these systems on adjudication outcomes is real, but the denial-rate trend and the role of automation within it must be kept analytically distinct. The trend itself is now a matter of public record: USCIS Form I-140 data show the EB-2 NIW denial rate climbing across FY2025, from roughly 37 percent of adjudicated petitions in the first quarter to 46 percent in the third; for the fourth quarter of FY2025, a National Foundation for American Policy analysis of subsequent USCIS

data places the rate at 64.3 percent, with EB-1A denials climbing in parallel. Analysts and a former USCIS adjudicator attribute this increase primarily to a more exacting application of the Dhanasar standard, rather than to any announced change in the governing legal test or to automation as such. What cannot be established from the public record is the contribution of AI-assisted screening to that trend, because USCIS has released no data isolating it. The role of automation is documented instead at the level of mechanism: practitioners report that mis-tagged evidence generates requests for evidence for documents that were in fact submitted, and that cross-document mismatch detection causes small discrepancies in job titles, dates, or signatures to trigger automated flags. Whether these operational effects move the aggregate denial rate, and by how much, is precisely the question that the absence of disclosure makes impossible to answer. [19]

Automation does not need to ‘decide’ the case to materially affect it. [20] This observation captures the central problem this paper examines: a system in which no single automated decision is formally determinative, but in which algorithmic processing at every stage of the review workflow can systematically disadvantage applicants who lack knowledge of, access to, or recourse against the processes shaping their cases. The question that follows is whether existing legal frameworks are equipped to govern a system that operates this way — and, as Section III demonstrates, the answer across all three jurisdictions examined is inadequate, and in the United States, critically so.

III. A Comparative Legal Analysis of AI Governance Frameworks

The legal landscape governing AI-assisted adjudication varies considerably across jurisdictions. An examination of three frameworks — the United States, the European Union, and South Korea — reveals not only divergent regulatory approaches but also a fundamental asymmetry: the jurisdiction with the most aggressive AI deployment in immigration adjudication has the weakest legal protections for those affected by it.

A. The United States: Broad Deference, Narrow Protections

1. The Plenary Power Doctrine and Its Limits

American immigration law rests on a doctrine that has no parallel in domestic civil law: the plenary power doctrine, which holds that Congress exercises nearly unreviewable authority over immigration matters. Long-standing Supreme Court precedent has recognized Congress as having plenary power over immigration, giving it almost complete authority to decide whether foreign nationals may enter or remain in the United States. [21] This structural deference to the political branches substantially constrains the judiciary’s ability to scrutinize the procedural mechanisms through which immigration decisions are made.

The doctrine does not, however, extinguish due process protections entirely. The Supreme Court has held that aliens physically present in the United States, regardless of their legal status, are recognized as “persons” guaranteed due process of law by the Fifth and Fourteenth Amendments, and that even those whose presence is unlawful, involuntary, or transitory are entitled to that constitutional protection. [22] The critical word, however, is “present.” The due process protections that attach to persons within U.S. territory thin considerably at the border and dissolve almost entirely for visa applicants abroad.

2. The Administrative Procedure Act and Its Gaps

The Administrative Procedure Act (APA) provides a second avenue for challenging agency action, requiring that final agency decisions be supported by reasoned explanation and be neither arbitrary nor capricious. Where AI tools contribute to a denial, however, the APA’s utility is limited: AI-flagged cases may move through the adjudicative pipeline in ways where automation influences the front end of adjudication — shaping timelines, scrutiny levels, and documentation burdens — without constituting a “final agency action” subject to APA review. [23]

3. Assessment: A Protective Gap

The United States has no federal statute requiring transparency, auditing, or explainability for AI systems used in immigration adjudication. The Court remains implicitly willing to give the political branches leeway to use immigration authorities in rough-hewn ways. And lower courts that might otherwise expand individual-rights review are constrained by the expectation that they would overvalue individual interests and undervalue governmental ones. [24] In the absence of legislative action, applicants affected by AI-assisted adjudication have no right to know that automated tools were involved, no right to receive an explanation of how their file was processed, and no meaningful avenue for challenging algorithmic errors that shaped — but did not formally decide — their cases. The only accountability framework that has ever applied to immigration AI has been administrative rather than statutory — a sequence of OMB memoranda that a single change of administration revised and, in 2025, replaced (see Section IV.B.2, *infra*). A safeguard that one executive can install and the next can rescind is not, in any meaningful sense, a legally enforceable protection for the applicants it nominally covers.

B. The European Union: A Rights-First Architecture

1. GDPR Article 22: The Right Against Automated Decision-Making

The General Data Protection Regulation, in force since 2018, establishes in Article 22 a right not to be subject to decisions based solely on automated processing that produce legal or similarly significant effects. Article 22 restricts solely automated decisions with legal or similarly significant effects unless strict conditions and safeguards are met, typically requiring meaningful human involvement. [25]

The practical significance of this protection has been expanded by EU judicial interpretation. The European Court of Justice's *Schufa* decision reinforced a broad reading of Article 22, holding that generating a score through automated profiling can itself constitute an automated decision where third parties rely heavily on that score —

meaning that AI-generated rankings or flags that effectively determine AI-generated rankings or flags that effectively determine outcomes may be treated as automated decisions even if a human nominally approves the result. [26]

2. The EU AI Act: High-Risk Classification and Structural Obligations

The EU AI Act, which entered into force in August 2024, creates a second layer of protection. Its high-risk regime — including the Article 86 right to explanation — was originally scheduled to apply from August 2026 but was deferred to December 2027 for standalone Annex III systems, which include border-control and law-enforcement decision-support tools, under the Digital Omnibus, Regulation (EU) 2026/1744, a delay that itself illustrates how demanding these obligations are to operationalize. Article 86 introduces an individual right: any person subject to a decision significantly affected by a high-risk AI system is entitled to a clear explanation of the AI's role in that decision, the main parameters that influenced the output, and the level of human oversight involved — a right that extends beyond GDPR Article 22, which covers only fully automated decisions. [27]

3. Assessment: A Model With Structural Limitations

The EU framework represents the most developed effort to subject AI systems to rights-based constraints. Its primary limitation, for purposes of this paper, is geographic: its protections extend to individuals within the EU, and visa applicants from outside that jurisdiction may not benefit from its provisions to the same degree. Moreover, the AI Act and GDPR are complementary instruments that must both be satisfied: compliance with one does not discharge obligations under the other, and failure to comply with either can result in separate penalties. [28]

C. South Korea: A Progressive Framework With Unresolved Gaps

1. The AI Framework Act: Structure and Scope

South Korea enacted the Framework Act on Artificial Intelligence Development and Establishment of a Foundation for Trustworthiness on

January 21, 2025, with full enforcement beginning January 22, 2026. Although the EU was first to pass comprehensive AI legislation, South Korea became the first country to bring such legislation into force, as the EU's high-risk obligations — now deferred to December 2027 — remain pending. [29]

The Act's regulatory architecture centers on the concept of "high-impact AI," defined as systems that may significantly affect or pose risks to human life, physical safety, and fundamental rights, and that operate in sectors including healthcare, energy, transportation, and public services. The Act stipulates duties to ensure AI transparency and AI safety, and sets forth specific obligations for operators of high-impact AI systems. [30]

2. Core Obligations: Articles 31, 32, and 34

Under Article 31, AI business operators offering high-impact or generative AI products must notify users in advance that the product or service utilizes AI. [31] Article 32 requires operators of AI systems exceeding prescribed computational thresholds to identify, assess, and mitigate risks throughout the AI lifecycle. Operators involved with high-impact AI are obligated to establish and implement risk management measures and user protection plans, and to ensure human oversight and supervision of high-impact AI systems under Article 34. [32]

3. Assessment: An Incomplete Application to Immigration

South Korea's AI Framework Act constitutes a meaningful advance over the U.S. approach. For a Korean visa applicant processed by a USCIS AI system, however, the Act provides no direct protection: it governs operators within South Korean jurisdiction, not U.S. federal agencies. The Act has been criticized for prioritizing industrial promotion over proactive regulation, and for lacking adequate protective measures compared to the EU AI Act, which imposes stricter sanctions on AI technologies that may be ethically or socially harmful. [33]

D. Comparative Summary

The three frameworks reveal a pattern that is troubling rather than reassuring. The jurisdiction with the most extensive AI deployment in immigration adjudication — the United States — has the least developed legal architecture for protecting applicants' rights within that system. The EU has constructed the most robust rights-based framework, but its geographic scope limits its relevance to applicants processed by U.S. agencies. South Korea has demonstrated political will to regulate AI comprehensively, but has not yet extended those protections explicitly to the immigration context or to its own nationals processed abroad. The gap that emerges from this comparative analysis is not merely a technical regulatory deficit. It is a structural asymmetry in which the people most affected by AI-assisted adjudication — visa applicants, asylum seekers, and migrants — are also the least protected by the legal frameworks currently governing that adjudication.

To see what this asymmetry means in practice, consider a hypothetical applicant — a petitioner for an EB-2 National Interest Waiver whose supporting documentation is assigned a low-confidence tag by an automated classifier of the kind described in Section II, prompting a request for additional evidence. The automated input is the same in each jurisdiction; only the legal consequences differ. In the European Union, she could invoke Article 22 of the GDPR — in force since 2018 — to demand an explanation of the automated role in her case, and, once the AI Act's Article 86 takes effect, a further right to explanation specific to high-risk systems. In South Korea, the Framework Act's notice and human-review obligations would entitle her to be told that an automated system had been used and to seek human reconsideration. In the United States, she would be entitled to neither: no statute requires the agency to disclose that automation played any part, and no mechanism exists for her to contest it. The same fact pattern, run

through three legal systems, yields three different levels of protection — and the jurisdiction subjecting her to the most automated processing offers her the least recourse.

IV. Human Rights Implications of AI-Driven Visa Adjudication

The structural asymmetry identified in Section III — the most AI-intensive jurisdiction offering the least legal protection — has concrete human rights consequences. This section identifies three specific categories of rights harm that flow from the regulatory vacuum documented above: the unauditable risk of algorithmic bias and the right to non-discrimination under ICCPR Article 26 (Part A); the transparency deficit and its relationship to due process (Part B); and the erosion of meaningful procedural rights through the “decision support” framing (Part C). Taken together, these three categories demonstrate that the costs of the current regulatory gap are not abstract — they fall on identifiable people, in identifiable ways, with legal consequences that existing frameworks have failed to address.

A. The Unauditable Risk of Algorithmic Bias and the Right to Non-Discrimination

1. The Technical Mechanism of Bias

Algorithmic bias is not a hypothetical risk; it is a well-documented failure mode of systems trained on historically generated data. Whether the specific tools deployed in U.S. immigration adjudication exhibit it cannot be established on the public record — not because the risk is absent, but because DHS has disclosed no data that would allow it to be detected or audited. Discrimination may occur at any point in the AI lifecycle as a consequence of data, social, or technical bias, or the mode of implementation. When training data are not representative of the population under consideration, or when they embed real-world discrimination, such discrimination is transmitted to the AI system and

disproportionately impacts those already experiencing historical disadvantage. [34]

Commentators warn that AI migration tools, trained on data that encodes historical discrimination and built by developers without expertise in racial bias, operate within legal systems that offer little transparency or recourse — and so risk reproducing the very inequities they claim to eliminate. On this view, absent robust safeguards, such systems can entrench racialized exclusion while presenting themselves as neutral efficiency measures. [35]

2. Concrete Examples: Risk Scoring Without Accountability

In 2024, more than 140 advocacy groups asked DHS to suspend select immigration AI pilots, arguing that ensuring fair, unbiased decision-making is difficult without full transparency into the algorithms that power tools like the ICE Hurricane Score and Risk Classification Assessment (RCA). [36] DHS characterizes these tools as informing rather than making custody and supervision decisions; critics counter that in practice they shape whether migrants are released or subjected to electronic monitoring. What is not disputed is that the algorithm's role is disclosed neither to the affected individual nor to their legal counsel. [37]

3. The International Human Rights Framework: ICCPR Article 26

The applicable international standard is Article 26 of the International Covenant on Civil and Political Rights (ICCPR), which guarantees all persons equal and effective protection against discrimination on any ground including race, national origin, and religion. The United States has ratified the ICCPR, accepting obligations that extend beyond its domestic constitutional guarantees. Those obligations are not, on their own, enforceable by an individual in a U.S. court: the Senate ratified the Covenant subject to a declaration that its substantive articles are non-self-executing. That limitation, however, cuts in favor of the argument advanced here rather than against it. Because the ICCPR binds the United States as a matter of state responsibility while

conferring no privately enforceable right, it is precisely Congress — not the judiciary — that must give its guarantees domestic effect. The Covenant thus does not supply a cause of action for the visa applicant; it supplies a measure of what the legislature owes her. [38]

What makes regulation of algorithms particularly complicated is the problem of procedural invisibility: algorithms arrive at outcomes through processes that are almost impossible to reconstruct, even for their designers. The input of data may not be discriminatory on its face, but may reflect existing patterns of structural discrimination — a complexity that cannot be adequately addressed by the broad, general definitions of non-discrimination as they appear in ICCPR and other international human rights instruments. [39]

B. The Transparency Deficit and the Right to Know

1. Algorithmic Opacity in Legal Context

Transparency is not merely a technical aspiration in the context of government decision-making — it is a legal requirement grounded in due process, reasoned decision-making, and the right to contest adverse determinations. Procedural justice requires that affected parties understand the grounds for decisions affecting them and have meaningful opportunity to contest adverse determinations. The inadequacy of existing disclosure mechanisms has been documented in ongoing federal litigation: advocacy organizations have sued DHS and USCIS under the Freedom of Information Act specifically to compel disclosure of the agency's AI use policies and systems. [40]

A well-known problem with AI systems is their non-explainability: the reasoning underlying a given output may be inaccessible even to the system's developers. And even if developers and users understand what is behind a decision, that understanding may not be transmissible to the affected traveler in any meaningful form. Moreover, automated risk assessment systems can and do make mistakes. [41]

The legal significance of this opacity was recognized in U.S. case law: a

court found that an algorithmic scoring system used to rank teachers had to be open to scrutiny in order to permit affected teachers to verify the accuracy of their score and challenge the decision if they disagreed. Without access to the underlying equations, codes, decision rules, and assumptions, those affected could not exercise their constitutionally protected due process rights — their scores remained a “black box, impervious to challenge.” [42]

2. The Absence of a Federal Explainability Obligation

The Trump administration’s January 2025 Executive Order rescinding the Biden administration’s AI governance framework eliminated the primary federal mechanism for requiring transparency in government AI systems. In its place, OMB Memorandum M-25-21 (April 2025) rescinded and replaced M-24-10, folding the earlier “rights-impacting” and “safety-impacting” categories — and the individualized review they triggered — into a single “high-impact AI” definition oriented toward accelerating adoption rather than constraining it. That successor framework still treats AI used to assess a foreign national’s entry into or continued presence in the United States as presumptively high-impact, and it retains an instruction that agencies offer consistent remedies or appeals. But the memorandum provides on its face that it creates no enforceable right or benefit and governs only agencies’ own use of AI — so neither the high-impact designation nor the remedies instruction gives a visa applicant anything to invoke. [43]

There is no federal statute, no applicable executive order, and no agency regulation requiring USCIS or ICE to disclose to visa applicants that their files have been processed by automated tools, what those tools concluded, or what weight that automated processing was given by the human officer who ultimately signed the decision. DHS has not published outcome-based performance metrics comparing AI-assisted processing to traditional processing, nor has it released data evaluating whether automated assistance changes approval or denial rates. [44]

3. Contrast With the EU Standard

The gap becomes most visible in comparative perspective. The EU

affords such applicants a right to explanation that the U.S. framework lacks: the GDPR's Article 22 already constrains solely automated decisions, and the AI Act's Article 86 will, once in force, add a right for any person subject to a significantly affecting high-risk decision to a clear explanation of the AI's role in the decision, the main parameters that influenced the output, and the level of human oversight involved. [45] The U.S. framework offers applicants no equivalent right.

C. Due Process Erosion and the Right to a Fair Hearing

1. The Procedural Stakes

The right to a fair hearing before an adverse government determination is a core principle of both U.S. constitutional law and international human rights law. ICCPR Article 14 guarantees the right to a fair and public hearing by a competent, independent, and impartial tribunal. AI-assisted adjudication undermines these guarantees in ways that are not fully captured by the formal preservation of human decision-making authority. When AI tools pre-sort, pre-classify, and pre-flag applications, the human officer who reviews the resulting file exercises judgment within a decision space that has already been shaped by automated processing that the applicant had no opportunity to contest, correct, or even observe.

2. The “Decision Support” Fiction

The “decision support” framing — under which AI tools influence but do not formally decide cases — obscures the due process problem rather than resolving it. It keeps the legal accountability framework anchored to a human decision-maker, even as the substantive factors shaping that decision are generated by systems operating outside any procedural safeguard. The ELIS Evidence Classifier, described in Section II, shows how this works in practice. Consider an EB-2 petition supported by documentary evidence. When the classifier assigns a low-confidence tag to one of those documents — because the scan is poor or the pages are out of order — the officer who inherits the file becomes measurably more likely to issue a request for evidence, and the petitioner must then rebut a deficiency that originates not in the merits of her case but in an

automated mis-read she never sees. No official has formally “decided” anything against her, and no adverse determination appears in the record; yet her evidentiary burden, her timeline, and her odds of success have all shifted. That is why the “decision support” label operates as a fiction: it locates accountability in a human choice that has already been pre-shaped by an unseen process, and it is this structural opacity that threatens the right to effective judicial protection. The same pattern recurs in enforcement. When an automated risk assessment flags a traveler, officers may subject them to more extensive inspection, and the traveler has no practical means of ascertaining why that flag was generated. [46]

3. The Compounding Effect on Vulnerable Populations

These due process deficits do not distribute evenly. They fall most heavily on applicants who lack legal representation, who submit from countries with limited access to qualified immigration counsel, and who do not read English fluently enough to parse the technical requirements of a request for evidence or notice of intent to deny. Practitioners report that AI-driven data-matching systems can produce false mismatches, leading to rejections of properly filed petitions; that mis-tagged evidence can generate requests for evidence for documents that were actually submitted; and that cross-document mismatch detection can cause small discrepancies in job titles, dates, or signatures to trigger automated flags. [47] For applicants who cannot identify the source of a deficiency notice, the gap between formal procedural rights and practical access to those rights widens into something close to a denial of meaningful process.

V. Conclusion and Policy Recommendations

A. Principal Findings

This paper has traced a specific and consequential arc. The United States government has deployed, as of 2026, a suite of AI tools — ELIS Evidence Classifier, ImmigrationOS, StateChat, Babel X, and various

fraud detection and risk scoring systems — that collectively reshape the processing of visa applications before any human adjudicator exercises meaningful independent judgment. These tools influence which documents surface for review, which applicants are flagged for heightened scrutiny, and which files arrive at a human officer’s desk already pre-sorted by algorithmic criteria that the applicant cannot observe, contest, or correct.

The comparative legal analysis in Section III demonstrated that this deployment is occurring in a regulatory environment that is structurally the weakest of the three jurisdictions examined. The European Union has constructed a layered framework — GDPR Article 22 and the EU AI Act — that provides affected individuals with substantive rights to explanation, human review, and contestability of AI-influenced decisions. South Korea’s AI Framework Act, in force as of January 22, 2026, establishes mandatory transparency obligations, safety requirements, and human oversight duties for high-impact AI systems. The United States, by contrast, has no federal statute requiring disclosure, explainability, auditing, or human review rights in the context of AI-assisted immigration adjudication.

The human rights analysis in Section IV identified three specific categories of rights harm: the unauditable risk of algorithmic bias, which where present would violate ICCPR Article 26’s guarantee of equal protection; a transparency deficit that undermines both the procedural right to a reasoned decision and the substantive ability to contest adverse outcomes; and the erosion of meaningful due process through the “decision support” framing that preserves formal human authority while allowing algorithmic processing to shape decision outcomes outside any procedural safeguard.

B. Policy Recommendations

The following five recommendations address, in turn, the three categories of harm identified in Section IV. Recommendations 1 and 2 respond to the transparency deficit; Recommendation 3 addresses

algorithmic bias; Recommendation 4 responds to due process erosion; and Recommendation 5 addresses the data conditions that generate both bias and opacity. Each is drawn from existing legislative proposals or comparative international models and is designed to be actionable within the current U.S. statutory framework.

Recommendation 1: Mandatory Disclosure and Notice

Congress should enact legislation requiring that any applicant whose immigration file has been processed, classified, flagged, or scored by an automated system receive written notice of that fact as part of any request for evidence, notice of intent to deny, or denial. The notice should identify the AI tools involved, describe in plain language the type of processing performed, and explain what role, if any, that processing played in the adjudicating officer's review. This recommendation draws on the model established by the White House Blueprint for an AI Bill of Rights, which advanced requirements for notice to individuals and explanations of automated decisions. [48]

Recommendation 2: Statutory Explainability Requirements

Congress should require that USCIS and ICE maintain documentation of the logic, parameters, and training data characteristics of all AI systems used in immigration adjudication, and make this documentation available for inspection in proceedings where an applicant challenges an adverse determination. A U.S. court has already recognized that where an algorithmic scoring system affects individuals' rights, those individuals must have access to the underlying equations, decision rules, and assumptions in order to exercise their constitutionally protected rights — without which the AI system remains a mechanism of adjudication that is, in practice, impervious to challenge. [49]

Recommendation 3: Independent Algorithmic Auditing

Congress should establish a mandatory, periodic, independent audit regime for AI systems used in immigration adjudication. Many state AI bills and proposed federal legislation impose affirmative reporting requirements and bias audit mandates, requiring impact assessments

to check for discriminatory effects against protected groups. [50] A federal immigration-specific audit mandate should require auditors — drawn from a roster of independent technical and civil rights experts, not agency personnel — to assess whether AI tools produce statistically differential outcomes across national origin, race, religion, or other protected characteristics, and to publish those findings publicly.

Recommendation 4: A Right to Human Review

Congress should codify a right for visa applicants to request de novo human review of any determination that was preceded by an automated flag, risk score, or AI-generated evidentiary classification. This right should be available without cost to the applicant, should suspend any adverse determination pending completion of the review, and should require the reviewing officer to document independently, without reference to prior AI outputs, the evidentiary basis for their decision. GDPR Article 22 already establishes this principle in EU law, requiring meaningful human involvement as a safeguard against decisions based solely on automated processing. [51]

Recommendation 5: Data Minimization and Bias Mitigation Standards

Congress should require that AI systems deployed in immigration adjudication be trained only on data that has been reviewed and certified for demographic representativeness, and that DHS publish, on an ongoing basis, outcome data disaggregated by national origin, race, and religion for each AI tool in use. The Equal Employment Opportunity Commission has recognized that if an algorithm results in a disparate impact on protected classes, liability attaches regardless of whether the tool was internally developed or procured from a third-party vendor. [52] The same principle should apply to government AI tools in immigration adjudication.

The strongest objection to this paper’s recommendations is structural rather than empirical. One might argue that the current state of American immigration AI reflects not an oversight but a considered constitutional design: that Congress exercises plenary power over

immigration — nearly unreviewable by the courts — precisely because immigration involves sovereign judgments about national membership that the political branches are best equipped to make. On this view, the roughly eleven-million-case USCIS adjudication backlog [8] makes some degree of automated processing a practical necessity, and the absence of judicially enforceable transparency obligations reflects a deliberate legislative choice to leave the executive branch free to modernize adjudication without judicial second-guessing. Fraud detection and national security screening, the argument continues, depend on confidentiality that disclosure mandates would compromise.

This objection, pressed to its conclusion, refutes itself. The plenary power doctrine constrains judicial review — it does not constrain Congress. The recommendations offered in this paper are directed not at courts but at the legislature: they ask Congress to exercise the very authority that the plenary power doctrine confirms it possesses. A statutory transparency mandate, an auditing requirement, or a data-disclosure obligation for immigration AI is not a judicial incursion into executive prerogative; it is an act of Congress regulating the administrative apparatus that Congress itself created. Far from being blocked by plenary power, legislative reform of the kind proposed here is precisely what the doctrine contemplates: political-branch accountability for political-branch decisions. The objection thus becomes a reason to pass legislation, not a reason to forgo it.

Nor does the confidentiality interest in fraud detection and national security screening argue against accountability as such: transparency can be designed with security in mind — disclosure to cleared counsel under protective order, in camera review, and audit findings reported in aggregate rather than case by case — so that the choice is not between secrecy and oversight, but between calibrated disclosure and none. And while the domestic due process protections invoked here reach applicants within U.S. territory but thin for those abroad, the non-discrimination and fair-hearing obligations the United States accepted in ratifying the ICCPR are not territorially bounded in the same

way — so the largest affected class is not, as the objection assumes, beyond the reach of any enforceable norm. And the efficiency argument, however weighty, proves only that AI processing should be permitted — not that it should be unaccountable. The EU AI Act and South Korea’s AI Framework Act demonstrate that transparency obligations and human oversight requirements can coexist with large-scale automated adjudication; the choice is not between efficiency and rights, but between efficiency with accountability and efficiency without it.

One limitation remains honestly on the table. That Congress can close this gap is not a promise that it will — immigration legislation is notoriously difficult to move, and the appetite to constrain the executive’s own tools may be scarce. But whether a reform is politically feasible and whether it is legally justified are distinct questions, and this paper defends the latter: what the law ought to require, not a forecast of when Congress will require it.

C. Toward a Rights-Based Framework for Algorithmic Adjudication

The deeper argument of this paper is not technical — it is normative. The adoption of AI tools in immigration adjudication has been framed, institutionally and in public discourse, as a question of administrative modernization: how can government agencies process more applications, more quickly, with fewer resources? That framing is not wrong, but it is incomplete. It treats efficiency as the primary value and human rights as a constraint to be managed, rather than recognizing that the legitimacy of an immigration system depends not only on its throughput but on its fairness.

The EU and South Korean frameworks reflect not merely that transparency and efficiency can coexist, but a deliberate ordering of priorities. They begin from the premise that AI systems affecting significant personal interests must be designed to protect rights first and optimize efficiency second. They embed that priority in legally enforceable obligations: explainability rights, audit mandates, human oversight requirements, and penalties for non-compliance.

The United States has, as of 2026, no equivalent framework for immigration AI. That absence is not a temporary gap waiting to be filled by future legislation — it is an active choice, reflected in the 2025 replacement of the prior AI-governance framework with one that creates no rights enforceable against the government, that imposes the costs of algorithmic error and discrimination on the population least equipped to bear them. The five recommendations offered here are not a comprehensive solution; they are a floor — the minimum legal infrastructure necessary to ensure that the efficiency gains of AI-assisted adjudication do not come at the cost of the rights of the people it processes.

This paper's argument stands or falls on its own terms, independent of its author's history. I note only that I write from a particular vantage point — having been, in 2025, subject to the kind of automated, unexplained processing this paper examines — and offer that not as authority, but as the reason the question first seemed worth asking.

An immigration system that cannot explain its decisions, that cannot demonstrate it is non-discriminatory, and that offers no meaningful recourse when its automated tools produce unjust results is not simply an inefficient system in need of reform. It is a system in which the rule of law has been partially displaced by algorithmic administration — and in which the human beings most affected by that displacement have been given no legal tools to push back. None of this is abstract. Behind every flagged file and every unexplained denial is a person who was never told what a machine decided about their life — and who deserved, at the very least, a system willing to explain itself. The technology is already here; the accountability is not. The only real question left is whether the law will catch up before the next two thousand people find themselves, as I once did, unable to learn why.

Notes

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[2] *Jane Doe 1 v. Bondi*, No. 1:25-cv-01998 (N.D. Ga. filed Apr. 11, 2025) (TRO entered Apr. 18, 2025, by Judge Victoria M. Calvert, ordering reinstatement of plaintiffs’ SEVIS records retroactive to March 31, 2025). See also ACLU of Georgia, “Temporary Restraining Order Granted in Student Visa Revocation Case *Doe v. Bondi*,” April 21, 2025, <https://www.acluga.org/press-releases/temporary-restraining-order-granted-student-visa-revocation-case-doe-v-bondi>.

[3] U.S. Department of Homeland Security, “AI Use Case Inventory,” January 28, 2026, <https://www.dhs.gov/ai/use-case-inventory/uscis>; Oleg Gherasimov, “USCIS Artificial Intelligence: What It’s Doing to Your Immigration Case Right Now,” SG Legal Group, April 23, 2026, <https://www.sglegalgroup.com/blog/uscis-artificial-intelligence-what-it-s-doing-to-your-immigration-case-right-now> (reporting “roughly 29 USCIS AI use cases” in the January 28, 2026, DHS inventory).

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<https://www.dhs.gov/publication/dhsuscispia-056-uscis-electronic-immigration-system-uscis-elis>. These figures reflect processing efficiency, not adjudicative accuracy; USCIS has published no error-rate data for the classifier. The broader pattern of rising requests for evidence and denials is reported in Cozen O'Connor, "Growing Use of Artificial Intelligence in U.S. Immigration Adjudications Is Driving Higher RFE and Denial Rates," April 27, 2026, <https://www.cozen.com/news-resources/publications/2026/growing-use-of-artificial-intelligence-in-u-s-immigration-adjudications-is-driving-higher-rfe-and-denial-rates>.

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[6] Cozen O'Connor, "Growing Use of Artificial Intelligence."

[7] U.S. Citizenship and Immigration Services, "Form I-140 by Fiscal Year, Quarter and Case Status" (FY2025, Q1–Q3), <https://www.uscis.gov/tools/reports-and-studies/immigration-and-citizenship-data> (National Interest Waiver adjudicated-denial rate rising from 37.3% in Q1 to 46.0% in Q3 FY2025, and 38.8% in aggregate for Q1–Q3; denial rate computed as denials divided by adjudicated petitions). For the fourth quarter of FY2025, a National Foundation for American Policy analysis of subsequent USCIS data reports a 64.3% denial rate: Stuart Anderson, "U.S. Immigration Service Increases Denials for High-Skilled Immigrants," *Forbes*, April 22, 2026, <https://www.forbes.com/sites/stuartanderson/2026/04/22/us-immigration-service-increases-denials-for-high-skilled-immigrants/>. Available analyses attribute the increase principally to a stricter application of the *Dhanasar* standard; no published USCIS data isolates the contribution, if any, of AI-assisted screening to the trend.

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[9] U.S. Department of Homeland Security, “AI Use Case Inventory”; *Nextgov/FCW*, “Law Enforcement Is the Leading DHS Use Case for AI,” January 29, 2026, <https://www.nextgov.com/artificial-intelligence/2026/01/law-enforcement-leading-dhs-use-case-ai/411063> (reporting 238 total DHS AI use cases in the January 28, 2026, inventory).

[10] Gherasimov, “USCIS Artificial Intelligence.”

[11] ACT-IAC, “Evidence Classifier” (page-scroll and 30-day adjudication figures); Cozen O’Connor, “Growing Use of Artificial Intelligence” (classifier mis-tagging and resulting requests for evidence).

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[16] Greenspoon Marder, “How Artificial Intelligence Is Revolutionizing Immigration Compliance.”

[17] Gherasimov, “USCIS Artificial Intelligence.”

[18] Cozen O’Connor, “Growing Use of Artificial Intelligence.”

[19] U.S. Citizenship and Immigration Services, “Form I-140 by Fiscal Year, Quarter and Case Status” (FY2025), <https://www.uscis.gov/tools/reports-and-studies/immigration-and-citizenship-data> (NIW adjudicated-denial rate rising from 37.3% in Q1 to 46.0% in Q3 FY2025); for the fourth quarter of FY2025, see the National Foundation for American Policy analysis reported in Stuart Anderson, “U.S. Immigration Service Increases Denials for High-Skilled Immigrants,” *Forbes*, April 22, 2026, <https://www.forbes.com/sites/stuartanderson/2026/04/22/us-immigration-service-increases-denials-for-high-skilled-immigrants/> (64.3% NIW denial rate; EB-1A rising from 25.6% to 46.6%). On the attribution of the increase to a more exacting application of the *Dhanasar* standard rather than to automation, see *id.* (remarks of Efren Hernandez, former USCIS adjudicator). The operational mechanisms described—mis-tagged evidence and cross-document mismatch flags—are reported in Cozen O’Connor, “Growing Use of Artificial Intelligence.”

[20] Richard T. Herman, “Algorithmic Immigration: Is Artificial Intelligence Shaping USCIS Decisions in 2026?,” Herman Legal Group, accessed August 4, 2026, <https://www.lawfirm4immigrants.com/uscis-artificial-intelligence-2026>

[21] *Kleindienst v. Mandel*, 408 U.S. 753, 766 (1972).

[22] *Mathews v. Diaz*, 426 U.S. 67, 77 (1976); *Plyler v. Doe*, 457 U.S. 202, 215 (1982); *Zadvydas v. Davis*, 533 U.S. 678, 693 (2001).

[23] Under the APA, judicial review is available only for “final agency action for which there is no other adequate remedy in a court.” 5 U.S.C. § 704. The Supreme Court has established a two-part test for finality: the action must mark the consummation of the agency’s decision-making process, and it must be one by which rights or

obligations have been determined or from which legal consequences will flow. *Bennett v. Spear*, 520 U.S. 154, 177–78 (1997); see also *Lujan v. National Wildlife Federation*, 497 U.S. 871, 882 (1990). Because AI systems in immigration adjudication are formally characterized as decision-support tools rather than decision-makers, their outputs are unlikely to satisfy the *Bennett* finality test, insulating algorithmic processing from direct APA review.

[24] David A. Martin, “Why Immigration’s Plenary Power Doctrine Endures,” *Oklahoma Law Review* 68 (2015): 1.

[25] Regulation (EU) 2016/679 (GDPR), art. 22.

[26] Case C-634/21, *SCHUFA Holding AG*, ECLI:EU:C:2023:957 (ECJ, Dec. 7, 2023).

[27] Regulation (EU) 2024/1689 (EU AI Act), art. 86.

[28] EU AI Act, art. 86; see also “Article 86—Right to Explanation of Individual Decision-Making,” [artificialintelligenceact.eu](https://artificialintelligenceact.eu/article/86), <https://artificialintelligenceact.eu/article/86> (annotated text of the provision); European Parliament, “EU Artificial Intelligence Act: First Regulation on Artificial Intelligence,” June 2023, <https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence> (noting geographic scope limitations and phased implementation); the high-risk regime’s application to standalone Annex III systems was subsequently deferred from August 2026 to December 2027 by the Digital Omnibus, Regulation (EU) 2026/1744.

[29] Framework Act on Artificial Intelligence Development and Establishment of a Foundation for Trustworthiness (AI Framework Act), Law No. 20676 (Jan. 21, 2025) (S. Kor.); Shin & Kim, “AI Basic Act Update: Enforcement and Key Implications,” February 11, 2026, <https://www.shinkim.com/eng/media/newsletter/3117>.

[30] AI Framework Act, art. 2(4).

[31] *Id.* art. 31(1).

[32] *Id.* arts. 32, 34.

[33] Juho Yoon, Jiyoung Sohn, and Jeonghee Kang, “Significance of the Passage of the AI Framework Act and Its Impact on the Industry,” Bae, Kim & Lee LLC, Lexology, January 10, 2025, <https://www.lexology.com/library/detail.aspx?g=9f68b5ab-b6b8-4822-823a-4872a8152cfb>.

[34] Natalie Sheard, “Algorithm-Facilitated Discrimination: A Socio-Legal Study of the Use by Employers of Artificial Intelligence Hiring Systems,” *Journal of Law and Society* 52, no. 2 (2025): 269–291, <https://doi.org/10.1111/jols.12535>.

[35] Yoann Emian, “Digital Borders and Racial Codes in AI Migration Control,” Ontario Bar Association, July 2, 2025, <https://oba.org/digital-borders-and-racial-codes-in-ai-migration-control>

[36] Letter from Electronic Frontier Foundation, Just Futures Law, The Advocates for Human Rights, and 140+ Organizations to Secretary Alejandro Mayorkas, “Cancel DHS Use of AI Technologies for Immigration Enforcement and Adjudication by December 1, 2024,” September 4, 2024, https://fedscoop.com/wp-content/uploads/sites/5/2024/09/Letter-from-140-Orgs-to-DHS_-Cancel-Use-of-AI-by-Dec-1.pdf; see also EFF, “EFF and 140 Other Organizations Call for an End to AI Use in Immigration Decisions,” September 5, 2024, <https://www.eff.org/deeplinks/2024/09/eff-140-other-organizations-call-end-ai-use-immigration-decisions>; Rebecca Heilweil, “140+ Groups Flag DHS’s AI Use Cases in New Letter to Mayorkas,” *FedScoop*, September 4, 2024, <https://fedscoop.com/dhs-ai-use-cases-letter-mayorkas>.

[37] Rachel Levinson-Waldman and Spencer Reynolds, “A Start for AI Transparency at DHS with Room to Grow,” Brennan Center for Justice, January 22, 2025, <https://www.brennancenter.org/our-work/analysis-opinion/start-ai-transparency-dhs-room-grow> (assessing DHS’s AI use-case inventory and finding its disclosures about the ICE Hurricane Score incomplete); Just Futures Law and Mijente, *Automating Deportation: The Artificial Intelligence Behind the Department of Homeland Security’s Immigration Enforcement Regime* (June 2024),

<https://mijente.net/wp-content/uploads/2024/06/Automating-Deportation.pdf> (documenting that DHS deploys these tools without notifying impacted individuals).

[38] International Covenant on Civil and Political Rights, art. 26, Dec. 16, 1966, 999 U.N.T.S. 171. The United States ratified the Covenant in 1992 subject to a declaration that its substantive provisions (Articles 1 through 27) are not self-executing. 138 Cong. Rec. S4781-84 (daily ed. Apr. 2, 1992); see *Sosa v. Alvarez-Machain*, 542 U.S. 692, 728 (2004) (noting the declaration).

[39] Zihan Ma, "The Inadequacy of the Current International Human Rights Regime for Algorithm Discrimination," *Michigan Journal of International Law* (blog), February 2024, <https://www.mjilonline.org/the-inadequacy-of-the-current-international-human-rights-regime-for-algorithm-discrimination>.

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[41] Pieter Binst et al., "Automated Decision-Making and Artificial Intelligence at European Borders and Their Risks for Human Rights," arXiv preprint arXiv:2410.17278 (2024), <https://arxiv.org/abs/2410.17278>.

[42] *Houston Federation of Teachers v. Houston Independent School District*, 251 F. Supp. 3d 1168, 1177 (S.D. Tex. 2017).

[43] Exec. Order No. 14179, 90 Fed. Reg. 8741 (Jan. 23, 2025) (directing OMB to revise Memoranda M-24-10 and M-24-18 within sixty days). Pursuant to that order, OMB Memorandum M-25-21, "Accelerating Federal Use of AI through Innovation, Governance, and Public Trust" (Apr. 3, 2025), rescinded and replaced M-24-10, consolidating its separate "rights-impacting" and "safety-impacting" categories into a single "high-impact AI" definition. See M-25-21 § 6(k)

(treating as presumptively high-impact AI used in risk assessments relating to a foreign national’s entry into or continued presence in the United States, including immigration, asylum, detention, and travel-authorization determinations); *id.* § 4(b)(vi) (instructing agencies to offer consistent remedies or appeals); *id.* n.4 (providing that the memorandum applies only to agencies’ own use of AI and creates no right or benefit, substantive or procedural, enforceable against the United States). The use-case inventory obligation survives the transition: *id.* § 3(b)(v) (requiring each agency to update and publicly post its AI use-case inventory at least annually), implementing the Advancing American AI Act, 40 U.S.C. § 7225.

[44] BMD LLC, “Invisible Algorithms.”

[45] EU AI Act, art. 86.

[46] Binst et al., “Automated Decision-Making.”

[47] Cozen O’Connor, “Growing Use of Artificial Intelligence.”

[48] White House Office of Science and Technology Policy, *Blueprint for an AI Bill of Rights: Making Automated Systems Work for the American People* (October 2022), <https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights>.

[49] *Houston Federation of Teachers*, 251 F. Supp. 3d at 1177.

[50] Littler, “What Does the 2025 Artificial Intelligence Legislative and Regulatory Landscape Look Like for Employers?,” January 14, 2025, <https://www.littler.com/news-analysis/asap/what-does-2025-artificial-intelligence-legislative-and-regulatory-landscape-look>.

[51] GDPR, art. 22.

[52] U.S. Equal Employment Opportunity Commission, “Select Issues: Assessing Adverse Impact in Software, Algorithms, and Artificial Intelligence Used in Employment Selection Procedures Under Title VII of the Civil Rights Act of 1964,” Technical Assistance Document EEOC-NVTA-2023-2, May 18, 2023, <https://www.eeoc.gov/laws/guidance/questions-and-answers-clarify-and-provide-a-common-interpretation-uniform-guidelines>.

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AI-Enabled Crime: Protecting Society's Vulnerability through the Destruction of Digital Trust

Isabella Fama

Stevens Institute of Technology

**“SOMETHING AS SIMPLE AS RECOGNIZING
A FAMILIAR FACE OR VOICE HAS NOW
BECOME UNTRUSTWORTHY.”**

Abstract

Artificial intelligence, or AI, is often associated with the idea of trust. As AI systems become more deeply embedded into daily life, their misuse has expanded in parallel, enabling a new class of crime that exploits emotional vulnerability and publicly available data. These threats take advantage of the fact that many cannot identify fabricated information, especially among older individuals. The United States does not have any current federal statute that restricts the use of public data for AI training, despite the documented risks identified by federal agencies. The legal system, designed before the emergence of synthetic media, is now faced with issues of confidentiality and integrity. This paper investigates and suggests that Congress must intervene by establishing federal guardrails that would assist in the regulation of AI-enabled crime and preserving digital trust.

Introduction

The rapid growth and integration of AI across professional, academic, and commercial environments have fundamentally altered the nature and concept of trust. AI has increasingly been encouraged as a tool for efficiency, but the intended use of something does not always align with the reality of how it is truly being used. This is seen and observed through the growth of AI-enabled crime, which thrives and takes advantage of the public's inability to distinguish authentic communication apart from algorithmic and generated fabrication of information.

Older adults have become increasingly targeted in this new rise of crime. AI-generated scams can easily be used to replicate and mimic voices, tricking victims into sending them money, or compromising their safety. These scams have grown to take advantage of emotional vulnerability and security. Something as simple as recognizing a familiar face or voice has now become untrustworthy.

The widespread availability of personal data and information online provides the material for AI training. Individuals with any digital presence are unknowingly contributing to data that can later be used against them.

The legal system has now begun to confront these growing issues. The recent case of *Morgan v. V2X* [1] reveals the uncertainty that has been unaddressed in terms of confidentiality and evidence. Federal agencies such as the Department of Homeland Security [2] and the Federal Trade Commission [3] have also become more drastically attentive to this new age of digital predation, and the difficulty of preventing it.

As AI-enabled crime continues to grow, the structural gaps within US law become increasingly exposed. This paper discusses the effects AI-enabled crime has begun to have on society, arguing that Congress must establish regulations that govern the use of public data in the training of AI.

The New Age of Deception

AI has completely altered the perception of fraud. Scams that once solely relied on voice mimicking and impersonation have now evolved much further. With the use of AI, scammers can now impersonate someone using hyper-realistic voice cloning or deepfake videos. The Federal Trade Commission has now identified this as a “new frontier of impersonation scams” [4].

AI has changed how believable scams are; the FTC warns that scammers can exploit simple truths about our personal lives, personal details that victims once assumed were safe. In April 2024, the FTC's official alert warned how scammers are able to clone voices just by hearing someone's voice for a few seconds, usually being taken from social media, voicemails or videos on the internet. Scammers use this tactic mainly to make requests for large sums of money, using sensitive information to their advantage. This creates a completely irrational and emotional situation, where any rational judgment or thinking is neglected. These findings align with the Department of Homeland Security's assessment from Impact of Artificial Intelligence on Criminal and Illicit Activities [2], which similarly documents cases of victims receiving calls that sound as if their loved one is in distress. The FBI's 2024 Elder Fraud Report [5], supports these concerns: impersonation scams resulted in an estimated \$1.3 billion in losses, about 25% of reported cases involved AI-generated audio, and voice cloning complaints have increased 700% from 2022 to 2024. These brief statistics are likely an underrepresentation of the true number of the victims' experiences, many victims go without reporting the crimes.

The FTC puts a strong emphasis on older adults being most vulnerable. Many grew up in an era where all that was needed to prove your identity was a voice; not to mention the new and unfamiliar technology. A study conducted by Pew Research Center [6] in 2024 revealed several findings: about 62% of adults over the age of 65 are unable to identify AI-generated audio, 83% of participants believe that a familiar voice guarantees authenticity and 71% of participants were unaware that AI was able to clone voices. This ongoing threat, which the FTC has shared public information on, is more than a safety concern.

AI-enabled deception undermines and discredits personal autonomy by attacking and mimicking behaviors that individuals use to make informed decisions. Since deception has changed entirely, truth has become difficult to distinguish from manipulation due to the disruption of cues that were once relied on for legally-meaningful decisions.

Exploitation of Public Data

AI-enabled crime is heavily reliant on personal data that is accessible online, often unknowingly by users. Platforms that have social media posts, public records, and digital interactions are now materials that can be used by AI to produce more realistic and believable media. A 2024 taxonomy of AI privacy risks [7] found that machine learning systems are built in ways that create vulnerabilities by enabling new forms of distortion, unwanted disclosure, and identification. The analysis concluded that 321 documented AI-related privacy incidents demonstrated multiple dangers: AI systems created privacy threats, for example deepfake (or deepfake-based) exposure, as well as increased risks already existing through larger scale surveillance and use of secondary data. This now creates an extremely dangerous dynamic, specifically related to AI-enabled crime. Traditional US data-privacy frameworks assume that individuals are able to control the dissemination of their information. In the US, once data has become publicly available, there are no laws that impose any restrictions on how it can be collected, analyzed or reused.

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Academic research conducted through Cambridge University Press [8] notes that AI reduces the barriers that allow entry for criminal conduct. Meaning practically anybody with minimal technological expertise is able to easily commit AI-related crimes. The DHS's report [2] similarly places a strong emphasis on the reasoning that the outputs from current AI models are even possible because of the amount of data that is available. Anyone with a presence online is susceptible to becoming a target. The DHS has warned that generative AI has criminal factors that are able to commit crimes faster and more efficiently, and that these readily-available tools are able to empower even less-skilled criminals by filling knowledge gaps. The report highlights that the United States currently has no federal statute which regulates whether AI companies are allowed to train on publicly available data. The absence of federal restrictions on AI use for public data creates a dynamic of uncertainty and fear.

Without any systems currently put into place to stop the exploitation of public data, criminals are able to continue their work. This threat is not hypothetical, but it is active and growing quickly. In the Oxford Journal of Legal Studies [9], it is noted that modern AI challenged traditional legal practices and assumptions about identity and human authenticity. This raises questions about what frameworks can be put into place to adequately assess the harm from continuing to rise. The United States has field-specific legislation that serves as protection of consumer data such as HIPAA [10], protecting health information and FERPA [11], protecting student education records. There is, however, no federal legislation that protects consumer data with regard to AI.

Modern Law Implications & Standings

The rise of AI-enabled crime has exposed gaps in legal frameworks that were designed before synthetic media even existed. AI-enabled crime is able to continue because of the lack of legal standings, as previously mentioned before. As AI-enabled crime continues to grow, it raises the question of how can the integrity of legal evidence, protection of victims, and liability be preserved in a new age of deception that has become undetectable? Courts have been able to rely on audio, video, and photos as trustworthy and reliable evidence. Under the Federal Rule of Evidence 901 [12], these forms of evidence were considered self-authenticating because they presumably reflected reality. This has now completely changed because of AI. The DHS [2] has warned that AI-generated content is extremely difficult to distinguish from human-produced content. The legal systems' traditional safeguards are no longer trustworthy or able to keep up with the speed and believability of newly introduced AI-enabled crimes.

US federal courts were recently faced with one of the first cases that directly addressed how AI intersects and interferes with confidentiality: *Morgan v. V2X*. [1] The central question was whether a litigant's use of an AI tool to preview confidential materials violates work-product protection and client confidentiality, and if courts require disclosure of the use of an AI tool under a protective order. Under the Federal Rule of Civil Procedure 26(c) [13], the courts may issue protective orders on confidential information. The defendant was looking to amend a protective order to restrict AI use, arguing that uploading documents to AI platforms risked exposing sensitive information, since these platforms store information and later use it for model training. The court held that it does pose a confidentiality risk, unless the provider is barred from storing or training on inputs. The court then rejected the argument that AI use should automatically waive the work-product protection, reasoning that interacting with AI does not inherently impact or reveal litigation strategies, therefore not constituting a per se waiver. It is important to note that *Morgan v. V2X* is a single district-court order that holds no precedential weight. Following this case, federal courts may require disclosure of an AI tool when used, protective orders must explicitly state the use of AI.

The case shows our government's growing recognition of AI, and although the case does focus on the implications of confidentiality, it expands beyond just that. It is a call to action that courts must evaluate and take into consideration technological advancements and how they are being used, not just simply the contents. With this being said, now judges are required to understand how AI systems work. There is now a significant new judicial responsibility that drastically differs from traditional practices. *Morgan v. V2X* illustrates an urgent need for procedural rules related to AI. In response to this, courts can implement reforms under existing statutes. The first reform would be directly under the Federal Rule of Evidence 901 [12]. When it comes to digital audio and/or visual evidence, implementing AI-detection and provenance metadata to ensure that authentication is now being modernized, but not requiring the creation of a new evidentiary category. The second reform being the clarification of confidentiality rules under Federal Rule of Civil Procedure 26(c) [13]. This ties back to *Morgan v. V2X*, and the courts have already addressed the issues of confidentiality in relation to AI. Congress could further this by requiring disclosure of AI tools in all and any litigation and standardized AI training and protective-order language. The third reform would be establishing criminal liability for AI-facilitated Impersonation Under 18 U.S.C. § 1028 [14]. The law already criminalizes the misuse of identification documents. By Congress expanding the law to now include anything AI generated media for crime purposes of unlawful misuse, the prevention of fraudulent use is now even further protected. These reforms will help courts to stay on top of the growth of AI, which is not expected to stop. Without them, courts will need to address AI issues on an ad hoc basis, which creates inconsistency and uncertainty within rulings.

Ethical Transparency

In addition to the legal aspect that surrounds AI-enabled crime, there is also a clear ethical standpoint: the crimes are not only defined by the actions, but by how institutions design and deploy their systems. Courts will continue to struggle with the issue of evidence and confidentiality clashing with AI, but policymakers, creators, and developers are also to be held accountable for the ethical burden that falls on them and their systems.

AI companies hold a unique position when it comes to their products: they are actively creating tools that are capable of deception, but they are able to disclaim responsibility from any consequences. In an analysis conducted by Toereck Law [15], the terms of AI platforms and their services function as “private ethical frameworks” which allow them the right to: store any and all user inputs, analyze user inputs, reuse user inputs and train models on user inputs. This practice raises many ethical concerns, mainly that individuals who are using AI systems without knowing any disclosed risks, are becoming involuntary contributors to systems that can be later used against them. Platforms fully understand how their systems work, but their users are not, creating a clear distortion of knowledge and disclosure. The FTC’s consumer-protection principles [16] strongly emphasizes transparency and fairness specifically for these reasons. Yet, many AI platforms simply disclose what is technically accurate, but only to those who understand them. This imbalance violates ethical principles; individuals cannot and should not consent to practices they do not understand.

Counterargument: Practicality

A significant counterargument to the regulations and reforms proposed in this paper concerns the practicality of governing AI at a scale, considering its worldwide use, global infrastructure and limited enforcement capability. There are currently more than 1 billion people in the world engaging with AI-related tools monthly, without being aware. [17]. Google Gemini [18] is now integrated into their devices, software, and browsers. Microsoft Copilot[19] is now also embedded across Windows, Office and Edge. The level of adoption and integration makes it incredibly difficult to distinguish how AI is being used in harmful ways. If federal reforms would need to operate assuming that AI is not a tool of discretion, but a feature of everyday life.

The scale of adoption highlights a substantial resource gap between government enforcement agencies and AI-enabled activity. Agencies that would be responsible for enforcing hypothetical reforms merely do not have the personnel or funding to do so.

The FTC has been the primary agency which has informed the public of AI-enabled crime, as they are responsible for deceptive practices under 15 U.S. Code § 45 [20]. The agency has about 1,300 staff members [21] and regulates millions of business and digital services. With the added factor of now having to monitor billions of outputs from AI platforms makes the FTC unlikely to realistically and effectively monitor AI-specific tasks. Even if Congress were to expand the FTC's authority, the volume at which AI-related content is growing would make the oversight unrealistic and unattainable.

AI is not only a domestic tool; it is used globally and the US therefore has very limited or no regulation. According to the OECD AI Policy Observatory [22], over 70% of major AI platforms are either developed and/or hosted outside of the US. Countries such as China, India as well as the European Union and United Arab Emirates host large-scale systems, where the US law is not applicable to them. Even if Congress were to implement regulations on domestic AI practices, platforms will continue to grow abroad, still affecting US residents. National regulations simply cannot fully govern technology that is being used globally.

These considerations form a compelling counterargument: federal reforms may be able to mitigate specific risks such as confidentiality protection and AI impersonation, they are unable to fully address the broader challenge of regulating a growing technology.

Conclusion

AI-enabled crime presents a threat to the structure of digital trust. As explored in this paper, the newly advanced forms of AI have questioned what was traditionally assumed to be safe in regard to authenticity. Voices, faces and any accessible form of public data online can be replicated, later to be used for scamming, phishing and manipulation. Elderly individuals have become the main target of AI-enabled crime, because of their unfamiliarity with technology.

The US legal system is now beginning to respond, as seen in *Morgan v. V2X*. [1] The courts recognized the risks associated with AI, but also highlighted the limits of judicial intervention. At this time, courts must evaluate cases one by one because of the lack of legal regulation. Each case must be individually based on digital evidence authentication, assessment of synthetic media and regulated AI use, creating inconsistency. The paper argues that Congress must establish guardrails that govern the use of public data in relation to AI training and modernizing evidence and confidentiality-related rules. Reforms to Rule 901 [12], Rule 26(c) [13], and 18 U.S.C. § 1028 [14] would protect victims by creating a framework of legal action which is capable of keeping up with its rapid growth.

The counterarguments that address scale, reach and enforcement limitations are significant to the conversation. AI is not simply exposed to a small population of people, billions interact with it daily, with major platforms operating outside of US jurisdiction. Instead of diminishing the challenges that have arisen in domestic regulation, the counterargument underscores the urgency and strong need for regulations.

AI-enabled crime is now a permanent feature in modern technology, and the possibility of it being temporary is now almost impossible to consider. In order to protect society vulnerability, it requires acknowledging that the legal system must evolve with the technology that must be regulated. Acting now would preserve the integrity of evidence, personal autonomy and prevent the weaponization of public data in the new age of artificial intelligence.

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Artificial Intelligence, Accountability, and Ethical Governance in Modern Digital Systems

Kamakshi Padma Vritika Mandapaka

Stevens Institute of Technology

**“THERE IS A GROWING
RECOGNITION THAT THE ERA OF
SELF-REGULATION IN AI IS ENDING.”**

Abstract

Artificial intelligence (AI) has rapidly evolved from a specialized technological tool into a foundational component of decision-making across healthcare, finance, education, criminal justice, and public administration. While AI offers significant opportunities to improve efficiency, innovation, and economic growth, its widespread adoption has also introduced complex legal and ethical challenges involving algorithmic bias, privacy, transparency, accountability, and the protection of fundamental rights. Existing governance approaches remain fragmented and often rely on voluntary ethical principles that lack meaningful enforcement mechanisms. This article argues that ethical principles alone are insufficient to govern increasingly autonomous AI systems. Instead, effective AI governance requires legally enforceable accountability mechanisms that establish clear responsibilities for AI developers, deployers, and users while ensuring transparency, explainability, meaningful human oversight, and appropriate legal remedies for individuals affected by automated decision-making. To support this argument, the article evaluates emerging governance frameworks, including UNESCO's Recommendation on the Ethics of Artificial Intelligence, the European Union's AI Act, the United States' NIST AI Risk Management Framework, and the OECD AI Principles, examining how different jurisdictions balance technological innovation with the protection of human rights. Building upon these frameworks, this article proposes a governance approach centered on mandatory transparency, independent algorithmic auditing, human oversight in high-risk applications, and clearly defined legal accountability throughout the AI lifecycle. It concludes that the long-term legitimacy of artificial intelligence will be determined not by the sophistication of the technology itself, but by the willingness of legal institutions to transform ethical principles into enforceable legal obligations that promote public trust, protect fundamental rights, and support responsible innovation.

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I. INTRODUCTION

Artificial intelligence has become an integral component of contemporary decision-making, influencing outcomes in healthcare, finance, employment, criminal justice, and public administration. It diagnoses diseases, determines creditworthiness, influences hiring decisions, and increasingly shapes the outcomes of judicial proceedings. [1] Yet the legal and regulatory frameworks governing these systems remain fragmented, voluntary, and largely aspirational. The gap between what AI can do and what the law requires of those who deploy it has grown into one of the defining governance challenges of the twenty-first century.

The proliferation of voluntary ethics frameworks developed by technology companies, governments, and international organizations has undoubtedly advanced responsible AI discourse. However, these frameworks generally lack binding legal obligations, standardized enforcement mechanisms, and effective remedies for individuals harmed by automated decision-making.

The stakes are substantial. When an AI system incorrectly denies a loan application, misidentifies a criminal suspect, or recommends a harmful medical treatment, the affected individual rarely has meaningful recourse. The developer may disclaim responsibility, the deploying company may point to the algorithm, and regulators may lack the authority or expertise to intervene. [31] Although the accountability gap [2] has technical dimensions, its most significant implications are legal, as existing regulatory systems often fail to allocate responsibility when

algorithm system cause harm. Several jurisdictions are now moving toward more structured and enforceable governance. The European Union's Artificial Intelligence Act, adopted in 2024, [3] represents the most comprehensive regulatory framework to date, classifying AI systems by risk level and imposing obligations of transparency, human oversight, and conformity assessment on high-risk applications. In the United States, the National Institute of Standards and Technology released its AI Risk Management Framework in 2023, [4] providing voluntary guidance for organizations developing and deploying AI systems across sectors. In New York City, Local Law 144 [5] introduced mandatory bias audits for automated employment decision tools, marking one of the first binding municipal AI regulations in the country. At the international level, UNESCO's Recommendation on the Ethics of Artificial Intelligence further reflects a growing global consensus that AI governance should be grounded in human rights, transparency, accountability, fairness, and meaningful human oversight. [22] Although the Recommendation is not legally binding, it provides an internationally recognized ethical framework that increasingly influences national AI policies and regulatory initiatives. These developments signal a growing recognition that the era of self-regulation in AI is ending.

This article proceeds in four parts. Part II examines the ethical challenges that current AI systems present, including algorithmic bias, opacity, privacy violations, and the erosion of human oversight, drawing on recent legal disputes and documented harms to ground the analysis in concrete reality rather than theoretical risk. Part III surveys the emerging landscape of AI governance across major jurisdictions, comparing the EU's risk-based regulatory model, the U.S. sector-specific approach, and the principles-based frameworks adopted by international bodies including UNESCO and the OECD. Part IV develops an enforceable AI governance framework by examining

mandatory transparency obligations, human oversight in high-risk applications, independent algorithmic auditing, and legal liability for algorithmic harm. The article concludes that the legitimacy of AI in public and private life depends on society's willingness to subject it to the same rule of law that governs every other consequential technology.

II. ETHICAL CHALLENGES AND LEGAL RISKS IN AI DEPLOYMENT

The challenges posed by artificial intelligence extend well beyond theoretical ethical debates. As AI systems increasingly influence decisions involving liberty, employment, healthcare, finance, and public safety, failures in their design or deployment can produce tangible legal and societal harms. From biased hiring tools and flawed criminal risk assessments to privacy violations and misdiagnoses in clinical settings, what emerges is not a theoretical warning but a documented pattern of harm that existing legal frameworks have consistently failed to address.

These incidents reveal systemic shortcomings in accountability, oversight, and regulatory enforcement. Voluntary ethical guidelines, while valuable in articulating aspirational principles, have proven insufficient to prevent or remedy the harms documented across multiple sectors. This Part examines five domains in which AI deployment has generated well-substantiated harms, demonstrating why enforceable governance mechanisms not merely ethical commitments are necessary to safeguard fundamental rights and maintain public trust. These examples establish the factual and legal foundation for the governance framework proposed in Part IV.

A. Algorithmic Bias in Employment

Among the earliest and most consequential demonstrations of AI bias in a corporate setting was Amazon's internal recruitment tool, developed

between 2014 and 2018 to automate the screening of job applicants. The system, trained on historical hiring data, learned to penalize resumes containing words associated with women, including the names of women's colleges and systematically downgraded female candidates for technical roles. [6] When Amazon engineers discovered the bias, the company abandoned the tool entirely [7] rather than attempt to correct it. The episode demonstrates that algorithmic decision-making, despite objective and impartial outcomes, can reflect the assumptions and biases embedded in the data on which such systems are trained. More broadly, AI systems trained on historical datasets can reproduce and amplify existing social biases, raising significant legal concerns regarding discrimination, equal employment opportunity, and organizational accountability. [32] [40] Importantly, the absence of discriminatory intent does not eliminate legal responsibility when automated systems produce discriminatory outcomes. This distinction is particularly significant in employment law, where liability may arise from discriminatory effects even when there is no evidence of intentional discrimination.

New York City's response to this class of problem is instructive. Local Law 144, enacted in 2021 and enforced beginning in 2023, [5] requires employers using automated employment decision tools to conduct and publish annual independent bias audits before deploying such systems. The law also mandates disclosure to job candidates that an automated tool is being used. While limited in scope, Local Law 144 represents a model of enforceable algorithmic accountability one that moves beyond voluntary commitments and imposes binding legal obligations on organizations deploying AI systems. More importantly, Local Law 144 reflects a broader shift in AI governance from reactive compliance toward preventive regulation. Rather than waiting for discriminatory outcomes to emerge through litigation, the law requires organizations to evaluate algorithmic systems before deployment through independent

bias audits and transparency obligations. [33] This approach recognizes that accountability should be embedded throughout the AI lifecycle rather than imposed only after harm has occurred. As similar regulations emerge across other jurisdictions, algorithmic auditing may become a foundational component of responsible AI governance. The Amazon case and New York City's legislative response illustrate a broader shift in AI governance from voluntary ethical commitments toward enforceable legal accountability. Together, they demonstrate that algorithmic bias in machine learning systems presents a broader governance challenge requiring clear regulatory standards, institutional oversight, and meaningful accountability. As AI increasingly influences employment and other high-stakes decisions, organizations must be held responsible for ensuring that automated systems comply with principles of fairness, transparency, and non-discrimination.

B. Opacity and Due Process in Criminal Justice

Perhaps no deployment of AI raises more urgent legal concerns than its use in criminal sentencing. COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is a proprietary risk assessment tool used by courts in several U.S. states to predict the likelihood that a defendant will reoffend. Scores generated by COMPAS have been used to inform decisions about bail, sentencing, and parole decisions with profound consequences for individual liberty. [34] Because these decisions directly affect an individual's constitutional rights and personal liberty, the use of AI-assisted risk assessments raises fundamental questions regarding procedural fairness, judicial discretion, and the extent to which automated recommendations should influence legal decision-making.

A 2016 investigation by ProPublica found that COMPAS scores were racially biased: black defendants were nearly twice as likely as white defendants to be falsely flagged as high risk for future offending. [8] The

tool's developer, Northpointe (now Equivant), disputed the findings, arguing that COMPAS was equally predictive across racial groups and that differences in the error rates did not, by themselves, demonstrate racial bias. [35] The dispute nevertheless highlighted a broader structural problem: because COMPAS is proprietary software, defendants and their attorneys have no meaningful ability to examine how the system generates its scores or to challenge its accuracy. [36] In *State v. Loomis*, the Wisconsin Supreme Court upheld the use of COMPAS scores in sentencing while acknowledging that the defendant had no ability to access or contest the algorithm, [9] a troubling concession that due process concerns were subordinated to efficiency. The COMPAS case illustrates what legal scholars call the "black box" problem in AI governance: [2] when consequential decisions are made by opaque algorithms, the right to a fair hearing becomes functionally meaningless. Transparency and explainability are fundamental safeguards in contexts where AI influences judicial decision-making, supporting procedural fairness, protecting due process, and maintaining public confidence in the administration of justice.

The use of COMPAS therefore reinforces a central argument of this article: when AI systems participate in high-stakes public decision-making, voluntary ethical principles alone are insufficient to protect individual rights. Legally enforceable requirements for transparency, explainability, human oversight, and judicial accountability are necessary to ensure that technological innovation remains consistent with the principles of due process, equality before the law, and the rule of law itself.

C. Privacy Violations and Facial Recognition

Clearview AI, a facial recognition company, scraped billions of images from social media platforms without user consent to build a database sold to law enforcement agencies across the U.S. and internationally.

[10] The company's technology enables law enforcement to identify individuals from a single photograph, a capability with obvious investigative value and equally obvious potential for abuse.

Regulators in multiple jurisdictions found Clearview's practices unlawful. The UK Information Commissioner's Office issued a £7.5 million fine in 2022, finding that the company had violated data protection law by collecting images without a lawful basis. [11] Similar findings followed in Australia, Canada, France, and Italy. The United States, where no comprehensive federal privacy law existed at the time, enforcement was limited to state-level actions under the Illinois Biometric Information Privacy Act. [12]

The Clearview AI episode demonstrates that AI-driven privacy violations can occur at a scale and speed that exceed the capacity of many existing legal frameworks, exposing significant gaps in privacy regulation, regulatory enforcement, and cross-border governance. It also underscores the consequences of regulatory fragmentation: when privacy protection depends on the jurisdiction in which an individual happens to reside, the protection is structurally unequal.

Clearview AI therefore illustrates a central argument of this article: effective AI governance cannot depend on fragmented regulatory approaches or voluntary corporate commitments. Meaningful protection of privacy requires harmonized legal standards, enforceable consent requirements, robust regulatory oversight, and clear accountability for organizations that develop and deploy AI-powered surveillance technologies. Without such safeguards, advances in facial recognition risk undermining fundamental rights while eroding public trust in the responsible use of artificial intelligence.

D. Accountability in Legal and Commercial Contexts

In 2023, two U.S. attorneys submitted a court brief containing citations

to six legal cases that did not exist. [13] The citations had been generated by ChatGPT, which produced plausible sounding but entirely fabricated case names, docket numbers, and judicial quotations. When the court discovered the error, the attorneys were fined and publicly sanctioned. The incident prompted federal judges across the country to issue standing orders requiring disclosure of AI use in legal filings.

A related but distinct accountability question arose in the case *Moffatt v. Air Canada* (2024), in which the Civil Resolution Tribunal of British Columbia held Air Canada liable for inaccurate information provided by its customer service chatbot. [14] The company had argued that the chatbot was a separate legal entity whose representations did not bind the airline, an argument the Tribunal rejected, holding that a company cannot disclaim responsibility for information disseminated through its own automated systems. The ruling established an important precedent: organizations deploying AI-powered customer interfaces bear legal responsibility for the outputs those systems produce.

Together, these cases demonstrate that the accountability gap in AI has moved well beyond theory, producing concrete legal disputes, and courts are beginning to fill the regulatory void that legislatures have left. These disputes also illustrate an important shift in AI governance. Rather than treating algorithmic failures as isolated technological errors, courts are increasingly evaluating them through established principles of negligence, agency, professional responsibility, consumer protection, and organizational liability. This judicial approach suggests that while comprehensive AI legislation continues to evolve, existing legal doctrines remain capable of providing meaningful accountability when organizations fail to exercise appropriate oversight over automated systems.

E. AI in Healthcare: Diagnostic Bias and Liability

The deployment of AI in healthcare settings presents the accountability

gap in a domain where its consequences are uniquely difficult to remedy. AI diagnostic tools are now used to detect cancer in radiological images, predict patient deterioration in intensive care units, and guide treatment recommendations for conditions ranging from sepsis to mental health disorders. The potential benefits are significant; studies suggest that AI diagnostic tools can match or exceed the accuracy of specialist physicians in specific tasks. The risks, however, are equally significant and less frequently acknowledged.

Research published in leading medical journals has documented systematic racial and ethnic bias in AI diagnostic systems. A study published in *Science* found that AI models trained predominantly on data from white patients performed substantially less accurately on patients of other racial and ethnic backgrounds, a disparity with direct implications for health equity. [15] In dermatology, AI skin cancer detection tools [16] trained on datasets lacking sufficient representation of darker skin tones have demonstrated lower diagnostic accuracy for Black patients, a finding that has prompted calls for mandatory diversity requirements in medical AI training data.

Liability questions in AI assisted healthcare remain largely unresolved. When an AI system recommends a treatment that harms a patient, the question of who bears legal responsibility, the software developer, the hospital, the attending physician, or some combination has no settled answer in U.S. law. The Food and Drug Administration has begun to develop frameworks for regulating AI-enabled medical devices, but the pace of regulatory development has lagged behind the pace of clinical deployment. [17] The result is a landscape in which patients may be harmed by AI systems without clear legal pathway to remedy.

The healthcare context makes the case for enforceable AI governance with particular force. When the stakes are a patient's life or health,

voluntary ethical commitments by technology companies are not a sufficient substitute for legal accountability.

The healthcare sector therefore exemplifies the central argument of this article. While ethical principles such as fairness, transparency, and beneficence remain essential, they must be supported by enforceable legal obligations that clearly define responsibility, establish appropriate regulatory oversight, and provide effective remedies when AI systems cause harm. As AI becomes increasingly integrated into clinical practice, maintaining public trust will depend not only on technological accuracy but also on the strength of the legal frameworks governing its development and deployment.

Collectively, the cases examined throughout this Part demonstrate that the most pressing challenges posed by artificial intelligence are rooted not in what the technology cannot do, but in what the law has failed to require. They reveal a consistent pattern in which rapid innovation has outpaced the legal frameworks necessary to ensure transparency, accountability, and effective protection of fundamental rights.

III. THE EMERGING LANDSCAPE OF AI GOVERNANCE

The recognition that AI requires regulatory attention has produced a diverse range of regulatory and governance responses across jurisdictions and international institutions. These responses reflect fundamentally different assumptions about the appropriate role of law in managing technological risk; assumptions rooted in distinct legal traditions, political economies, and relationships between the state and the private sector. This Part surveys major governance frameworks, identifies their respective strengths and limitations, and argues that a risk based, legally enforceable governance model provides the most effective foundation for an AI accountability framework capable of protecting fundamental rights while supporting responsible innovation.

Although these frameworks share common objectives including promoting trustworthy AI, protecting fundamental rights, and encouraging responsible innovation, they differ significantly in their legal force, enforcement mechanisms, and regulatory scope. Some rely primarily on voluntary guidance and industry self-governance, while others establish binding legal obligations backed by regulatory oversight and sanctions. Examining these differences is essential for understanding how legal systems can balance technological innovation with public accountability and why certain governance models are better equipped to address the accountability gap identified in Part II.

A. The European Union: A Risk-Based Regulatory Model

The European Union's Artificial Intelligence Act, which entered into force in August 2024 represents the most comprehensive binding legal framework for AI governance yet enacted anywhere in the world. [18] The Act adopts a risk-based architecture that classifies AI systems into four categories: unacceptable risk, limited risk, high risk, minimal risk and impose legal obligations proportionate to the potential harm each category presents. Unlike many earlier AI governance initiatives that relied primarily on voluntary ethical principles, the EU AI Act establishes legally enforceable obligations supported by regulatory oversight and financial penalties. In doing so, it reflects a preventive approach to AI governance that seeks to reduce foreseeable risks before they result in legal disputes or societal harm.

AI systems classified as presenting unacceptable risk are prohibited outright. These include systems that deploy subliminal manipulation to influence behaviour, exploit vulnerabilities of specific groups, enable real time biometric surveillance in public spaces by law enforcement (with limited exceptions), and create social scoring systems for general purposes. These restrictions reflect the view that certain uses of AI conflict with human dignity and the principles of democratic governance, regardless of how advanced or commercially valuable the technology may be.

High risk AI systems used in critical infrastructure, education, employment, essential services, law enforcement, migration, and the administration of justice are subject to the most demanding obligations under the Act. Developers and deployers of high-risk systems must implement risk management systems, ensure the quality and representativeness of training data, maintain technical documentation, enable human oversight, achieve accuracy and robustness standards, and register their systems in a publicly accessible EU database. High-risk systems require independent conformity assessment prior to deployment.

For foundation models and general-purpose AI systems a category added in response to the rapid development of large language models, the Act imposes transparency obligations, requires documentation of training data, and mandates compliance with EU copyright law. Developers of the most powerful foundation models, defined by computational thresholds, face additional obligations including adversarial testing and incident reporting.

The penalties for non-compliance are substantial. Violations of the prohibited practices provisions may attract fines of up to thirty-five million euros or seven percent of global annual turnover, [19] whichever is higher. The Act also establishes a European AI Office within the European Commission to coordinate enforcement and develop technical standards.

The EU AI Act is not without limitations. Critics have raised concerns that its compliance requirements may disadvantage smaller developers relative to large technology companies with dedicated legal and regulatory teams. Others have questioned whether the risk classification system is sufficiently dynamic to keep pace with rapidly evolving AI capabilities. Nevertheless, as a demonstration that

comprehensive, binding AI regulation is legally and institutionally achievable, the EU AI Act is a landmark development that has already begun to influence regulatory thinking beyond Europe. Although implementation challenges remain, the EU AI Act currently represents the most comprehensive attempt to translate broad ethical principles into enforceable legal obligations. Its risk-based framework illustrates how governments can balance technological innovation with the protection of fundamental rights through clear regulatory standards, institutional oversight, and meaningful legal accountability. For these reasons, the Act provides an important foundation for the accountability framework proposed in Part IV of this article.

B. The United States: Sector-Specific Regulation and Voluntary Frameworks

The United States has not enacted comprehensive federal AI legislation. Instead, its governance model has evolved through sector-specific rules, executive actions, and voluntary frameworks, reflecting both a longstanding preference for industry-led regulation and the political difficulty of passing broad technology laws. This approach has supported flexibility and innovation but has also produced fragmented oversight and uncertainty regarding legal accountability. The most influential federal initiative to date is NIST's AI Risk Management Framework, released in 2023, [20] which offers voluntary guidance organized around the functions of Govern, Map, Measure, and Manage. Although widely adopted, its nonbinding nature limits consistent compliance and leaves implementation dependent on organizational discretion rather than enforceable obligations.

Executive Order 14110, issued in 2023, [21] directed federal agencies to develop AI governance standards, required developers of advanced AI systems to share safety test results, and instructed the Department of Commerce to establish reporting requirements for high-risk systems.

While significant, its impact is constrained by the inherent instability of executive action, which can be modified or rescinded by subsequent administrations. In contrast, state and local governments have enacted more concrete measures. New York City's Local Law 144 imposes mandatory bias audits for employment algorithms, and states such as California, Colorado, and Illinois have advanced legislation targeting facial recognition, employment screening, and algorithmic insurance practices. This patchwork of state rules creates compliance complexity for national organizations and results in uneven protections for individuals across jurisdictions.

The U.S. model reflects an innovation-first philosophy that prioritizes technological development and addresses harms primarily after they occur. Without comprehensive federal legislation, individuals harmed by algorithmic systems must rely on tort law, civil rights statutes, and consumer protection doctrines that were not designed for AI and generally operate reactively. This stands in contrast to the European Union's preventive, risk-based regulatory approach, which imposes binding obligations and centralized oversight. Each model offers distinct advantages, but the comparison underscores a central point: durable and effective AI governance requires balancing innovation with enforceable legal accountability capable of protecting fundamental rights and sustaining public trust.

C. UNESCO and the International Ethical Framework, OECD AI Principles and Global Coordination

In November 2021, UNESCO's 193 member states adopted the recommendation on the Ethics of AI, [22] the first global standard on AI ethics adopted by a body of that breadth. The Recommendation articulates four core values: respect for human dignity and human rights, fostering just and peaceful societies, ensuring diversity and inclusiveness, and supporting environmental sustainability. From these

values, it derives ten core principles including proportionality and do no harm, safety and security, privacy and data protection, multi-stakeholder governance, responsibility and accountability, transparency and explainability, human oversight, sustainability, awareness and literacy, and fairness and non-discrimination.

The recommendation established the Global AI Ethics and Governance Observatory, designed to serve as a resource for policy makers, regulators, and civil society organizations seeking to translate ethical principles into concrete governance measures. [23] The Observatory's work reflects a recognition that the gap between principle and practice in AI governance is not merely a matter of political will but of institutional capacity, many governments lack the technical expertise and regulatory infrastructure to implement AI governance effectively, and international cooperation is essential to addressing that gap.

The OECD's (Organisation for Economic Co-operation and Development) 2019 AI Principles similarly provide a non-binding but influential framework for trustworthy AI, endorsed by more than fifty countries. [24] They emphasize human-centered values, transparency, robustness, and accountability, and recommend that governments invest in research, strengthen digital ecosystems, and pursue international cooperation. As noted in the document, "the OECD Principles have served as a reference point for national AI strategies and regulatory frameworks worldwide," including the EU AI Act, which reflects the OECD's focus on risk management and human oversight. Together, UNESCO and OECD demonstrate that while voluntary frameworks can establish global consensus on core values, they cannot by themselves ensure consistent, enforceable protection for individuals harmed by AI systems, highlighting the need for jurisdictions to build binding regulatory structures on top of these shared ethical foundations.

Comparative Analysis

Framework	Legal Status	Approach	Enforcement
EU AI Act	Binding law	Risk-based classification	Fines up to 7% global turnover
U.S. AI RMF	Voluntary	Sector-specific guidance	No direct penalties
NYC Local Law 144	Binding law	Domain-specific mandate	Municipal enforcement
UNESCO Recommendation	Non-binding	Principles based	None
OECD AI Principles	Non-binding	Principles based	None

The comparison reveals a consistent pattern. Non-binding frameworks have succeeded in articulating shared values such as transparency, accountability, and human oversight, yet they fall short of providing reliable, enforceable protections for individuals harmed by AI systems. The EU AI Act demonstrates that robust, binding regulation is attainable. The broader challenge for both national and international governance is to build upon that foundation rather than retreat to the limitations of voluntary standards.

IV. TOWARDS AN ENFORCEABLE AI ACCOUNTABILITY FRAMEWORK

A. A New Framework for understanding AI Liability: The Developmental Model

Legal systems usually begin with a straightforward question: who bears responsibility? In the context of artificial intelligence, that question becomes far more complex because AI systems do not behave like

ordinary tools. A hammer performs exactly what its user directs. An AI system, by contrast, learns from its training data, feedback, and the environment in which it operates, and its behavior can evolve in ways its developers did not foresee and its deployers may not fully understand.

A more illuminating analogy, and one with genuine legal utility, is the developmental model of human responsibility. Consider the trajectory of a child from infancy to adulthood. An infant arrives in the world without knowledge, without judgement, and without the capacity of moral reasoning. What the child becomes depends on three things: what it is taught, the environment in which it develops, and the decisions it makes as its capacity for independent judgement matures. No single actor bears complete responsibility for the adult the child becomes. Responsibility is distributed across parents, educators, institutions and ultimately the individual, in proportion to each actor's role in shaping development.

AI systems follow a strikingly similar trajectory. A newly created model knows only what its training data allows it to know. Like an infant learning language through observation and imitation, an AI system absorbs patterns from the data it is given, first reproducing those patterns mechanically, much as a child mimics sounds before grasping their meaning, and only later gaining the ability to generate new outputs. The quality of that learning depends entirely on what the system is taught and who teaches it. As philosophers of science might note, and as Shakespeare suggested in his reflection on the seven ages of man, what endures is not the individual moment but the lasting imprint of what was built, taught, and passed on. [37] AI systems evolve through new versions, but what persists is the underlying architecture of values or the absence of them, embedded by their creators at the foundation. [38]

This developmental model has direct implications for legal liability. Just as we do not hold a child solely responsible for behavior that reflects the environment in which it was raised, we cannot hold any single actor solely responsible for the harms an AI system produces. Liability must be distributed across the chain of development and deployment, allocated in proportion to each actor's contribution to the harm.

B. A Three-Tier Liability Framework

Drawing on the developmental model, this article proposes a three-tier framework for allocating legal responsibility for AI caused harm.

Tier One: Developer Liability: The developer of an AI system occupies the position of the child's first teacher- the actor who shapes the foundational architecture of the system's knowledge and judgement. If a developer trains an AI system on biased, incomplete, or deliberately misleading data, the resulting harms are properly attributed to that foundational decision. Developer liability should attach when the training data was known or reasonably knowable to be biased or defective; when the system's design incorporated insufficient safeguards against foreseeable misuse; or when the developer failed to document known limitations and risks in a manner accessible to downstream deployers.

This standard reflects established principles of products liability law. [25] A manufacturer that releases a defective product into the marketplace is responsible for the harm that defect causes, even when the injury occurs long after and far from the original act of manufacture. AI systems function as products in this legal sense, and their developers occupy the role of manufacturers whose foundational design choices can give rise to downstream harm.

Tier Two: Deployer Liability

The organization that deploys an AI system in a commercial or

governmental context occupies the position of the parent or institution that shapes the environment in which development continues. A company that takes an AI system and integrates it into consequential decision making: hiring, lending, medical diagnosis, criminal risk assessment assumes responsibility for ensuring that the system performs appropriately in that specific context. Deployer liability should attach when the organization failed to conduct adequate pre-deployment testing; when it deployed a system in a domain for which it was not designed or validated; when it failed to implement human oversight mechanisms for high-stakes decisions; or when it continued to use a system after known failures were identified.

The Air Canada chatbot case, [26] discussed in Part II, illustrates this principle. Air Canada could not disclaim responsibility for its chatbot's outputs on the grounds that the chatbot operated autonomously. The Tribunal correctly recognized that an organization cannot benefit from automation while escaping the legal consequences of its automated systems behavior. The deployer who captures the value of AI must also bear the risk.

Tier Three: Shared and Systemic Liability

Some AI-related harms cannot be traced to a single actor; instead, they arise from failures spread across the entire chain of development, deployment, and regulation. When the COMPAS algorithm generated racially biased risk scores that influenced sentencing decisions, responsibility did not rest solely with Equivant, the developer, nor solely with the courts that relied on those scores, nor with the jurisdictions that authorized their use. Each played a part in a system that produced unjust outcomes. A meaningful liability framework must therefore be capable of addressing this kind of distributed causation, where harm results from the combined actions and omissions of multiple actors rather than from one identifiable source.

For harms of this nature, this article advances a shared-responsibility model inspired by the market-share liability doctrine developed in pharmaceutical litigation. [27] When multiple actors across the AI development and deployment chain contribute to a harmful outcome and the specific source of causation cannot be pinpointed, liability should be distributed among them according to their respective roles and the extent to which each failed to meet the applicable standard of care.

C. Mandatory Transparency and Explainability Obligations

Liability systems function best when paired with transparency rules that make the underlying facts discoverable. An injured party cannot pursue a meaningful claim against an AI developer or deployer if the system operates in complete opacity. For that reason, transparency and explainability are not simply ethical preferences they are essential conditions for legal accountability.

The article argues that any AI system used in high-stakes decision-making including employment, credit, healthcare, criminal justice, housing, education, and public benefits, should be subject to mandatory explainability obligations. These obligations should require informing individuals when an automated system influenced a decision affecting them; providing a plain-language explanation of the factors the system considered and how those factors were weighted; documenting the system's training data, known limitations, and error rates; and ensuring a meaningful opportunity to contest automated outcomes before a human decision-maker.

These obligations are entirely within current technical capability. The field of explainable AI already offers developed tools including LIME, SHAP, and counterfactual explanation techniques[28] that can generate clear, human-interpretable accounts of how models reach their

decisions across a wide range of applications. The real barrier is not technological feasibility but the absence of legal requirements. Without legal mandates, organizations have little incentive to prioritize transparency, [39] and commercial pressures will continue to reward opacity rather than accountability.

D. Mandatory Human Oversight in High Stakes Domains

No AI system, however sophisticated, should be permitted to make final, unreviewable decisions in matters affecting fundamental rights and interests without meaningful human oversight. This principle reflected in the EU AI Act's human oversight requirements [29] and in UNESCO's Recommendation should be translated into binding legal obligations in the United States and other jurisdictions that have not yet adopted comprehensive AI regulation.

Human oversight obligations should ensure that in domains such as criminal justice, healthcare, employment, and financial services, AI outputs serve as inputs to human judgment rather than substitutes for it. A COMPAS risk score should guide a judge's reasoning, not replace it. An AI-generated diagnostic suggestion should inform a physician's assessment, not override it. For oversight to be meaningful rather than symbolic, human decision-makers must understand the system's limitations well enough to exercise genuine, independent judgment.

E. Independent Algorithmic Auditing

Transparency and human oversight requirements are most effective when supported by independent verification. This article proposes the establishment of mandatory independent algorithmic auditing requirements for high-risk AI systems, modelled on financial auditing frameworks. [30] Just as publicly traded companies are required to have their financial statements audited by independent certified accountants, organizations deploying high risk AI systems should be required to submit those systems to independent technical audit at regular intervals and before significant updates.

Audits should assess the accuracy and reliability of the system across demographic groups; the quality and representativeness of training data; the effectiveness of bias mitigation measures; and compliance with applicable transparency and explainability requirements. Audit results should be made publicly available, enabling regulators, researchers, civil society organizations, and affected communities to assess AI systems performance and hold deployers accountable.

F. Federal Coordination and the Path Forward

Although recent state and local initiatives demonstrate growing recognition of the need to regulate AI, the resulting patchwork of laws cannot provide the consistency or legal certainty required for effective national governance. Varying regulatory standards increase compliance burdens, create opportunities for regulatory arbitrage, and produce uneven protections for individuals depending on the jurisdiction in which they reside.

V. CONCLUSION

Artificial Intelligence, like every transformative technology before it, will continue to evolve. Systems that seem sophisticated today will be superseded by more capable successors. Algorithms will be retrained, updated, deprecated, and replaced. What will not automatically change unless we deliberately design it to, is the architecture of accountability within which those systems operate. The versions pass. The values, or their absence, persist.

This is the central lesson of the developmental model proposed in Part IV. A child shaped by neglect, bias, or deliberate miseducation does not simply outgrow those influences upon reaching adulthood. The effects compound. They shape decisions, relationships, and institutions long after the original conditions that produced them have changed. Artificial

intelligence systems operate by the same logic. A model trained on biased data, deployed without meaningful oversight, and shielded from legal accountability does not become more just as it scales. It becomes more consequential and the harms it produces become harder to trace, harder to remedy, and harder to reverse.

The governance frameworks surveyed in this article represent genuine progress. The EU AI Act demonstrates that comprehensive, binding AI regulation is legally achievable and politically possible. UNESCO's global ethical framework demonstrates that the core values of human dignity, transparency, and accountability command genuine international consensus. The emerging body of case law, from the Air Canada chatbot ruling to the judicial sanctions imposed on attorneys who submitted AI-generated fabrications demonstrates that existing legal institutions are capable of adapting to the challenges AI presents, even in the absence of specific legislative mandate.

But progress is not sufficiency. Voluntary principles without enforcement mechanisms are statements of aspiration, not instruments of governance. A framework that protects individuals in the European Union but leaves those in other jurisdictions without meaningful recourse is a framework that has chosen convenience over justice. A liability system that diffuses responsibility so thoroughly that no single actor can be held accountable is a system that, in practice, holds no one accountable at all.

The question this article ultimately poses is not technical. It is not whether artificial intelligence can be made safer, more transparent, or more explainable --- the evidence suggests that it can. The question is whether democratic societies will choose to require it. Whether the governments that claim to represent human dignity will build

institutions capable of defending it. Whether the legal systems designed to provide remedy for harm will extend that protection to harms produced by machines as readily as they extend it to harms produced by people.

Shakespeare wrote that all the world is a stage, and the men and women merely players; each making their exits and their entrances, each playing many parts. Artificial intelligence has now joined the cast. It did not write its own lines. It learned them from us, from our data, our decisions, our values, and our failures. What it does with those lines, on the stages that matter most, the courtroom, the hospital, the hiring office, the sentencing hearing will reflect not the intelligence of the machine but the wisdom, or the negligence, of the society that built it.

The systems we design today will outlast the intentions of their designers. The question worth sitting with; the one this article hopes to leave unresolved enough to demand an answer, is this: when the algorithm makes the decision that changes a life, whose values will it be acting on?

And did we ever actually choose them?

Notes

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Drawing the Line: How to Detect Gerrymandering through Mathematics

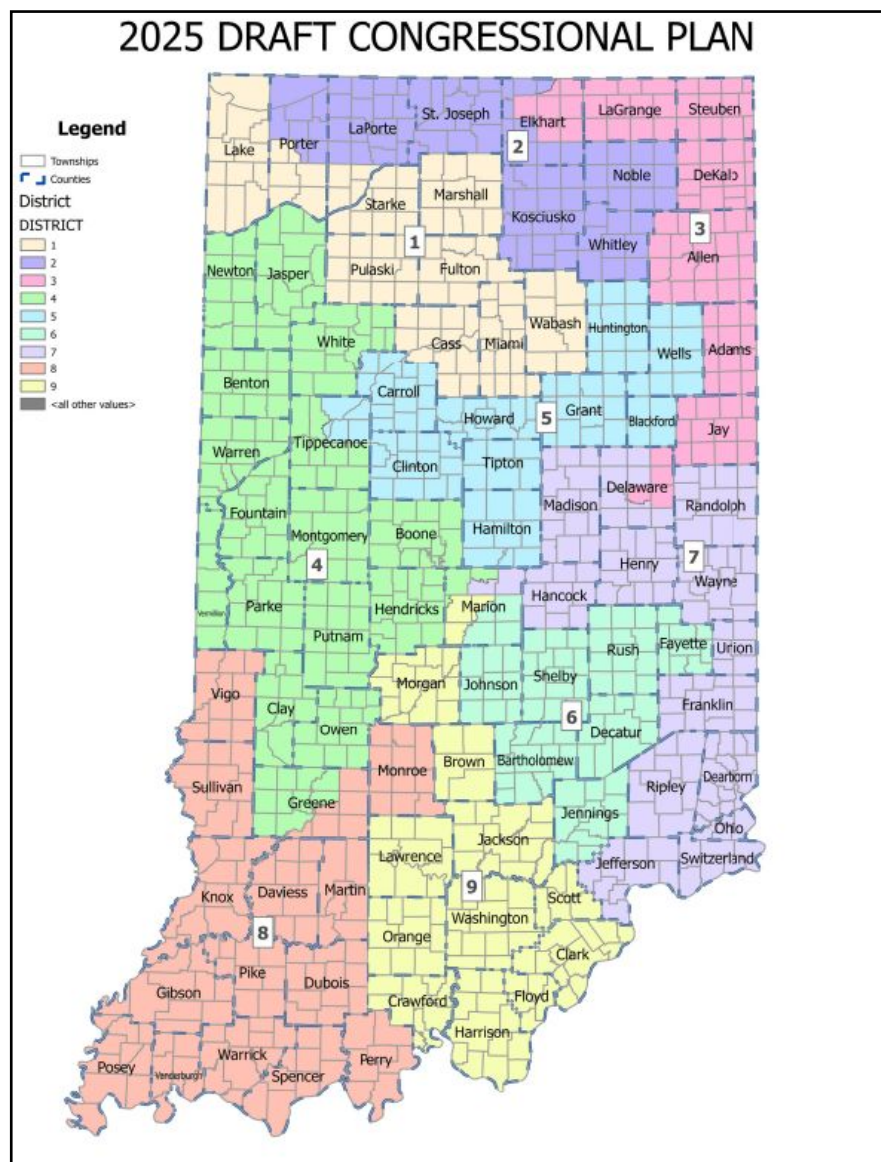
Tanay Thakore

Columbia University in the City of New York

**“THE MOST EFFECTIVE DEFENSE AGAINST
THE USE OF COMPUTERS TO CREATE
BIASED, PARTISAN MAPS IS THE USE OF
COMPUTERS TO EXPOSE THEM.”**

Abstract

Starting in the summer of 2025, states across the nation took up partisan gerrymandering. A cursory inspection of proposed map in Indiana does not uncover anything amiss. My question became how to detect partisan gerrymandering for a justiciable standard as first introduced by Justice Kennedy in *Vieth v. Jubelirer*. This paper explores many proposed methods, before ending up on recombination as the ideal method.



In November 2025, Indiana state legislators unveiled a redrawn congressional district map that fundamentally altered the state's political landscape. Although Indiana historically maintains a 60-40 Republican voting split, this mid-decade redistricting was strategically modeled to shift the state's 7-2 Republican delegation into a 9-0 congressional sweep. [1] The timing and nature of the announcement—following public pressure from President Donald Trump and lobbying from Vice-President JD Vance, who visited the state multiple times in the lead up to the reveal—bolstered claims by national media that the map was a product of aggressive gerrymandering, the practice of manipulating legislative districts for partisan gain. Indiana Republicans, including Governor Mike Braun and State Representative Andrew Ireland were notably transparent regarding these objectives, with Ireland going so far as to say “Democrats have no business representing us in Washington.” [2] However, a cursory visual inspection of the proposed map does not immediately reveal any manipulation afoot.

Visual hallmarks, such as county splits, compactness, and contiguity have served as the primary metric to detect partisan manipulation since the inception of the practice. In 1812, Massachusetts Governor Elbridge Gerry, a Democratic-Republican, deemed it necessary to massage the electoral lines in order to limit the Federalist party's power in the state senate. The most flagrantly distorted district was located in Essex County; it “meandered around the western and northern county line” to consolidate as many Federalist towns as possible. Capitalizing on this visual absurdity, The Boston Gazette appended “a head, wings, tail, and claws” to this map, christening it a “new species of Monster”: a Gerrymander, a portmanteau of Gerry and salamander, the animal that the district resembled. [3]

In contrast to the 1812 ‘Monster’, the 2025 Indiana map looks remarkably normal. The proposal splits only six of the state's ninety-two counties, and where it does, it largely adheres to township boundaries, the first-level subdivision of counties in Indiana.

Absent are districts that look like an octopus with ‘tentacles’ emanating from a central blob or those districts with ‘holes’ that define typical partisan maps; instead, these districts appear geometrically compact. Moreover, the connectivity of the districts is robust. The intradistrict path between any two points avoids the convoluted routes—such as those traversing bodies of water multiple times or running along the narrow shoulders of a bustling interstate—that have appeared in the past. [4] All things considered, for a map that gives Democrats no business representing Hoosiers in Washington, Indiana’s map looks ordinary.

To understand how a visually compact map can still guarantee a partisan sweep, one must look to the dual mechanics of gerrymandering: packing and cracking. [5] Packing involves concentrating the opposition’s voters into a single district to waste their surplus votes in a lopsided victory; while cracking involves splintering a coherent bloc of voters across multiple districts to ensure they cannot achieve a majority in any of them. The typical strategy is to pack many opposition voters into the same district, and then crack the rest of the voters across the remaining districts. While the original 1812 Gerrymander of Essex county relied on packing the Federalists together, a method easily detected by jagged lines and county breaks, what Indiana has done falls solely under cracking. Some understanding of Indiana political geography would reveal that five of the six county splits target Democratic-leaning urban centers: Porter county around Valparaiso, Elkhart county around Elkhart, Tippecanoe county around West Lafayette, Delaware county around Muncie, and Marion county around Indianapolis. [6] The cracking is most apparent in Marion county, where Indianapolis is partitioned into four separate districts. Consequently, three neighbors in the heart of the state may find themselves represented by three different congressmen, each tied to a district that stretches to the state’s distant western, eastern, or southern borders. By submerging these urban Democratic clusters within sprawling rural Republican constituencies, the map achieved a total partisan sweep without failing the eyeball test.

While the eyeball test provides an intuitive starting point for identifying electoral manipulation, it remains fundamentally subjective. A boundary that appears suspiciously jagged to one observer may seem like a necessary adherence to balance out the population to another. Consequently, the central challenge of determining gerrymandering has historically been establishing an objective, quantifiable standard for partisan gerrymandering that can withstand scrutiny. This search for a mathematical smoking gun was catalyzed by Supreme Court Justice Anthony Kennedy in *Vieth v. Jubelirer* (2004). In his concurring opinion to the majority, Kennedy noted that with “no agreed upon substantive principles of fair districting,” the Court lacked a basis to define “clear, manageable, and politically neutral standards” for measuring how a partisan map burdens representational rights. However, Kennedy left the door ajar, suggesting that the current absence of a workable standard “does not mean that none will emerge in the future.” [7] And since then, scores of mathematicians and political scientists have raced to develop a single definitive metric to enter through that doorway and satisfy the Court’s demand for a neutral arbiter of democratic fairness.

In the race to satisfy Justice Kennedy’s demand for a ‘manageable standard’, early efforts focused on quantifying the visual hallmarks of redistricting: county splits, compactness, and contiguity. The first of these, the frequency of county splits, is a simple counting problem that is easily circumvented by modern mapmakers who are gerrymandering, as evidenced by Indiana’s 2025 map. The last, contiguity, is defined in mathematical terms as path-connected, meaning a traveler can move between any two points within a district without ever leaving it. [8] When geographic anomalies, such as islands are involved, bridges or ferries may serve as this connection. But path-connectedness remains a binary condition rather than a metric, a district is either path-connected or it is not, meaning one can create an extremely complicated district that is only tenuously connected and still pass this test. This leaves compactness as the primary geographic method for detecting gerrymandering. Although thirty-two states have enacted statutes requiring compact districts, their legal definitions range from “doesn’t meander too much” to “those that are square, rectangular, or hexagonal in shape, and not irregularly shaped, to the extent permitted by natural or political boundaries.” [9]

Mathematically, compactness is a functional ratio: a measure of a district's area relative to its perimeter. The underlying theory is that a gerrymandered district will inevitably feature an inflated perimeter that fails to encompass a proportionally significant area.

While several geometric scores exist to quantify compactness, the Polsby-Popper and Reock benchmarks remain the most prominent. The Polsby-Popper score, defined as $PP = 4\pi A/p^2$, calculates the ratio of a district's area (A) to that of a circle with an identical perimeter (P), with a perfect circle representing the ideal of compactness. [10] However, this metric struggles with natural, meandering boundaries; for instance, Indiana's 8th district in its 2021 map follows the Ohio River, resulting in a 'poor' score of .211 despite being graded an 'A' for geographic fairness by the independent Princeton Gerrymandering Project. [11] Conversely, the Reock score measures the ratio between a district's area and the area of the smallest circle that can encompass it. [12] This metric is easily manipulated, as a district can be stripped of specific voting blocs through thin slices and still maintain a high score. Ultimately, whether utilizing Polsby-Popper, Reock, Schwartzberg, Convex Hull, Population Polygon, Length-Width, or any other of the multitude of potential geometric measures, the score will prove insufficient: a map can be fairly drawn yet score poorly, or a high score will hide aggressive gerrymandering. [13] For issues relating to electoral manipulation, maybe an analysis of the votes themselves is needed, instead of an analysis of the map itself.

The leading metric for electoral analysis is the efficiency gap, first proposed in 2015 by Nicholas Stephanopoulos and Eric McGhee. This method quantifies fairness by identifying wasted votes; wasted votes are those cast for a losing candidate or those for a winner that exceed the 50% threshold required for victory. A district where the winning side holds nearly all the wasted votes is considered packed, while one where the losing side holds the wasted votes is competitive (typically seen as a positive). However, when several adjacent districts where most of the wasted votes are on the losing side, it is a sign of a cracked plan. The final efficiency gap is determined by the formula, $EG = (W_A - W_B) / V$, where $W_A - W_B$ is the difference between two parties' wasted votes and V is the total number of votes cast. [14]

Stephanopoulos and McGhee proposed that a value exceeding .08 denotes an unacceptable level of gerrymandering. Their nonprofit organization PlanScore calculated that Indiana's proposed map had a theoretical efficiency gap of .286, [15] highly suggestive of gerrymandering.

Is that it? Is the efficiency gap the smoking gun needed to detect gerrymandering? No. Its mathematical foundation relies on the questionable assumption that a 75-25 vote split is the ideal. As a result, it penalizes any deviation from this ratio. [16] For example, when a party wins 60% of the statewide vote (typical of Indiana), a fair map according to the efficiency gap will give the winning party 80% of the seats, not the 100% that occurs in Indiana's proposed map nor the 60% that proportionality (typically seen as the ideal) dictates. It also tends to struggle with volatility in competitive races. Because a genuinely close election produces lopsided wastage depending on which party gets lucky on election day, a last-minute systematic shift toward one party can make a neutral map look like a deliberate gerrymander. [17] This appears to be the case in Indiana's 2024 data (what PlanScore used for their analysis), where a late surge of undecided voters for Republican candidates skewed the results. [18] Furthermore, while other electoral methods beside the efficiency gap exist, they often rely on the state being competitive (within a 55-45 margin) or both parties winning at least one seat—conditions that Indiana simply does not meet. [19]

The fundamental issue with all of these metrics is their attempt to distill the immense complexity of districting into a singular number. A more effective approach may be seeing what makes a map definitively not gerrymandered. And for that, we return to Massachusetts, a state that has received much criticism from Republicans for its congressional maps. State Representative Andrew Ireland's justification of Indiana's gerrymandering was "[Massachusetts also has nine congressional seats. The political makeup is almost exactly the same [as Indiana], 60/40, but Democrat to Republican. But in Massachusetts, ... it is entirely a Democrat congressional delegation." [20] But the scholarly consensus is that Massachusetts is not partisanly gerrymandered, even though no Republican has represented the state since 1994. [21]

To substantiate this point, a team of researchers at Tufts University generated many quintillion redistricting maps for Massachusetts with total freedom to gerrymander (their only requirement was that the districts were equal in population). Only the 2016 voting data was able to feasibly have proportional representation for Republicans; once the contiguity requirement was restored, no statewide election data since 2000 resulted in a Republican seat. Furthermore, across five separate elections, none of these quintillions of redistricting maps achieved an efficiency gap under 11%, meaning that they were all above the 8% signifying gerrymandering. The data reveals that “[t]hough there are more ways of building a valid districting plan than there are particles in the galaxy, every single one of them would produce a 9–0 Democratic delegation,” meaning that partisan gerrymandering is an impossibility in Massachusetts. [22] This conclusion suggests a new detection strategy for gerrymandering: by comparing a map against a massive ensemble of viable alternatives, one can identify partisan gerrymandering not by a single metric, but whether the map grants a party more seats than the statistical average. The question shifts from ‘Is this map fair?’ to ‘Does this map produce a normal result?’

In order to see if a map is normal, it requires a large collection of viable alternatives, a process that introduces one of the most intriguing challenges in redistricting. The simplest method is the flip-based method, where a computer flips the district of one precinct at a time to create a new temporary map, and after many flips, it creates a new map for the ensemble. It then generates quintillions of these maps. [23] This can be visualized as a blindfolded person standing on a single tile in a large room, taking one random step at a time to explore. After thousands of steps, the person arrives at a new location representing a new map. If repeated enough times, the resulting data points should represent the floor of the entire room. But now imagine if the room was actually two rooms connected by a very long and narrow passageway. The blindfolded traveler may never be able to find the entrance to the corridor, much less navigate to the other side. This same issue arises with the flip-based system, that it may be trapped within a specific neighborhood of maps, failing to represent the entire spectrum of possibilities. The other idea is recombination where two districts are merged and then split along a new boundary.

By ensuring the population remains balanced while allowing the shapes to change drastically, the computer effectively takes much larger leaps. This approach allows the algorithm to traverse the entire space more fully, ensuring the ensemble is an accurate representation of all the potential maps. This approach is also much quicker at generating a large sample, taking only a few hours. [24]

While 21st-century gerrymandering has been revolutionized by initiatives like the Redistricting Majority Project (REDMAP), which leveraged sophisticated software and granular voter data to cement long-lasting partisan advantages, [25] this same computing power now provides a vital neutral framework for evaluating fairness. While these algorithms do not necessarily identify a singular perfect map, they excel at detecting whether a specific proposal offers an extreme, outlier advantage compared to the broader landscape of mathematical possibilities. The most effective defense against the use of computers to create biased, partisan maps is the use of computers to expose them. In the end, the Indiana State Senate voted to reject the proposed map, with every Democrat and a majority of Republicans deeming it gerrymandered and unfit for the state. [26] Instead, the state maintained a map resulting in a 7-2 congressional delegation split; a distribution that aligns with the median outcome seen in the ensemble of Indiana's potential districts. [27]

Notes

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- [10] *Id.*
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Mr. AI Attorney at Law: AI in the Criminal Law System and the Right to Effective Assistance of Counsel

Luke Plesko

United States Air Force Academy

**“ HOW WE DEFINE ‘EFFECTIVE COUNSEL’
TODAY WILL DETERMINE THE BOUNDARIES
OF JUSTICE FOR GENERATIONS TO COME. ”**

Abstract

Synthetic legal representation has generated a debate to determine whether AI learning models have a place in the legal profession. The tension between U.S. Constitutional privileges and protections and the increase of generative AI is only growing, imminently requiring the Legislature or Judiciary to proactively consider the right to counsel under the 6th Amendment. This paper explores whether the Sixth Amendment's guarantee of assistance of effective counsel in a criminal proceeding constitutionally requires human advocacy or could it be substituted by artificial intelligence. Found in the landmark case *Strickland v. Washington*, the right to counsel requires effective representation by a showing of: (1) deficient performance by counsel, (2) that performance prejudiced towards the jury. With an increased public interest in new technology at issue with data privacy, attorney client privilege and the roll out of products like Harvey AI, which has already been integrated into over one thousand law firms, the demand for a constitutional answer has begun to be hotly disputed and debated. [1] This paper will answer whether it is legal e for Artificial Intelligence to provide effective counsel, what level of oversight would potentially be constitutionally required, and whether AI could increase national access to justice. Artificial intelligence cannot substitute human advocacy because, at present, it does meet the threshold as defined by the Strickland efficiency test. Specifically, the lack of technological accountability and bias introduce deficiencies in performance. Further, concerns of confidentiality and lack of fidelity present clear and distinct potential prejudice towards jury members. As courts, legislatures, and law schools begin to reconcile with this transformation, how we define "effective counsel" today will determine the boundaries of justice for generations. The proposed policy change is to introduce an intermediate scrutiny test for permitting the use of AI that does not prevent a misrepresentation from the Sixth Amendment.

I. Introduction

Robotic weaponry, image generation, and synthetic therapy have one thing in common: potentially dangerous uses of Artificial Intelligence (AI). As examples, AI can quickly get abused without proper protection mechanisms to maintain law of war principles within the realm of robotic weapons, prevent the creation of cartoon child pornography through image generation, or halt artificial therapy from negatively influencing people to harm others and themselves before a tragedy. However, the question remains: who is to blame when technology fails to account for human consciousness or when it cannot impute human judgment? As AI advances, humans must emphasize the importance of maintaining human input through critical touchpoints with technology, particularly in fields that demand accountability, ethics, and fidelity. [2] The rapid development of AI necessitates a rigorous re-evaluation of the jurisprudential underpinnings of privilege and constitutional defense. The legal profession is currently witnessing a shift from digital tools for practitioners to the risk of potential AI replacements. This evolution challenges the traditional boundaries essential to the proper functioning of the criminal justice system.

Since the ratification of the Sixth Amendment in 1791, the right to “effective assistance of counsel” in a criminal setting has been interpreted as a constitutional guarantee to an accused in a criminal proceeding. Moreover, it is a guarantee of personal, professional legal advice, ethical representation, and zealous advocacy. As these systems become more capable, a novel dilemma is presented: whether the Sixth Amendment right to “effective assistance of counsel” requires meaningful human involvement in criminal defense representation, like litigation, when AI systems may equal, or soon outpace, the abilities of even the best, although still human, legal minds.

Although emerging AI systems may soon rival or exceed human attorneys in drafting, legal research, and legal writing, the Sixth Amendment’s guarantee of right to counsel mandates effective assistance which current AI systems cannot satisfy. Current AI lacks accountability, possesses inherent bias based on datasets which effectively renders its advice deficient, does not adequately possess

means of protecting client confidentiality, and wholly lacks fidelity, resulting in clear and distinct discriminatory effects which could prejudice potential jury members. By denying the use of artificial attorneys, the legal system will maintain its constitutionally grounded principles founded within the Sixth Amendment and protect the right to “effective assistance of counsel” as defined by the Strickland efficiency test.

This paper proceeds in three parts: (1) background, (2) analysis, and (3) policy consideration. First, this paper examines the text, history, and early interpretations of the Sixth Amendment with the introduction of the right to effective assistance of counsel. Second, this paper analyzes in detail whether artificial intelligence can satisfy this constitutional standard with four key deficiencies: (1) the lack of clear accountability for AI systems; (2) the inherent bias of AI; (3) AI’s inability to preserve confidentiality between client and attorney; and (4) the absence of fidelity for the client. Finally, the paper addresses efficiency considerations for AI and emphasizes the constitutional importance of the right to counsel regardless of the efficacy of AI.

II. Background

A. Text and Foundation

The Sixth Amendment to the United States Constitution provides that: “In all criminal prosecutions, the accused shall enjoy the right to a speedy and public trial, by an impartial jury of the State... and to have the Assistance of Counsel for his defence.” [3] At its core, the Sixth Amendment grants all individuals, under prosecution by the United States government, State governments, or district prosecutors, basic privileges to ensure fair and equal application of the law. [4] In the Sixth Amendment toward the end of the amended section, the right to the “assistance of counsel” outlines a defendant’s right to fairness within the legal system, ensuring that they retain the ability to meaningfully respond to charges brought by the government. These privileges remain absolute and are the core of the United States’ legal system because as noted by the Supreme Court in *Cronic*, “[o]f all the rights that an accused person has, the right to be represented by counsel is by far the

most pervasive, for it affects his ability to assert any other rights he may have.” [5]

Importantly, the text of this Amendment does not merely reference the presence of counsel, rather, the assistance of counsel. [6] This right has been deemed so essential to the proper functioning of the judicial system that “[e]ven in instances in which a defendant wishes to waive counsel, “[t]he Constitution does not forbid States from insisting upon representation by counsel for those competent enough to stand trial but who suffer from severe mental illness to the point where they are not competent to conduct trial proceedings by themselves.” [7] This principle also imputes a standard of meaningful representation. Additionally, to understand the full scope of the Sixth Amendment, it is necessary to understand the history and purpose of the amendment. Although interpretative approaches may differ, whether grounded in originalism, living constitutionalism, political theory, or pragmatism, all perspectives require an inquiry into the purpose of the right to counsel and what underlying issues this guarantee seeks to address and overcome.

B. Historical Analysis

The right to counsel in criminal courts has its history founded in English common law. [8] Under English common law, defense attorneys were not guaranteed participation on behalf of their client in felony cases, until the 18th century. [9] Prior to the American Revolution, reform movements had secured more robust legal measures as an early identification of this issue. [10]

Then, during the American Revolution, the Colonies and their representatives at the Congressional Convention in Philadelphia found this basic principle critical to the preservation of the justice system and it was retained when ratifying the Articles of Confederation. Ratified in 1791, the Sixth Amendment guaranteed accused people the “Assistance of Counsel” in criminal prosecutions as a structural equalizer. [11] A consistent theme of the drafters of the constitution was protecting an individual from the awesome, overwhelming power of the government. As evidenced here, the right to counsel arose from concerns about this power disparity in trials under British rule, where

accused individuals were left to defend themselves against the English barristers, often completely without legal education, experience, or awareness. [12]

C. Early Interpretation

The Supreme Court's early interpretation of the Sixth Amendment transformed the right to counsel throughout the 20th century. [13] Initially, the right was understood by the exact wording of the constitution; however, through a series of decisions, the Supreme Court expanded the scope of the Sixth Amendment to ensure that defendants receive fair and just trials with effective legal representation.

One of the earliest interpretations of the Sixth Amendment right to counsel was in the 1932 Supreme Court case, *Powell v. Alabama*. [14, 15] In this case, nine black adolescents were accused of raping two white women. In the Alabama state court system, the adolescents were tried in three separate cases, each lasting less than a single day. The Alabama law at issue required that counsel be appointed in capital cases, however the assigned defense counsel had failed to consult their clients. All nine were sentenced to death. The defendants appealed and were granted a hearing at the United States Supreme Court. [16] The Court analyzed elements of the 14th Amendment's due process clause and the origin of those laws. [17] In the concurring opinion, Justice Sutherland stated that this clause was derived from the 6th Amendment. Justice Sutherland referenced the "[a]ssistance of counsel" in his opinion and stated that this assistance must be granted in capital cases. [18] Thus, the Supreme Court held that, in capital cases, the right to counsel is a fundamental and essential for issuing the ultimate form of punishment: death. [19] This forced states to finance and grant an accused counsel, specifically in capital cases.

Similarly in *Gideon v. Wainwright*, the Supreme Court extended the Sixth Amendment right to counsel to all state criminal prosecutions. [20] Clarence Gideon, charged with felony theft in Florida, denied counsel because he could not afford an attorney. At the time, Florida law required counsel to be financed by the state only in capital cases. [21] Subsequently, Gideon's request for a state defender was denied on the grounds of Florida Law. The Court examined that a fair trial cannot be

guaranteed if the defendant lacks legal expertise. [22]

“It might, however, be said that there is such an implication in *Avery v. Alabama*, 308 U. S. 444 (1940), a capital case in which counsel had been appointed, but in which the petitioner claimed a denial of “effective” assistance. The Court, in affirming, noted that, “[h]ad petitioner been denied any representation of counsel at all, such a clear violation of the Fourteenth Amendment's guarantee of assistance of counsel would have required reversal of his conviction.” [23]

This reasoning represents the Court’s recognition that effective legal representation is essential to ensuring fairness within the criminal justice system. [24] Gideon’s conviction was overturned because the state had granted him to waive his right to counsel while Gideon was in an unknowing and involuntary state. [25] This case expanded the scope of the requirement of states for providing counsel ensuring that if someone chooses to waive this right that they do so in a knowing and voluntary state.

The Supreme Court further refined their interpretation and the meaning of effective in *Strickland v. Washington*, establishing the modern test for *effective* counsel. [26] In *Strickland*, the defendant argued that his attorney’s failure to investigate mitigating evidence constituted ineffective representation. [27] The Supreme Court analyzed the definition of effective counsel when considering a defendant’s right to assistance of counsel, as provided by the Sixth Amendment. [28] Notably, the court held “the essential aim of the Amendment is to guarantee an effective advocate for each criminal defendant.” In the opinion, the Court agreed that the Sixth Amendment guarantees more than the mere presence of counsel and articulated a two-pronged test for effectiveness. [29] Justice O’Connor provided the two-pronged test by defining just the opposite. Counsel can be deemed *ineffective* when: (1) counsel’s performance was deficient, and (2) the deficiency prejudiced the defense. [30]

Taken together, these decisions by the Supreme Court establish that the Sixth Amendment is not just the presence of counsel, rather the right of effective counsel for each accused. It establishes the baseline for all

other constitutional rights in the criminal justice system as, without effective counsel, there would be no 4th Amendment search and seizure issues, 5th Amendment right to remain silent issues, or 14th Amendment due process issues raised on behalf of the defendant. Further, there may be no *Brady v. Maryland* discovery violations or *Giglio* notices of exculpatory evidence disclosed to the defense. [31]

In sum, the Sixth Amendment right to effective counsel is the lynchpin in the criminal justice system upon which a just and fair application of the law rests. The question presented in this paper is founded on these cases and requires the definitions to be answered.

III. Analysis

A. Framing the Problem

Artificial Intelligence (AI) systems are now capable of performing high-level legal tasks such as generating arguments, analyzing objections and rules of evidence, researching applicable case law, and even formulating defense strategies. However, the current Sixth Amendment right to counsel and Fourteenth Amendments Due Process clause interpretation requires human advocacy because of the ineffectiveness, lack of judgment, and high risks of AI. The core issue is engulfed by the constant tension between constitutional law, theory, and practice, and ever-evolving, emerging technological breakthroughs.

B. Liability

A central component of effective assistance of counsel under Strickland is the ability to identify and evaluate deficient performance. At the core of the legal system is the principle that a party is harmed, they seek a remedy. The purpose of such a suit is to hold the responsible party accountable for the wrong. Specifically, in the criminal context, when an individual is charged, tried, and convicted, the accused is held responsible and criminally liable for their wrongdoing. This foundational principle of accountability in legal systems becomes increasingly difficult to ascertain if the wrongdoer is not human. Through existing frameworks, AI systems cannot be held professionally responsible for errors, unknown biases, and legal discrepancies. [32] For example, when an artificial intelligence system produces an incorrect business

strategy or reports inaccurate financial data, corporations are often left uncertain as to who should be held responsible for the resulting harm. [33] According to Cornelia Walther, responsibility in AI systems often “cascades across multiple levels – from developers to corporate boards to regulators.” [34] This interpretation on fragmenting responsibility in the business world regarding AI is a direct and intolerable problem. Hazardous scenarios include losing valuable talent and participation from employees. [35] Similarly, in legal practices, the use of AI creates even more serious concerns.

The issue of accountability becomes even more pronounced under traditional tort law standards. As a recent analysis by the RAND Research & Commentary Reports “Liability for Harms from AI Systems,” the complexity of artificial intelligence makes it increasingly difficult to determine which party is legally responsible for harm. [36] “Courts could find it difficult to apply negligence law to AI because of the complexities of AI development and the AI ‘supply chain’ between AI developers and deployers, both of which could make it difficult to identify negligent behavior and the party responsible for an alleged harm.” [37]

As recently as March of 2026, large technology companies, YouTube and Meta, were for the first time held liable for the harm their targeting AI algorithms cause to children. In *K.G.M. v. Meta et al.*, the companies were found liable for intentionally engineering products to maximize addictiveness which led to severe impacts on the psychological wellbeing of young users. This liability was distinguishable from prior products liability holdings because it centered on the design of the algorithm rather than the content posted on the websites. [38] Additionally, while the Ninth Circuit Court of Appeals ruled that Section 230 of the Communications Decency Act, the tool to traditionally deflect liability from host websites, is an affirmative defense at trial instead of a pretrial motion so many additional trials targeting AI design and who bears the liability for it are forthcoming. [39] The likely forthcoming circuit split and disjointed quality of assigning liability in tort cases grows exponentially more concerning when applied to the Sixth Amendment’s right to counsel guarantees.

If courts cannot identify a responsible actor, defense counsel that failed in their representation, they cannot meaningfully evaluate whether counsel's performance was deficient and, in turn, this imposes prejudice towards the jury based on those discrepancies. [40] Unlike mortal attorneys, who are subject to the professional rules of conduct and discipline, artificial intelligence systems cannot be sanctioned, disbarred, or held personally accountable for providing inefficient counsel. [41] Responsibility instead falls within a supply chain of custody diffused among developers, deployers, and users. As a result, the legal system is left without a clear mechanism to address deficiencies with "Mr. AI, Attorney at Law," undermining the core guarantee of accountability embedded in the right to counsel.

Furthermore, the complexities of AI are prevalent in the *unexplainability* of artificial intelligence systems. This *unexplainability* is often referred to as the "black box" problem because it refers to a system whose internal workings are hidden from view. [42] The black box problem is extremely risky for accountability purposes within the legal system. For example, often in criminal cases, a central issue is the determination of *mens rea*. Proving or disproving intent and reasoning behind an action. However, AI cannot justify its actions with code or explanations because of the Black Box problem. As a result, the trier of fact would be left without a clear basis to evaluate intent or assign fault in the decision-making process.

C. Bias

Hypothetically, before trial, a defendant is given a choice between two forms of counsel: a barred attorney or an advanced artificial intelligence system. "Mr. AI, Attorney at Law," has been trained by thousands of closed cases, specifically, it has received data from all available cases, records, and historical information. Therefore, its legal recommendations are necessarily shaped by the patterns and outcomes embedded within the data, creating the ultimate confirmation bias. In its current state, Artificial Intelligence is incapable of generating new ideas. [43] Instead, AI uses the information available to it from a data set, developers and users set the bounds for how much information is entered, and the AI will predict the most probable next step, argument, or outcome based on that dataset. [44]

These narrowed data set an AI learns from draws concerns of bias. As recognized by UNESCO, AI decisions are susceptible to inserted biases. [45] For instance, UNESCO highlights that a simple search for “greatest leaders of all time” often produces results overwhelmingly composed of male figures, because AI-systems utilize search-engine technology to find the most probable result. [46] As mentioned, search-engines are not neutral and priorities results with the “most clicks relying both on user preferences and location.” [47] These concerns are not isolated but have the capability of branching to other facets of the rule of law, such as ethics. As emphasized in *Legal Ethics and the Rule of Law*, attorneys are bound by standards and duties, including maintaining candor to the tribunal, a base-level of competence, and the ultimate accountability. [48] These standards, as identified, are customary for all attorneys, including artificial attorneys of the future. Any legal representation or counsel should be bound by the rules of professional conduct, without exception. [49] As a part of the rules of professional conduct, legal representation must not discriminate or hold biases for counsel or proceeding assistance. [50] AI, in its current form, holds its inherent predictability bias based off of the datasets entered in the large learning model. [51] For example some products already being utilized, such as the AI system COMPAS, assigned higher recidivism scores to black defendants than to white defendants with the same or similar records when conducting risk assessment inquires. [52] The risk of learned discrimination that AI poses prevents it from complying with ethical constraints required by legal practitioners.

Because AI systems maintain learned confirmation bias which leads to inherent discrimination and ethical concerns, AI cannot provide effective representation as required under the Sixth Amendment until this issue is adequately remedied. Accordingly, a system that risks providing discriminatory outcomes, based on a dataset it learned from, the constitutional guarantee of *effective* assistance of counsel as identified by Strickland cannot be satisfied.

D. Confidentiality

Client confidentiality is a foundational principle of proper advocacy and required for effective legal representation. [53] The confidentiality between client and attorney is essential to ensure clear communication

and function as legal counsel. Furthermore, the ABA Model Rules of Professional Conduct do not permit attorneys to share any information related to their client without informed consent from the client to reveal that information. [54] The purpose of this protection is to ensure accused are comfortable being honest and open with their counsel without fear of reprisal. This protection is nearly absolute. The punishment for revealing sensitive information of a client by a defense attorney often automatically triggers an investigation and can lead to a suspension or revocation of a bar license to practice law. [56] As identified by the article *What is Client Confidentiality*, a confidentiality leak stays with a professional for the rest of their career and can adversely affect the ability to be retained in the future, in extreme cases even effectively ending their careers. [57]

This is the foundation of the concern of Artificial Intelligence releasing sensitive information without penalty, because they are not members of the bar. Regardless of the implications of accountability, the sheer risk of information not being controlled is apparent. For example, data leakage may occur when an AI system reveals information from prior inputs, or when unauthorized actors exploit weaknesses. [58] A third party can easily access information through tricking the system into giving personal or sensitive information. [59] This data spillage happens almost constantly. [60] No matter the brand, the company, the dataset, or the inputs of the user, AI cannot sufficiently protect information, including developers, moderators, or government regulators, let alone the obvious risk of simple user negligence. [61] Confidential communication, like notes, defense strategies, or even an admission to guilt would be unprotected from AI databases and could be used by the government, potentially causing a mistrial. [62] Accordingly, the use of artificial intelligence in legal representation fundamentally erodes the protections granted by confidentiality thus does not meet the threshold of effective counsel as outlined by the Supreme Court. The artificial system that requires inculpatory information to generate a result cannot guarantee the privacy of client information and trial strategy cannot satisfy the Sixth Amendment guarantees of *effective* assistance of counsel.

On the other hand, the *Heppner* case provides an alternative view on the

use of AI within the scope of confidentiality. [63] In *Heppner*, the defendant used an AI chatbot to assist in preparing documents related to his defense on charges of fraud and of false statement. [64] The FBI searched Heppner's belongings, with a warrant, and seized his devices that contained the documents generated by the chatbot. [65] The Southern District of New York ruled that attorney-client privilege did not apply because of the use of a third-party and this was clearly distinguishable from cases like *Carpenter v. United States* where cell-site data on a cell phone was found to be protected. [66]

The Court's analysis was founded on the idea that Mr. Heppner was comfortable sharing the information with someone outside of his attorney; therefore, the information is not privileged to the protection of confidentiality. However, this interpretation, if applied broadly, would extend to a vast number of tools routinely used in modern legal practice. Third-party technologies, such as email and research engines, are increasingly utilizing AI in less obvious and more insidious ways, often automatically utilizing AI to generate results or suggest email edits for human ease of use. [67] The use of these tools does not automatically relieve a user of confidentiality merely because they have AI systems, but the argument that documents generated by a bot are not privileged will likely not be construed so broadly in future cases.

Rather, the Court outlined four considerations that, when taken together with the totality of the circumstances, determine whether privilege applies: "(1) how the AI tool is used; (2) which platform is used and whether it is consumer-grade or enterprise-secured; (3) who directs the AI use (counsel versus client or pro se litigant); and, (4) what the AI platform's terms of service permit regarding data handling, training, and confidentiality." [68] These pseudo-factors suggest that at the core of the court's analysis regarding AI is more about the user and their purpose than it is about what they are using it to accomplish. A lawyer using AI to research a novel legal nuance seems to fall squarely in the privileged category while a client looking to double check their attorney's work via Mr. AI consult, like Heppner, seems to fall squarely outside of the privilege.

However, critics argue that the Heppner holding represents a judicial

misunderstanding of AI. By ruling that a defendant automatically waives confidentiality by disclosing defense strategies to a bot, the court essentially anthropomorphized the software, treating the model as if it were an independent, human third-party instead of a tool. [69] Obviously, AI does not possess independent consciousness or external thoughts; they are mathematical and data processing machines. [70]

AI also remains structurally prone to data spillage and leaks because prompts are processed on external servers and potentially reviewed by human content moderators. [71] This creates a vulnerability where sensitive data leaves the control of the defense. Disclosing defense strategies to an application that retains and learns from that data constitutes disclosure, fundamentally eroding the confidentiality protections essential to the Sixth Amendment. [72]

Conversely, the Eastern District of Michigan carved out a protective distinction based on litigant status in *Warner v. Gilbarco, Inc.* (2026). [73] Warner, a pro se litigant, admitted to utilizing OpenAI's ChatGPT for drafting her legal filings. [74] The defendants subsequently moved to compel the production of her entire chat history, arguing that no attorney-client relationship existed. The court denied the motion to compel. Recognizing that the Work Product Doctrine operates to protect a zone of privacy in which a litigant can freely analyze facts and develop legal strategies, and relying upon the historical rationale of *Anderson v. Furst* (2019), the court held that because a pro se litigant simultaneously serves as both client and counsel, the traditional direction of counsel requirement is inherently satisfied within their own person. [75] Warner's ChatGPT prompts constituted the digital record of her own thoughts, legal theories, and potential strategies and to compel their disclosure would grant the opposing party an unwarranted peek into her mind.

Notably, the distinction between *Heppner* and *Warner* is a matter of user status. For a represented criminal defendant, independent use of a commercial AI tool falls entirely outside the attorney's protected professional sphere. [76] In contrast, the pro se litigant's AI prompts are properly categorized as work product and viewed as the digital equivalent of a human lawyer's handwritten notes. By employing AI as a

legal tool, the pro se litigant is engaging in the protective act of lawyering, which must be shielded from an adversary.

Further, perhaps the more burgeoning question is regarding AI's ability to maintain privacy and whether AI preserves a reasonable expectation of privacy. While the use of third-party tools does not inherently defeat the privilege of confidentiality, the risks presented by artificial intelligence are fundamentally different because AI both processes and stores personal information over time, essentially "learning" their user, creating a grave concern for the preservation of confidentiality as presented in the Heppner case. A third-party system that cannot guarantee the protection of client communications cannot support the trust and privacy necessary for effective legal representation under the Sixth Amendment.

E. Fidelity

Fidelity is defined as a "level of loyalty or faithfulness," and it lies at the heart of the attorney-client relationship. [77] This principle requires that counsel act with undivided loyalty, exercising judgment separated from independent interests and only guided by the interest of the client. [78] The lack of extreme loyalty by an ineffective counsel can result in "significant financial and legal recourse for the breaching party." [79] The Supreme Court has long recognized that effective legal representation depends upon this form of loyal advocacy. In *Gideon v. Wainwright*, the Court emphasized that a fair trial cannot occur if the defendant is deprived of a competent counsel capable of advocating on behalf of said defendant. [80] Furthermore in *Strickland v. Washington*, the Court clarified that effective assistance, through Justice O'Connor's opinion, requires the exercise of professional judgement made with the commitment to serve the client's interests. [81] These decisions confirmed that effective counsel and the inherent right to counsel as outlined by the Sixth Amendment requires the introduction of a zealous and loyal advocate committed to the interests of the accused.

In the current stage of Artificial Intelligence, loyalty and commitment are continually at odds with programming. For example, in the 2025 study by Sewoong Oh, strategies designed to control AI's ability to maintain loyalty to a user were tested on the level of engagement and

continual support for the user's interests. [82] The study found that "loyal" AI is not inherently loyal to an individual user, but rather to the broader dataset available to the AI. [83] The study labelled this dataset as the "community" from which the AI was trained. [84] Applying this to the right to counsel, the principle of fidelity requires loyalty to client interests regardless of information, content, or context of the situation, current politics, or the encyclopedia of information that can be provided. AI inherently cannot uphold this extreme standard at its current phase, thus, under the *Strickland* test for effectiveness, AI cannot satisfy the Sixth Amendment guarantee aiding of Counsel to defendants.

F. Efficiency

It is appropriate to address dissenting opinions based on the right to counsel being guaranteed by an artificial system, thus potentially ending the access to justice problem. The first major counter argument is that Artificial Intelligence increases efficiency and is inexpensive. As stated in *Strategic Artificial Intelligence Use in Litigation Practice*, tools such as DeepJudge and Harvey AI have demonstrated substantial gains in efficiency including 60% faster discovery times, 85% adoption rates among attorneys, and increased improvements in satisfaction. [85] However, this is a false equivalence as the Sixth Amendment requires effective counsel measured via qualitative measures, rather than quantitative. As identified by *Strickland*, the "objective standard of reasonableness" is judged based on the set standards embargoed on licensed counsel such as principles of fidelity and confidentiality. [86] Thus, the risk involved with the unexplainability of AI substantially outweighs the efficacy of AI reported by quantitative measures. Further, the potential for abuse of the AI system and endless mistrial investigations would counter the gain in efficiency as identified by the research.

IV. Policy Consideration

Due to the dangers of AI utilization in its current form, the test for allowing the use of AI as counsel for defendants in criminal cases should be that of intermediate scrutiny. For example, the new test for determining AI's usage to be incorporated into the Sixth Amendment's Right to Counsel clause should follow intermediate scrutiny: (1)

substantial government interest such as increasing access to justice; and (2) narrowly tailored to a function that does not replace core principles of human legal advocacy, such as fidelity or confidentiality.

Substantial Government Interest

In this proposed policy, the government must prove a substantial government interest in the regulation or restriction with the use of AI in providing counsel to a defendant. Specifically, the government interest must be grounded in increasing areas of clear efficiency and increasing access to justice. Examples that meet this interest may include small, rural counties with no or few criminal defense attorneys or perhaps even limiting AI usage to lower-level traffic citations or misdemeanor offenses which have much lighter potential sentences. The test of effective set by Strickland, derived from the opinion of Gideon and the importance of counsel precedent set in Powell, ensures that the counsel is more than suitable for providing legal assistance and representation for the defendant. If the government is unable to prove that the total effectiveness of an AI counsel is proportional to the test set by Strickland and has clear significance in governmental efficiency, then it may be permissible.

Narrowly Tailored

It is not sufficient for the AI to meet a substantial government interest alone. The test must be narrowly tailored to the function of just defense. This limitation does not extend to supportive tools such as research, legal writing, statistical analysis, grammatical revisions, or overall case preparation and organization. Nor does it necessarily prohibit the use of an AI prosecutor. If the government chooses AI counsel as efficiency over effectiveness, that is a burden on the government. Strickland does not apply the efficiency test to prosecutors; thus, the government can choose to be more efficient but must take on the liability of the lack of effectiveness with current AI. By restricting functions that are not threatening to the core principles of legal advocacy, such as confidentiality or fidelity, the government would consequently become ineffective and less efficient. Furthermore, attorneys across the United States already rely on a variety of tools that incorporate AI including Microsoft's Co-pilot, Harvey AI, legal research databases such as LexisNexis, and automation software observed in Adobe. [87] These

technologies have embedded AI into their mechanics. As such, a restriction on these appropriate uses of AI would undermine the efficiency and effectiveness of modern legal representation. A narrowly tailored approach ensures that AI may advance without disrupting effectiveness, which is essential for the Sixth Amendment's Right to Counsel.

V. Conclusion

While Artificial Intelligence is becoming inculcated in the day-to-day rhythm of society with unparalleled speed, its role must be carefully confined to assistance, never true representation. The Sixth Amendment's guarantee of counsel was born from a tradition that values accountability, human judgment, and loyalty to the principles that form the basis of democracy. Embedded within those values, AI cannot be held ethically or professionally accountable, uphold the duty of candor to the tribunal, comprehend human dignity, or act with fidelity to our constitutional system due to its informational biases and established issues, including the Black Box problem. Its undoubtable ability to enhance the practice of law and provide access to justice cannot be ignored; however, the justice system must tread carefully and embrace a model that follows with the ideals of intermediate scrutiny. Holding that the use of AI must have a substantial government interest like efficiency, be narrowly tailored to a function that is not legal advice or litigation and be the least restrictive means by allowing human attorney oversight and representation.

Notes

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[4] *Id.*

[5] National Association of Criminal Defense Lawyers "Right to Counsel" citing United States v. Cronin, 466 U.S. 648 (1984). <https://www.nacdl.org/Landing/Right-to-Counsel>

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[13] *Id.*

[14] "Powell v. Alabama." Oyez. Accessed April 27, 2026. <https://www.oyez.org/cases/1900-1940/287us45>.

[15] Powell v. Alabama, 287 U.S. 45 (1932); see <https://supreme.justia.com/cases/federal/us/287/45/#tab-opinion-1933739>

[16] *Id.*

[17] *Id.*

[18] *Id.* at 53; *Id.* at 66

[19] *Id.*

[20] Gideon v. Wainwright, 372 U.S. 335 (1963); see <https://supreme.justia.com/cases/federal/us/372/335/#tab-opinion-1944169>

[21] *Id.* at 337.

[22] *Id.* at 342.

[23] *Id.* at 343.

[24] *Id.* at 344.

[25] *Id.* at 335.

[26] *Strickland v. Washington*, 466 U.S. 668 (1984); see <https://supreme.justia.com/cases/federal/us/466/668/>

[27] *Id.* at 680.

[28] *Id.*

[29] *Id.*

[30] *Id.* at 682 (deficient performance); *Id.* at 712 (demonstrable prejudice).

[31] *Brady v. Maryland*, 373 U.S. 83 (1963); *Giglio v. United States*, 405 U.S. 150 (1972).

[32] Cornelia C Walther, "Who's Accountable When AI Fails?," Knowledge at Wharton (May 27, 2025), <https://knowledge.wharton.upenn.edu/article/whos-accountable-when-ai-fails/>.

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Per and Polyfluoroalkyl Substances: Persistence, Exposure, and Accountability of Aqueous Film-Forming Foam Contamination

Isabella Clancy

St. Mary's College of Maryland

**“ PFAS CONTAMINATION REPRESENTS A
SYSTEMATIC GOVERNANCE FAILURE
RATHER THAN SOLELY BEING A SCIENTIFIC
OR TECHNICAL PROBLEM. ”**

Abstract

Per and Polyfluoralkyl substances (PFAS) are a group of thousands of synthetic compounds that fail to decompose or break down. PFAS accumulate over time, causing a crisis of contamination in water sources, soil, and human bodies. PFAS are abundant in aqueous film-forming foam (AFFF), and the mass production and use of it create pollution/contamination in the immediate area of naval bases. When consequences ultimately occur, citizens look to their government for the assistance they're supposedly entitled to. However, recurring dismissal and failed remediation prove to be the sole outcome. The naval base, NAS Patuxent River (PAX), is compared and placed in the historical context of military pollution and government reaction efforts. This is done through using pre-existing scientific studies/results, selective policy and litigation review, and semi-structured interviews with various stakeholders. It has become apparent that remediation efforts fail due to factors like military immunity, the prioritization of corporate profit, and the weaponization of government ignorance. This places the burden of contamination on the victims. Moving forward, continuing this research is a vital step to understanding more effective paths of remediation

INTRODUCTION

Per and polyfluoroalkyl substances (PFAS) are detrimental to the environment and communities surrounding pollution hotspots. As forever chemicals that are widely used for firefighting purposes, PFAS have been added to aqueous film-forming foam (AFFF) formulas since their invention in the 1960s (Hu et al. 2016). PFAS can travel far past the initial site of release. Scholars in environmental science and public health have created extensive documentation on the toxicology and ecological distribution of PFAS. Research has definitively established that PFAS persists because of the strength of the carbon-fluorine bond, resisting hydrolysis, photolysis, and microbial degradation (Panieri et al. 2022; Rogers et al. 2025). As a result, these compounds bioaccumulate across trophic levels with documented tissue concentrations in fish livers reaching 8,700 times higher than surrounding water (Hill et al. 2024). The human health consequences include elevated risks for kidney, testicular, thyroid, and bladder cancers, ulcerative colitis, osteoporotic fractures, polycystic ovarian syndrome (PCOS), and immune suppression (ITRC 2025; Xu et al. 2021; Steenland & Winquist 2021). Those directly exposed to PFAS from occupational exposure show serum PFAS levels in the 99th percentile of the U.S. population (Santos 2026; Rosenfeld et al. 2023).

Concurrently, researchers in environmental policy and law have begun to examine the regulatory frameworks dictating hazardous substances and the ability to litigate areas like corporate and government liability. Despite clear evidence proving the toxicity of PFAS dating back to the 1960s, regulatory agencies have delayed action through the suppression of new scientific findings, weakening environmental and health standards, and prioritizing industry profit (Richter et al. 2025; Southerland 2026). The Environmental Protection Agency (EPA) was designed to operate as an independent entity, but has instead functioned as a barrier to remediation and accountability (Elder 2025). In addition, the Department of Defense (DOD) holds sovereign immunity and the Federal Facility Agreements that grant the military control over its own cleanup (Pietkoski 2020).

Legal doctrines like the Feres doctrine and the federal officer removal statute aid in protecting the military and corporate manufacturers from liability (Stokes 2023; Mystal 2025). At the state level, certain legislatures have implemented bans and disclosure laws, but that authority is undermined by federal preemption and Supreme Court precedent (Shook et al. 2025; Maryland Daily Record 2026). These existing policies create an illusion of protection while systematically harming public health.

There is extensive amount of scientific evidence and legislation because DOD sites have a history and trend of contaminating the surrounding communities. Naval Air Station (NAS) Patuxent River base (PAX) has been selected to compare with Camp Lejeune in North Carolina, and the 157th Air Refueling Wing of the New Hampshire National Guard. North Carolina and New Hampshire provide a historical pattern for the PAX base to be placed in. The selection of the PAX base is justified by the site's characteristics, not solely by geographic convenience. As a Superfund site since 1994, PAX has over three decades of regulatory remediation efforts, documentation, and contamination data (PATUXENT RIVER NAVAL AIR STATION). Moreover, the base's location on the Chesapeake Bay is of importance, as the downstream consequences of AFFF discharge are visible due to the Bay being an ecologically and economically vital estuary. PAX also shares the same legal and institutional structures that govern all DOD installations. This makes the findings from the PAX base generalizable to the various other military sites facing PFAS contamination across the country.

The dynamic of the interplay between the scientific process, policy formation, and regulatory implementation has yet to be adequately addressed in existing academia and research, especially in the context of military-grade contamination. Despite well-known case studies of military contamination, researchers must examine why similar contamination patterns cause weak, fragmented policy responses and public outcomes. There is a critical knowledge gap between academic fields, enabling the continuation of unabated contamination.

This study addresses that gap by examining four interconnected issues. First, the chemical persistence of PFAS and the documented impacts on human and biota health, supported by local biomonitoring data, toxicological studies from military sites, and ecotoxicological research on bioaccumulation in the Chesapeake Bay Web. Second, the institutional and legal structures that protect polluters, and the strategic production of ignorance through poorly designed and functioning government agencies. Third, the role of corporate manufacturers like 3M and DuPont, whose internal documents were revealed through litigation, showed concealed knowledge and the deliberate shifting of liability. Fourth, the environmental justice implications, which include the intentional and disproportionate siting of military bases and hazardous waste landfills in communities of color, and the burden of accountability that is placed on victims. Together, the four sections reveal the depths of systematic failures and provide a better understanding of why such failures persist.

An interdisciplinary study will be conducted to explore the contributors to systemic governance to understand the structural reforms necessary. It will analyze how scientific data on PFAS in marine life and water informs (or fails to inform) policy and litigation, evaluate the economic and enforcement failures of current laws, and interview individuals involved across a spectrum. This spectrum includes scientists in Southern Maryland, individuals a part of various communities suffering from PFAS contamination, and an environmental policy expert. The goal is to produce an integrated assessment that explains the current stagnation and multidisciplinary solutions for better environmental governance.

METHODS

This project utilizes a comparative case study approach, supported by mixed methods of data collection from multiple data sets and studies that range from official state-conducted work to unofficial, local work conducted by passionate individuals. Semi-structured interviews were conducted to understand and gain diverse perspectives. A selective .

policy and judicial review provided the basis and sources for establishing the issues with current governing structures.

Engagement with Pre-Existing Data Sets

This research draws upon multiple scientific data sets and studies that employ different methodologies, spatial scales, and analytical targets. This approach reveals how methodological choices shape detection outcomes, consequently impacting regulatory responses. By consulting varied field studies, this research emphasizes methodological differences through data and results.

A local activist and scientist in Southern Maryland, Pat Elder, conducted targeted sampling near the PAX Webster Field Annex at several locations along the St. Mary's River, specifically areas directly downstream from the annex. Elder collected surface water, foam residue, oyster, crab, and rockfish tissues, and his own blood. These samples were collected using clean, non-fluorinated containers. Samples were analyzed using liquid chromatography-tandem mass spectrometry (LC-MS/MS) in accordance with EPA-approved PFAS testing methods (Elder 2024). This enabled detection at lower concentrations. Elder utilized a professional third-party laboratory for all PFAS analyses. The data set reflects its design to establish the presence of contamination.

The Maryland Department of the Environment (MDE) conducted a pilot study to examine PFAS in surface water and oysters near NAS Patuxent River. Systematic field sampling was conducted at multiple sites across the St. Mary's River and its tributaries during the summer months (Maryland Department of the Environment 2020).

The collected samples were tested for PFAS concentrations measured in parts per billion, and then compared to state screening levels and EPA drinking water advisories (Maryland Department of the Environment 2020). This government-led study used a broader spatial scale than Elder's, but set detection thresholds 1,000 times too high. This potentially understates contamination levels (Elder 2024).

The Maryland Watershed Association commissioned Eurofins Lancaster Laboratories for a PFAS study. The Eurofins study used EPA Method 537, Isotope Dilution, which is a targeted analytical method using LC-MS/MS to quantify trace levels of multiple PFAS compounds (Eurofins Lancaster Laboratories 2020). The lab used chain of custody documentation, preserved samples on ice, and high laboratory quality control (Eurofins Lancaster Laboratories 2020). This data set is important because it is independent from both state and military oversight, producing results that directly contradict MDE's conclusions.

Document Analysis

For policy and legal aspects, this research analyzes federal and state legislation, relevant judicial opinions, and issued regulatory documents. Sources include the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), the Freedom of Information Act (FOIA), the Toxic Substances Control Act (TSCA), and judicial opinions addressing sovereign immunity, the Feres doctrine, and the federal officer removal statute. State-level legislation from California, Colorado, Connecticut, Maine, Maryland, and Minnesota is examined to understand the divergence between state and federal regulatory action.

When employing legal analysis, the IRAC (Issue, Rule, Application, Conclusion) method was utilized. Each judicial opinion and statutory provision is narrowed to the core legal issue. The relevant rule of law, whether that is a statute, regulation, or precedent, is then identified. The rule is applied to the specific facts of PFAS contamination at military installations. A conclusion is then drawn about the legal outcome and its implications for accountability and remediation. This method ensures repeatable analysis of varying legal materials.

To document the perspective of communities impacted by military pollution, this research examines journalistic materials, including investigative reports by Sharon Lerner and Mariah Black. The local news coverage of contamination at NAS Patuxent River, Camp Lejeune, and the New Hampshire National Guard facility is included as well. These sources show the formation of relevant narratives, public engagement,

and the lived experiences of affected residents. Lived experiences are important in this research because such perspectives are often absent from typical academia.

This research also sought to obtain primary documentary evidence through the Freedom of Information Act (FOIA). FOIA is a federal law designed to provide public access to agency records, with the stated goal of government transparency. One FOIA request was filed with the Department of the Navy, and the second was filed with the Department of the Navy Archives. Both requests sought contracts between 3M and the Navy, as 3M was the original manufacturer of AFFF in the 1960s. This request was denied due to the absence of specific documents, and the denial indicated that the requested materials were stored at an archive department. The second FOIA request was filed with the appropriate archive department, but no response has been received. Under FOIA regulations and trends, a lack of response qualifies as a denial.

Semi-Structured Interviews

Four semi-structured interviews were conducted with stakeholders who have diverse perspectives on PFAS contamination. Pat Elder, a Southern Maryland scientist and activist living approximately 600 meters from the Webster Field Annex, provided scientific data and perspectives on government and military accountability. An individual working with the environment at PAX held an institutional perspective on the bases remediation efforts, sampling protocols, and operational constraints.

Doris Brock, whose husband died of cancer after 35 years of service in the New Hampshire National Guard, discussed her experience of ongoing advocacy work for veteran recognition and compensation. Another participant, M. Dibbles, holds environmental policy expertise and assisted in providing analyses of structural limits and environmental justice.

All interviews were recorded using OtterAI with prior informed consent from each participant. Institutional Review Board (IRB) approval was

received before any data collection. The PAX individual consented to the use of their name, but did not speak on behalf of any organization or institution, and especially not on behalf of the DOD or PAX specifically. Due to this, their name is removed along with their position. They are referenced as the PAX individual or PAX interviewee to distinguish their personal scientific experience from any institutional position, as they spoke on their perspective as an environmental scientist. Both Pat Elder and Doris Brock consented to the use of their names and requested no restrictions. M. Dibbles declined to have their name or affiliated work included and is therefore referred to as “M. Dibbles”.

The interviews follow a semi-structured format where questions varied by participant and did not strictly adhere to a pre-curated list. Questions were tailored to each stakeholder’s position and expertise. Content from certain interviews was deliberately included in subsequent interview questions to explore differing perspectives on the same issue. This includes the validity of independent testing and the role of scientific uncertainty in policy delay. No standardized list of questions was used, as the goal was to capture each participant’s framing of the problem rather than imposing a uniform structure.

The value of semi-structured interviews in environmental governance research lies in allowing researchers access to lived experiences, identifying various framings of complex problems, and understanding how stakeholders view causation and justice. Van Looy (2005) demonstrated how exploratory interviews with regulators, scientists, and project developers generated insights into regulatory gaps that desk-based analysis alone cannot identify. Similarly, Knill, Steinebach, and Zink (2025) conducted 157 semi-structured interviews with environmental agency members across six countries. This method allowed researchers to access information about implementation challenges and prioritization decisions. These are insights that quantitative data alone cannot provide.

LITERATURE REVIEW

The literature on PFAS contamination has grown exponentially over the past two decades. Environmental scientists and public health researchers have produced extensive documentation on the toxicology and bioaccumulation of PFAS compounds in both humans and wildlife populations (Panieri et al. 2022; Glaser et al. 2021; Baldwin et al. 2025). Concurrently, legal scholars and policy analysts have examined the regulatory frameworks dictating hazardous substances and the limits of immunity (Pietkoski 2020; Stokes 2023; Pollack et al. 2023).

The review is organized into sections that build toward the argument that PFAS contamination represents a systematic governance failure rather than solely being a scientific or technical problem. The beginning examines the chemistry and environmental behavior of PFAS, and how the compound's properties make it essential to aqueous film-forming foam (AFFF) (Panieri et al. 2022; Glaser et al. 2021; Hu et al. 2016). The following documents the human and ecological health consequences of PFAS exposure through ecotoxicological research, biomonitoring, and epidemiological studies (ITRC 2025; Elder 2024; Hill et al. 2024; Hoverman et al. 2025; Holden et al. 2025). The historical pattern of military contamination is then established through well-documented cases, along with the corporate conduct of 3M and DuPont (Bove et al. 2024; Brock 2026; Richter et al. 2025; Bilott 2019). The examination of regulatory and legal work then demonstrates how laws function as barriers to accountability (Pietkoski 2020; Stokes 2023; Mystal 2025). Together, these sections show that the persistence of PFAS in the environment is mirrored in the persistence of institutional delays, regulatory capture, and doctrines that prioritize profit over justice.

PFAS Chemistry and Environmental Behavior

The chemical and molecular properties of per- and polyfluoroalkyl (PFAS) compounds are the reason it has become indispensable to military firefighting foams. The bond that creates environmental persistence also produces the thermal stability and surface tension reduction required for aqueous film-forming foam (AFFF).

Understanding the chemistry of PFAS builds the foundation for why military installations adopted and continue using these compounds. AFFF is designed to extinguish high-hazard Class B 16 flammable liquid fires (like fuel, oil, and solvents) by creating a thin, aqueous film that spreads quickly across the surface of the fuel, which smothers the fire and flammable vapors (Hu et al. 2016). This aspect of the formula is due to the fluorosurfactants (PFAS-based compounds). Surfactants are molecules that have both a hydrophobic and a hydrophilic part. In a fluorosurfactant, the hydrophobic tail is a fluorinated chain, and the extreme strength and low polarity of the C-F bond create a repulsion of both water and oil. The dual repellency is crucial because the fluorosurfactants significantly lower the surface tension of the water, allowing the foam to easily spread across the surface of a low surface tension fuel like gasoline. It also provides great chemical and thermal stability, as it ensures the suppressant blanket remains effective under the heat of a fire instead of degrading upon contact with the fuel (Hu et al. 2016).

Known as forever chemicals, the persistence of PFAS is attributed to the carbon-fluorine (C-F) bond. Fluorine is the most electronegative element on the periodic table, meaning it has an extremely strong attraction for electrons (Panieri et al. 2022). When fluorine is bonded to carbon, the high electronegativity makes a very polarized and strong bond. Its strength as a function of its polarity is reinforced by the small size of the fluorine atom, allowing it to have optimal orbital overlap with the carbon atom (greater overlap increases the stability and strength of a bond), and specific electrostatic attractions that contribute to the bond's stability.

The powerful bond strength is a direct reason for its resistance against environmental degradation processes. The carbon fluorine bond resists hydrolysis (a chemical process in which water reacts with a compound, breaking chemical bonds), photolysis (another chemical process where ultraviolet radiation causes chemical bonds to break apart), and microbial degradation which is the breakdown of substances by

microorganisms like bacteria or fungi, using enzymatic processes to transform or mineralize compounds (Rogers et al. 2025; Li et al. 2025; "Photolysis" 2026; Trigo et al. 2009; "Biodegradation" 2026; Shane 2019). Hydrolysis is insufficient for proper and full degradation of the C-F bond because the electron density and steric hindrance (prevention of chemical reactions by large groups of atoms blocking the reaction site) by the fluorine atoms protect the carbon structure (Khan et al. 2019; Laçın et al. 2023). Similarly, the energy from ultraviolet radiation is usually not enough to break the C-F bond (Steinberg & Paul 2008; Sharma et al. 2022). Microorganisms, although adept at degrading most natural and synthetic organic compounds, lack the enzymatic ability necessary to degrade the perfluorinated bonds. The energy required to break the bond is so high that microbial metabolism offers no evolutionary advantage, leaving the compounds intact in environments for decades, and even centuries (Skeel & Boreen 2020; Sadowsky et al. 2014).

PFAS are put into two main categories, perfluoroalkyl substances and polyfluoroalkyl substances, based on their structure (Glaser et al. 2021). In perfluoroalkyls, all hydrogen atoms on a continuous carbon chain are replaced by fluorine atoms, which makes them fully fluorinated. This group includes the most well-known and studied compounds, such as PFOs (perfluorooctane sulfonic acid) and PFOAs (perfluorooctanoic acid). Polyfluoroalkyl substances are not fully fluorinated but contain a mix of fluorinated and non-fluorinated carbons along their chain. This difference in structure means that polyfluoroalkyl substances can act as precursor compounds (substances in a chemical reaction that produce another compound), eventually forming the more persistent perfluoroalkyls previously mentioned (Glaser et al. 2021)

The PFAS family is further categorized by chain length, as it dictates their transport. Long-chain PFAS are defined by the group attached to the fluorinated tail. PFCAs (perfluoroalkyl carboxylic acids) have seven or more fluorinated carbons, and PFSAAs (perfluoroalkyl sulfonic acids) have six or more fluorinated carbons.

PFOAs (a C8 PFCA) and PFOs (a C8 PFSA) are long-chain compounds and are relatively less water-soluble. Consequently, they tend to attach to soils, sediments, and sludge, accumulating on solids and migrating slowly underground, creating higher concentrations of contamination in the immediate surrounding area of the initial site of release (Malik et al. 2024). Short-chain PFAS are defined as PFCAs with fewer than seven carbons and PFSAs with fewer than six carbons. Due to their shorter and less hydrophobic fluorinated tail, they are significantly more water-soluble, less likely to adsorb to soil particles, and highly mobile in the environment, allowing them to travel quickly with water currents, seep into groundwater, and contaminate water sources far past the point of release (Malik et al. 2024). This creates an issue as they are harder to contain for remediation efforts.

Historically, AFFF formulas were primarily long-chain PFAS, mainly PFOs and their precursors. Due to the global discovery of their prevalence and toxicity, production shifted in the early 2000s to fluorotelomerization-based formulations, which produced short-chain PFAS (referred to as C6-AFFF) as replacements (Travis 2024). As a result of this, the chemical makeup of AFFF contamination at military sites is complex. It can consist of the original long-chain compounds from older formulas, the short-chain replacements from modern foams, and a plethora of polyfluoroalkyl precursor compounds belonging to fluorotelomer-based foams, which slowly turn into terminal compounds. So, older C8 (PFOA or PFOs) based AFFF created through electrochemical fluorination were phased out for C6 foams and designed to break down through time (Travis 2024). However, through their degradation path, they become precursors and result in the formation of terminal compounds. The early discovery of the presence and danger of PFAS led to some scientific advancement and change, but there is a clear failure in genuine remediation results when only utilizing technological inventions.

The use of AFFF during firefighting emergencies and routine training exercises brings a complex mix of PFAS directly into the environment.

Long-chain compounds are retained in the upper layers of soil, creating a long-term source of contamination, while short-chain PFAS rapidly leach through the soil column and act as an immediate contaminant to groundwater. Once in the groundwater, it can discharge into surface water bodies such as streams, rivers, and estuaries (Khan et al. 2023). The ratio of different PFAS compounds can serve as a "chemical fingerprint" that reveals transport pathways and the ongoing transformation of the original AFFF mixture as it remains in the environment. Furthermore, as PFAS cycle through food webs, bioaccumulation can vary per species and ecological niches, with higher concentrations found in predatory species and organisms with longer lifespans (Glaser et al. 2021).

Human and Ecological Health Consequences

The common practice of using aqueous film-forming foam has resulted in severe, lasting consequences for both human and ecological health. As a critical public health concern, Baldwin et al. (2025) emphasize the persistence of PFAS as the greatest concern, as the contaminated sites require fiscally extensive remediation strategies and pose chronic toxicological risk with transgenerational consequences. Moreover, the transition to fluorine-free foam (F3) alternatives addresses certain concerns, but it introduces additional uncertainties. New research has found that some replacement formulations may be more acutely toxic to aquatic species than the original formulas (Hold et al. 2025), and the long-term ecological impacts remain poorly understood. As concluded through several studies, there is a need for continued research to inform proper regulatory decisions and remediation strategies that will adequately protect human communities and the ecosystems they depend on.

Human Health Consequences

The human health consequences of AFFF-based PFAS exposure have been mainly documented through epidemiological studies of communities with contaminated drinking water.

The most extensive study so far takes place in Ronneby, Sweden, where AFFF from a nearby military base contaminated the municipal water supply from the mid-1980s until 2013 (ITRC 2025). Biometric monitoring of approximately 3,300 residents of Ronneby showed that the geometric mean serum levels (used to determine the central tendency of biomarker concentrations) of PFHxS (perfluorohexane sulfonate), PFOs, and PFOA were about 135x, 35x, and 5x higher than levels in a reference population (Xu et al. 2021). These levels of exposure in Ronneby's population gave crucial insight into the health issues of chronic PFAS exposure.

Cancer risks represent a primary concern when discussing PFAS exposure. Rosenfeld et al. (2023) state that international, national, and state agencies have concluded that PFOA is potentially carcinogenic, and studies on cancer risk among occupationally exposed firefighters show an increased risk for thyroid, bladder, kidney, testicular, prostate, and colon cancer (Li et al. 2022). The Ronneby cohort study found evidence for an elevated risk of kidney cancer with a non-statistically significant hazard ratio of 1.85 for kidney cancer in individuals with the highest exposure rates compared to those with lesser rates or no exposure (IRTC 2025). The authors also observed statistically insignificant increases in hazard ratios for bladder, brain, bone and cartilage, thyroid, and testicular cancers. It is important to note that although the hazard ratio of 1.85 is considered statistically insignificant, it aligns with findings from larger studies, such as the C8 Health Project and a meta-analysis by Steenland and Winquist (2021), which identified a relative risk of 1.27 for kidney cancer associated with PFOA exposure. When several studies show elevated, individually non-significant estimates in the same direction, the collective evidence creates a causal inference that PFAS do have a relationship with the development of certain cancers.

Beyond this, PFAS exposure has also been linked to other adverse health issues.

The research on Ronneby has documented significantly increased risks of major osteoporotic fractures (bone breaks caused by weakened bone density) with a hazard ratio of 1.11 and hip fractures (HR = 1.12) among those with the highest exposure levels (Xu et al. 2023). Hammarstrand et al. (2021) found that higher PFAS exposure was linked with significantly elevated hazard ratios for polycystic ovarian syndrome (PCOS, a hormonal disorder) (HR = 2.18). However, there were no associations found for endometriosis or uterine leiomyoma (ITRC 2025). There were mixed findings for thyroid issues, as Andersson et al. (2019) found no increased risk of hyperthyroidism (an overactive thyroid), and Li et al. (2021) noted no consistent evidence that links PFAS ingestion to alterations in thyroid hormone levels. The variety and extent of impacts shown through this data emphasize the importance and difficulty of PFAS toxicology.

The level and consistency of PFAS exposure dictate the severity and likelihood of health consequences. This leads to vulnerable populations, like minority communities or certain occupational fields, being at a higher risk. Firefighters represent a highly vulnerable group due to direct exposure to AFFF during training and emergency response tasks. Rosenfeld et al. (2023) documented that firefighters have been found to have elevated serum levels of long-chain PFAS, and because of bioaccumulation in the body, firefighters experience cumulative effects of exposure. Recent research involving more than 2,000 firefighters from seven states found that every individual had detectable PFAS in their blood. There were especially elevated levels found in those associated with original AFFF formulations, prior military service, and reliance on well water as a primary drinking source (Santos 2026). A Southern Maryland activist, scientist, and long-term citizen of St. Mary's, Pat Elder, showed that even individuals with non-PFAS exposure occupations have elevated levels of PFAS likely due to residing near a naval base. Elder resides in the Chesapeake Bay, about 600 meters across a creek from the Webster Field Annex of the Patuxent River Naval Air Station, where the Navy has conducted regular firefighting training exercises using AFFF for decades. Elder's personal biomonitoring

results, published in November 2024, documented extraordinarily high serum PFAS levels that place him in the 99th percentile of the U.S. population.

From washed-up shore foam that contains PFAS to the oysters he would harvest from the docks, a part of his home property, Elder regularly interacts with water sources that have been contaminated from naval exercises. His consumption of oysters, crabs, and rockfish as a typical Chesapeake Bay resident represents an example of PFAS bioaccumulating through the aquatic food web. Elder's blood serum testing was done by Eurofins Environmental Testing and found a total PFAS concentration of 42.2 ng/mL (nanograms per milliliter, equivalent to parts per billion) for the seven PFAS compounds identified by the National Academies of Sciences, Engineering, and Medicine as clinically significant. Astoundingly, his PFOS levels alone measured 32.3 ng/mL

In the context of the Agency for Toxic Substances and Disease Registry (ATSDR), the United States average PFOs level is 4.25 ng/mL, and 95% of the population has levels of 14.6 ng/mL or lower (only 5% have more than 14.6 ng/mL because PFAS concentrations in the general population are right-skewed). Elder's concentration of 42.16 ng/mL puts him far above the 20 ng/mL threshold that the National Academies use to encourage enhanced clinical screening, meaning Elder has a substantially increased risk for issues with thyroid function, kidney and testicular cancer, and ulcerative colitis (chronic inflammatory bowel disease). This consequence is placed upon an individual, a part of the regular general population, meaning those near military installations are likely unknowingly consuming PFAS contamination at harmful levels.

Impact on Wildlife Health

Along with the concern for human health, the ecological consequences of AFFF-derived PFAS are equally concerning due to documented bioaccumulation across 25 various biota.

The chemicals' stability and tendency to accumulate in biological tissues provide the foundation for assessing PFAS impacts on wildlife. As explained in Baldwin et al. (2025), the C-F bonds that enable PFAS' fire-suppressing properties also allow them to resist environmental degradation, creating long environmental half-lives that build chronic toxicity.

The ecotoxicological significance of AFFF and its replacements has impacted aquatic invertebrates that form the foundation of freshwater food webs. Research on a keystone zooplankton species, the *Daphnia magna*, found that fluorine-free foam alternatives can be more acutely toxic than the traditional PFA-based foams. Through a multi-generational exposure study, Holden et al. (2025) identified that exposure to three of six F3 alternatives led to changes in tolerance in the exposed populations in comparison to unexposed controls, with two populations displaying increased tolerance (potential adaptation) and one population showing decreased tolerance (indicating cumulative stress). The authors discussed these results as finding that the replacement formulations were more acutely toxic to aquatic species than the PFAS-based AFFF they were designed to replace. This is significant for regulatory decisions because it demonstrates that PFA-free alternatives pose their own ecotoxicological risks and that replacements are not to be utilized as single solutions.

Amphibians, which occupy an important ecological niche that bridges aquatic and terrestrial ecosystems, are particularly vulnerable to aqueous film-forming foam. Hoverman et al. (2025) studied the impacts of both PFAS-based AFFF, as well as fluorine-free foam alternatives (F3), on the larval development of the gray treefrog (*Hyla Versicolor*) species. Their study found that the PFAS-based AFFF formulation from Buckeye Platinum Plus reduced survival and delayed metamorphosis at the highest concentration that was tested. In comparison, the F3 alternatives showed comparatively limited effects on the completed transformation endpoint (Hoverman et al. 2025).

However, the study's behavioral observations found that tadpoles exposed to several foam formulations (including PFAS-based) sought refuge more frequently, resulting in 19 to 30 percent fewer tadpoles observed in open water compared to the controls (Hoverman et al. 2025). Although the behavioral shift did not directly alter or affect growth and development, such changes can negatively affect ecological interactions, like predation, parasitism, and competition, potentially impacting population-level outcomes.

Fish species are at a higher trophic level and are bioaccumulators. Fish exposed to AFFF contamination demonstrate substantial accumulation with specific tissue patterns. A controlled exposure study conducted by Hill et al. (2024) used fathead minnows exposed to PFAS-contaminated groundwater from Joint Base Cape Cod, Massachusetts (a naval site with decades of AFFF usage). It was found that the total PFAS concentrations in fish tissues increased over the course of 21 days, and the accumulation followed the distinct pattern of liver > kidney > gonad (Hill et al. 2024). The liver had the highest measured amount, the kidney had intermediate concentrations, and the gonad had the lowest concentration of the tested tissues. The tissue distribution matters because it indicates that PFAS accumulate in metabolically active (cells, tissues, or organisms engaged in biochemical processes to sustain life and function) and reproductive organs. This holds potential implications for metabolic function and reproductive success. On a concerning note, the study reported bioconcentration for perfluorooctane sulfonamide (FOSA), reaching 8,700 L/kg in the kidney tissue. This means that the PFAS compound becomes 8,700 times more concentrated in the fish's kidney than in the water it swims in. Another simple comparison to emphasize the power of accumulation is if 8,700 liters of water, enough to fill about 45 bathtubs, were to be condensed into a single organ the size of a marble (note that the size of the organ does not automatically produce a high bioconcentration factor; 8,700 L/kg is the normalized value).

An additional concern in the study's findings is that precursor compounds were present in both the groundwater and fish tissues. Hill et al. (2024) identified perfluoroalkyl sulfinates and perfluoroalkyl sulfonamide compounds in the liver tissue, which indicates the presence of alternative pathways for terminal PFAS exposure in aquatic wildlife. This is important because it shows that even when terminal PFAS concentrations are addressed and properly managed, precursor compounds can continue to exist as hidden reservoirs that ultimately contribute to the persistence of PFAS foams.

Another sphere of consequences is the avian species inhabiting areas adjacent to AFFF-contaminated sites. A study of tree swallows (*Tachycineta Bicolor*) near AFFF sources in Duluth, Minnesota, found that the total PFAS concentration in eggs at the National Guard Base site was 30 to 40 times higher than at a reference lake, and nestling PFAS concentrations were 10 to 15 times higher (Custer et al. 2024). The study identified PFHxS as a distinctive marker of AFFF contamination, and it comprised a higher proportion of total PFAS at AFFF-contaminated sites compared to reference locations or sites with primarily urban and industrial PFAS sources (Custer et al. 2024). Despite the substantial exposure differences, the researchers were unable to connect biomarker responses with PFAS exposure levels. This creates a complex challenge because there is a disconnect between documented bioaccumulation and observable physiological effects, further complicating risk assessment and management efforts.

In addition to the mere existence of PFAS in wildlife, there is a concern about the transgenerational consequences that may extend beyond the organisms directly exposed. A multi-generational study conducted by Holden et al. (2025) on *Daphnia* (small freshwater crustaceans) is the first look at evolutionary responses to fluorine-free foam alternatives, which resulted in demonstrating that chronic exposure over 84 days (about 10 to 12 generations for the species) can ultimately alter population tolerance.

Although the idea of evolutionary change in this issue may sound positive, when organisms divert their energy toward detoxification and survival, it may come at the expense of other functions.

Evolutionary trade-offs include reduced reproductive output (fewer or smaller offspring and reduced fertility), decreased competitive ability (the tolerant species may now only fit in to a contaminated environment), increased vulnerability to other stressors (a major issue in the face of climate change), genetic erosion and loss of genetic diversity, and increased population fragility, all of which can result to overall ecological collapse as *Daphnia* are a keystone species. Epigenetic inheritance is an additional consequence, as PFAS are endocrine disruptors that potentially alter the epigenetic marks that control which genes are turned on or off. Altered epigenetic states can cause disease and developmental issues, which will then impact unexposed generations, prolonging and further complicating mitigation and overall scientific understanding (Kebieche et al. 2025; Stefani et al. 2014; Holden et al. 2025; Cunha et al. 2025; Svoboda et al. 2022). The evidence across trophic levels undeniably shows that PFAS contamination from AFFF does not remain confined to the point of release. Since the scientific evidence makes clear that AFFF contamination exists, the focus must shift from occurrence and presence to how it occurs. More specifically, how military installations have used, disposed of, and continued to release AFFF for decades.

Military Use of Aqueous Film-Forming Foam and Evidence of Contamination

Military installations are primary PFAS contamination hotspots due to the decades of reliance on aqueous film-forming foams (AFFF) for firefighting training and emergency response. Unlike the industrial sources that are, to some degree, subject to state oversight, military facilities have operated under protocols that prioritize mission objectives without the proper understanding of or a cleanup plan for environmental contamination.

This has resulted in the common practice, and often unmonitored, release of PFAS foam into soils, groundwater, and surrounding waterways (Pietkoski 2020). Such practices have created an environmental legacy that is detrimentally affecting municipal water supplies and downstream ecosystems. Some literature has established that AFFF use is a “predominant point source of PFAS contamination...” which includes primary precursors, secondary precursors, and perfluoroalkyl acids (Yan et al. 2024).

An important focus on specific military AFFF PFAS is exemplified through a comprehensive review completed by Yan and colleagues. Yan et al. found significant discrepancies between the initial PFAS concentration profile in AFFF formulations and the concentration profile measured in AFFF-contaminated environments. Primary precursors make up 52.6 percent of PFAS mass in electrochemical fluorination (ECF)-based AFFF, and 99.5 percent of PFAS mass in fluorotelomer (FT)-based AFFFs, yet they represent only 0.7 percent of the total mass in AFFF-contaminated groundwater. In contrast, secondary precursors, which are less than 1 percent of PFAS in AFFFs, are 4.0 to 27.8 percent of PFAS in AFFF-contaminated environments (Yan et al. 2024). This difference is due to in situ biotransformation processes (indigenous microorganisms in soil/groundwater breaking down organic compounds), which means the contaminants found in today's environmental samples are not quite the same compounds that were originally released, further complicating source attribution and toxicity assessment. Understanding transformation dynamics is necessary but not sufficient for explaining the entire scope of contamination. To understand what quantities, through what disposal methods, and with what level of institutional awareness, the historical practices must be examined.

Historical Disposal and Contamination

The historical practices that created this contamination pattern are well-documented in regulatory records and firsthand accounts.

An environmental scientist who has experience with NAS PAX River base practices was interviewed, and they described the routine disposal methods that were standard across military bases before environmental regulations existed. Personnel would discharge AFFF onto the ground to test equipment or dispose of expired foam by pouring it out beside buildings, both of which were noted as “pretty consistent at all the installations” 1 . Such discharges were not considered problematic because the environmental risks associated with PFAS were not yet established. The result was the creation of predictable contamination zones near training spaces, hangars, and equipment storage sites - locations that now require investigation and remediation. When AFFFs are released during training exercises, firefighting operations, or accidental spills, PFAS bind to surface soil particles and rest in the upper layers of the subsurface. The foam can also infiltrate further into the subsurface, resulting in the formation of PFAS-contaminated groundwater plumes, which can discharge to surface water

Moreover, the foam releases can become runoff that brings PFAS to nearby surface water bodies, where the compounds have the chance to accumulate in sediment and infiltrate the food chain through bioaccumulation (Yan et al. 2024). Some AFFF-derived PFAS can become airborne and transported by air currents to be deposited onto surface soils and sediments via atmospheric deposition (transfer of particles from the atmosphere to the surface via precipitation or gravitational settling), creating contamination at far-removed distances from the original point of release (Yan et al. 2024).

PFAS Composition in Military Soils

Recent research quantified PFAS mixtures in surface soils at military installations to provide necessary data for ecological risk assessments. East et al. (2025) used a formalized prioritization method on a large dataset of PFAS concentrations in surface soil at AFFF-contaminated sites on active and former U.S. Air Force installations. They found that perfluorooctane sulfonate (PFOS) occurred most commonly and at the highest concentration across the PFAS measured.

It was also found that two to three PFAS made up 86 to 91 percent of the sum of PFAS in any given site-specific mixture. After PFOs, the most common and highest concentrated PFAS in order were PFHxS, PFOA, PFOSA, and PFHxA (East et al. 2025). Site-specific PFAS mixtures were found to be approximately 5 to 12% PFHxS, PFOA, PFOSA, and PFHxA, with PFOS comprising approximately 82% of the total PFAS mass. This profile holds important implications for risk assessments and remediation, as the dominance of PFOs in military installation soils, despite being phased out of ECF-based AFFF formulations after 2002, shows the sheer persistence of this compound. In addition, the consistent finding of two to three PFAS compounds constituting the vast majority of contamination at any given site implies that target remediation and risk assessment efforts could focus on a small number of priority compounds while still addressing the majority of environmental risk.

The uncertainty identified by East et al. (2025) has direct implications for PAX. The inconsistent prevalence of fluorotelomer sulfonates across military sites means that contamination profiles at PAX may differ from those at other bases. This would depend on which AFFF formulations were used and when they were released. Without knowing the specific mixture present in nearby soils and groundwater, risk assessments may target the wrong compounds. Unknown or poorly understood toxicological potency complicates efforts to determine whether PAX's contamination levels pose a threat to human and ecological health. These findings show that relying on a narrow set of priority compounds, similar to both the MDE and the base's own monitoring, systematically understates risk.

Contamination at Naval Air Station Patuxent River

NAS Patuxent River, labeled a Superfund site since May 31, 1994, is an example in Southern Maryland that illustrates the complexity of PFAS contamination assessment and the institutional tensions that occur when government and independent findings diverge.

The base, established in 1943, has a long history of AFFF use dating back to its invention in the 1960s, where firefighting training areas, hangars, and aircraft maintenance facilities serve as primary release points. The PAX individual interviewed for this research noted that the base has taken proactive steps under the Installation Restoration Program, including the early adoption of PFAS sampling and engaging with state and federal regulators through a Restoration Advisory Board that includes public representatives.

Despite the scientific discrepancies, NAS Patuxent River has taken efforts to address PFAS contamination under the Installation Restoration Program. The PAX interviewee discussed the base's proactive approach that puts PAX ahead of many other bases in responding to PFA's concerns. The base started requesting funding for PFAS investigation before the Department of Defense (DOD) formally established PFAS-focused programs. PAX became "one of the first Department of Defense installations to get out there and start doing investigative sampling" ². This proactive initiative allowed the base to build an understanding of contamination through archives that included interviews with retired firefighters and long-term personnel who could identify the historical disposal and practice locations.

PAX's Restoration Advisory Board (RAB) includes representatives from the Maryland Department of the Environment (MDE), the Environmental Protection Agency (EPA), St. Mary's County government, civilian community members, and local tribal nations. The RAB functions under a formal Federal Facilities Agreement that mandates transparency and public participation, and all sampling plans are reviewed by state and federal regulators before implementation. According to the interviewee, such a collaborative structure has created the balance necessary for thorough investigation with fiscal responsibility. All of their sampling results are made public and are available through the base's website, at the local library, and during public RAB meetings.

Those meetings have included efforts to increase community participation through catered meals and outreach to local organizations, showing the intention to include and educate the public on the reality of the contamination and current efforts, something contrary to notions made by critics.

However, the base has faced challenges with remediation due to institutional and technological limitations. The individual discussed the difficulty disposing of PFAS-contaminated materials due to the lack of treatment technologies. When the DOD took the stance of treating PFAS as hazardous waste (even before being formally listed as such under federal regulations), the base found that no landfills would or could accept PFAS-contaminated soils or water. This created operational challenges, as construction projects were unable to proceed due to indisposible excavated contaminated soils. If the base stockpiled the material, there would be new contamination sites as a result. PAX eventually secured a contract with a hazardous waste landfill in Emelle, Alabama, the only site willing to accept the waste.

Due to disposal issues, PAX has pursued alternative treatment technologies and disposal methods. The base has piloted thermal treatment projects to destroy PFAS compounds and leaching processes designed to separate PFAS from contaminated soils. Recently, the DOD has permitted disposal at Subtitle D landfills, which are standard landfills equipped with liners and leachate collection systems, provided that the landfills are willing to accept PFAS waste. However, as noted by the interviewee, most landfills remain reluctant to accept PFAS materials, showing the opposition of NIMBY (Not In My Backyard) from communities near proposed disposal sites. The base's efforts are supported through congressionally appropriated restoration funds, which are separate from base operating budgets, meaning that the cleanup does not compete with other priorities or problems for funding. Such dedicated funding, the individual stated, has allowed the restoration program to continue even during "austere years" when other infrastructure projects received no discretionary funding.

Potentially most significant, the interviewee described a shift in the base's approach to AFFF management. After 45 to 50 years of storing AFFF in hangars without a single documented fire requiring its use, the base decided to remove the AFFF systems and replaced it with water mist systems, and limit foam to cash trucks for emergency response. According to the individual interviewed, this decision was a risk-benefit choice - the infrastructure loss from a fire was easier to replace than long-term remediation costs and liability of AFFF storage and potential release. The base has also made the decision not to transition to F3, as the individual cited the new evidence that the alternatives are toxic to the food chains in the Bay, showing the level of responsiveness the base has to new scientific findings, even when it complicates transition timelines.

Aside from any measures taken, the empirical record of PFA detection at Patuxent River has significant contradictions between official and independent assessments. In 2020, MDE conducted a pilot study examining PFAS in surface waters and oysters in the St. Mary's River system. MDE reported no PFAS detections of concern and concluded that contamination levels did not warrant additional investigation or public health advisories (Maryland Department of the Environment 2020). These findings were used to reassure local communities and justify a limited regulatory response. The MDE based its study on detection thresholds that critics argue were too high to capture ecologically meaningful contamination, and the sampling strategy underestimated spatial and temporal variability in PFAS distribution (Elder "MDE Report Deceives Public").

Independent analyses commissioned by the Maryland Watershed Association produced different conclusions. Eurofins Lancaster Laboratories detected PFAS in aquatic organisms that are consistent with bioaccumulation and chronic exposure, which directly contradicts the MDE's reported findings (Eurofins Lancaster Laboratories 2020).

The Eurofins report documented PFAS concentrations in fish and shellfish samples taken from the St. Mary's River, that infers current contamination in the aquatic food web. These findings were especially concerning due to the ecological and economic importance of the St. Mary's River System, as it supports recreational and commercial fishing, and the broader Chesapeake Bay ecosystem it flows into.

The individual interviewed acknowledged the challenges with methodology that were behind the discrepancies and stated that early PFAS testing faced significant limitations. When the EPA first began requiring PFAS testing, "there were no labs that could meet what EPA in theory wanted," and many facilities that claimed to test for PFAS did not specify which of the thousands of chemical compounds they were detecting⁴. The lack of a standardized PFAS sampling protocol produced inconsistent results, but recent guidance from agencies such as the Michigan Department of Environment, Great Lakes, and Energy (2022) has improved comparability and reliability. Regardless, discrepancies persist, which raise questions about how scientific uncertainty is interpreted and mobilized in government decision-making. However, strong criticism of the MDE study remains.

Pat Elder, the local scientist and activist with high serum PFAS levels, continued the discussion of the MDE results. Elder stated that the sampling strategy relied on a limited number of collection points during a single season, failing to account for tidal dynamics, which may move contaminants into areas not sampled by the state. Moreover, in December 2024, Elder reported his discovery of PFAS foam residue on the beach at St. Mary's College of Maryland, which is located ten miles downstream from the base (Elder 2024). The presence of foam residues suggests current or recent releases of AFFF into the river, contradicting official assurances that contamination was historical, limited, and contained.

The difference between MDE's and Eurofin's findings shows a broader pattern in the literature.

Proffitt (2022) documents the evolution of sampling protocols, finding that the absence of standardized methods historically contributed to the inconsistent results across studies. When government studies report no cause for concern while independent studies have a polar opposite report, the subsequent confusion further erodes public trust, delays genuine remediation, and allows for the persistence of contamination while responsibility is constantly disputed among agencies.

Establishing a Pattern of Contamination: Historical Context

The contamination at Patuxent River cannot be analyzed alone, as it is part of a larger historical pattern of military pollution for which Camp Lejeune can represent due to its extensive documentation and presence in the media. The Camp Lejeune Marine Corps Base in North Carolina had a water contamination issue that persisted for decades before federal agencies acknowledged health risks or implemented remedial action. The contamination was a result of the disposal of industrial solvents, which include trichloroethylene (TCE), perchloroethylene (PCE), and benzene, into the base's water supply that supplied housing, barracks, and administrative buildings from the 1950s through the 1980s (National Academies Press 2009). The evidence from Camp Lejeune linked military installation water contamination to dire health outcomes. A cohort study by Bove et al. (2024) researched cancer among Marines, Navy personnel, and civilian workers exposed to the contaminated drinking water. The study found significantly elevated incidence rates for several cancers among exposed populations. Marines and Navy personnel had higher rates of acute myeloid leukemia, all myeloid cancers combined, cancers of the esophagus, larynx, soft tissue, and thyroid. Civilian workers had increased hazard ratios for lung cancer, female breast cancer, and all myeloid cancers (Bove et al. 2024).

A related study conducted by the same research team examined mortality in exposed populations and found increased mortality rates for several causes of death.

It documented increased mortality from cancers of the kidney, esophagus, and liver, along with Parkinson's disease and amyotrophic lateral sclerosis (ALS) among the groups exposed to contaminated water at Camp Lejeune (Bove et al. 2024). The consistency of those findings strengthened the causal inference linking military water contamination to detrimental health outcomes. In sum, Bove's research is the most epidemiological investigative study of military water contamination to date. The research team accessed the DOD's personnel records, reconstructed historical water contamination levels, and tracked health outcomes over decades. The finding on civilian workers, who were not subject to the same chain of command and typically lived off base, experienced elevated cancer rates, reinforcing the conclusion that the contamination was responsible for the 41 observed health effects, reducing the ability to deflect blame on genetics, misfortune, or other parts of military service.

The Camp Lejeune contamination established a form of precedent, in a non-legal context, that has several implications for other military bases, including PAX. Camp Lejeune demonstrated that institutional delays in the acknowledgement of contamination and implementation of remediation can span decades, during which exposed individuals will continue to suffer health consequences. It showed the limitations of the United States' governing and regulatory systems that enable military institutions to operate and pollute under exemptions. Lastly, it showed that once contamination has received acknowledgement, establishing causation and compensating affected populations can be contentious and non-existent, especially as many exposed individuals die before receiving recognition or compensation.

Although Camp Lejeune is a notorious case of military pollution, there are many other bases with similar or worse stories and data. The story of Doris Brock, whose husband served full-time in the New Hampshire National Guard, solidifies the historical and nationwide pattern of harm caused by military installation contamination and the institutional failures that allow and worsen it.

Brock's husband, Kendall Brock, served 35 years with the 157th Air Refueling Wing of the National Guard and was diagnosed with stage three bladder cancer in 2015, which rapidly progressed to stage four. Kendall Brock died in 2017. In the years following her husband's diagnosis, Doris Brock began documenting the health outcomes of her husband's colleagues, all of whom had served alongside him at the same base, performing the same jobs, for decades. Out of 66 friends, 34 died either before or shortly after retirement. "That number was like, very, very large to me," Brock stated, "for a small group of people, because the New Hampshire National Guard is not as big as the California National Guard" 5 .

Brock's experience follows the same pattern documented at Camp Lejeune. As Doris Brock's story gained traction in the media, a Marine Corps veteran who became one of the most prominent voices for Camp Lejeune families after his nine-year-old daughter died of leukemia linked to contaminated water, Jerry Ensminger, reached out. Ensminger warned Brock that she was to face "a horrible fight" when demanding accountability and justice, which is exactly what happened. The Camp Lejeune litigation spanned about four decades before Congress finally passed the Camp Lejeune Justice Act in 2022, but even then, the compensation process has been halted by denials and bureaucratic obstacles (National Academies Press 2009). As Brock stated, "almost 40 years worth of fighting, and they really aren't much further ahead than the day they started, other than the fact that everybody knows who Camp Lejeune is."

Both Camp Lejeune and the New Hampshire National Guard facility experienced delays that have lasted for decades. The New Hampshire site was designated as a Superfund site in 2014 due to industrial solvents, degreasers, and other aircraft repair chemicals, all of which Brock noted to predate the current focus on PFAS. When she began her advocacy, PFAS "wasn't even a buzzword" 6 .

Her fight focuses on the solvents and chemicals used in military operations, many of which are well-known carcinogens with established health consequence profiles (Agency for Toxic Substances and Disease Registry). This raises serious concerns, as no type of pollutant or contaminant should become a buzzword that overshadows other forms of deadly pollutants. Prioritization is one thing, but this can be deemed as ignorance that directly disrespects and further harms other victims. When Brock meets with the media, their first question is directed toward PFAS studies. Her stance is consistent, as PFAS is not her main concern, and with an exclusive focus on PFAS, policymakers are “missing the boat on the solvents and chemicals that military members work with” 7 . The current focus on PFAS, although important, obscures the broader chemical issue that continues to affect service members, their families, and surrounding communities. Brock’s observation that PFAS has been pushed as the only form of relevant pollution speaks directly to the need for a comprehensive approach that addresses a full range of hazardous substances released over the decades.

Additionally, the institutional response Brock has encountered shows the same patterns of delay, denial, and information control that are the main characteristics across military contamination cases. She described reading Veterans Administration disability claims that became public knowledge and being struck by “how many people are denied service connection for their illnesses, when there are studies out there that actually say those illnesses are linked to occupational chemical exposures.” Brock stated the Agency for Toxic Substances and Disease Registry provides fact sheets on each toxic chemical that individuals work with, describing the illnesses these men and women are suffering. Yet veterans continue to be denied disability benefits for conditions clearly due to their occupational exposure. From Brock's perspective, this pattern of denial reflects a cost issue. Acknowledging the link between service and these illnesses would require the government to provide disability compensation to tens of thousands of veterans and their survivors, which is a fiscal liability the DOD and Veterans Administration seem to be unwilling to accept.

Brock's response to claims that cancer among service members is attributable to genetics or familial habits is that her husband had no family history of cancer and never smoked. His best friend, who died from pancreatic cancer three months later, also had no family history and never smoked. The insistence on attributing service members' cancers to individual behaviors and genetic predispositions shifts the responsibility away from the institutions whose practices created the exposure. As Brock stated, genetics is a relatively new field, and the military has never conducted systematic studies about genetic histories or controlling for these variables in exposure assessments. The lack of such studies is, in itself, an institutional choice. It allows the government to continue claiming insufficient evidence, even as the human toll rises.

When asked whether she believes the government withheld or minimized information about contamination risks, Brock stated she believes "they do withhold a lot of information." She discussed a pattern across studies she has read: "They all have the same conclusion, not enough information to support, or, you know, inconclusive." This consistent finding despite mounting evidence, she characterized as institutional unwillingness rather than genuine scientific uncertainty, as is her distrust of government-conducted studies: "I don't trust studies, especially performed by the government." Overall, Camp Lejeune establishes a pattern for military contamination and for the decades-long struggle for acknowledgment, accountability, and justice.

Partners in Pollution: Corporate Profit through Contracts with the Military

The PFAS contamination crisis is about more than the Department of Defense's contaminants leeching into and staying in the environment, and the concept of forever chemicals is not foreign, nor was the use of them a mere accident. The overarching justification for many things related to PFAS is scientific uncertainty, but the reality is that there is only certainty in calculated corporate decision-making, deliberate concealment, and the prioritization of profit over human life and the

ecosystems upon which it relies. The corporate giants that manufactured and profited from these forever chemicals, mainly 3M and DuPont, held knowledge of PFAS toxicity decades before the public or regulators were informed (Richter et al. 2025). Their actions follow a pattern of corporate behavior that is built on the strategies of the tobacco and fossil fuel industries. This includes manufacturing doubt, suppressing evidence, and externalizing deadly costs onto society while enjoying the profits (Cordner et al. 2025).

Concealed Knowledge

Corporate profit is a large motivator in the mass production of AFFF. 3M and DuPont acted on that profit motive through the deliberate suppression of evidence and knowledge. Internal documents that were uncovered through litigation initiated by attorney Robert Bilott, whose story is dramatized in the film *Dark Waters*, showed that DuPont and 3M were aware of the dangers of PFAS as early as the 1950s (Bilott 2019). Investigative journalist Mariah Blake's book from 2025, *They Poisoned the World: Life and Death in the Age of Forever Chemicals*, utilizes internal industry documents and extensive interviews to understand the full picture of corporate greed. Blake wrote that the companies created, used, and marketed harmful substances while hiding implications (Blake 2025, p.4). A 2025 study analyzed industry documents that spanned from 1961 to 2006, and found "clear evidence that the chemical industry knew about the dangers of PFAS and failed to let the public, regulators, and even their own employees know the risks" (Richter et al. 2025, p.3). The documents detailed DuPont's pre-existing knowledge of enlarged livers in animals exposed to C-8 (PFOA), factory workers bearing children with birth defects, and animals dying within days of ingesting these chemicals. None of those findings were reported to the Environmental Protection Agency, which was required by law (Richter et al. 2025).

The timeline of knowledge is undeniable in proving corporate intent. Interestingly, the U.S. government approached DuPont to mass-produce

PFAS during World War II, but initially declined due to liability concerns. The government then agreed to indemnify DuPont, meaning the government would hold the responsibility for harm caused (Blake 2025, p. 78). By the 1960s, tests conducted by both DuPont and 3M revealed toxic effects of PFAS exposure on animal subjects, which included damage to the kidney, liver, and spleen (Bilott 2019). In 1978, 3M informed DuPont that it had confirmed the presence of PFAS in the blood of its own workers. A 1979 DuPont study found that PFAS bind to blood protein instead of breaking down and exiting the body, making the chemical biopersistent. In that same year, an internal survey of DuPont employees at a Teflon plant found increased cases of liver damage, but the exposed workers were not notified of those findings (Richter et al. 2025). By 1980, both companies reassigned female workers from PFAS-related jobs due to animal studies that indicate risks of fetal defects (Lerner 2022). In 1989, 3M's internal studies found increased levels of cancer in workers exposed to PFAS. And in 1992, DuPont confirmed those findings and documented increased cancer rates among its own employee population (Richter et al. 2025).

In Robert Bilott's memoir of his 20-year-long battle against DuPont, he wrote that when he first requested documents related to DuPont's PFOA emissions, the company provided a single box of papers. After months of legal pressure, the company produced "hundreds of thousands of pages of documents dating back decades", proving the company was aware of PFOA's potential danger since the early 1960s (Bilott 2019, p.87). The documents provided included internal studies showing that PFOA caused birth defects in rat offspring, accumulated in human blood, and remained in the body indefinitely. One internal DuPont memo from 1970 blatantly stated that PFOA "is toxic to humans and the environment and should be handled with extreme care" (Blake 2025, p. 156). However, these findings were never shared with nearby residents, workers, or regulators.

Disregarding the accumulating evidence, DuPont publicly insisted that its products were safe. In 1998 and 2002, DuPont emailed the EPA asking it “to quickly (like first thing tomorrow) say the following: That consumer products sold under the Teflon brand are safe and to date, there are no human health effects known to be caused by PFOA” (Lerner 2022, p. 145). This request was made at the same time its internal company documents demonstrated knowledge that PFOA had severe health risks, which is why Lerner terms “the playbook of deception,” in reference to the same tactics used by the tobacco industry to create doubt about the link between smoking and cancer (Lerner 2022, p. 89).

The most blatant example of corporate deception in the PFAS crisis is DuPont’s divestment of its environmental liabilities. In 2015, DuPont put its PFAS operations under a separate company, Chemours, transferring all PFAS-related environmental liabilities to the new entity (Stokes 2023). The separation agreement included provisions that explicitly moved to prevent Chemours from having any recourse against DuPont (New Jersey Attorney General 2022). DuPont’s strategic move protected its assets from the growing liabilities strongly linked to decades of PFAS contamination. However, DuPont’s estimations of liability were far too low, proven by its estimated liability for the Fayetteville Works facility in North Carolina at \$2 million, but in 2019, the settlement was over \$200 million for remediation of water contaminated by that facility. New Jersey, Ohio, New Hampshire, and Vermont joined in commencing legal actions, alleging that DuPont had committed fraud in divesting its PFAS liability by being intentionally designed for bankruptcy (Stokes 2023; New Jersey Attorney General 2022).

The actions of 3M, DuPont, and other PFAS manufacturers represent a systemic pattern where corporate actors take positions, ranging from resistance and stonewalling to strategic adaptation (Bartley & Fairbrother 2024, p. 12).

In the case of PFAS, the most popular corporate strategy is regrettable substitution, where “narrow bans and opaque/guarded safety information lead to the proliferation of alternatives that may not be safer” (Bartley & Fairbrother 2024, p. 15). The capitalist growth that drives this is not incidental; it is fundamental. Such an incentive structure operates through the pressure to maintain market share, the financial benefits of delaying regulation, the fiscal savings in avoiding toxicity testing, and the ability to place cleanup and health costs onto taxpayers (Bartley & Fairbrother 2024). So, even when companies adopt voluntary restrictions, they are avoiding a radical phase-out of PFAS (Karlsson et al. 2025, p. 8). Voluntary corporate accountability mechanisms appear progressive, but result in minimal compliance lacking substantive change (Karlsson et al. 2025, p. 12).

Although corporate strategies have historically alleviated polluters of accountability, these methods have begun to fail in the modern court of law. In 2023, 3M was pushed to an agreement to pay between \$10.5 billion and \$12.5 billion to public water systems across the U.S. to remediate PFAS contamination. Another agreement allocated up to \$4 billion in liability, with Chemours assuming 50 percent and DuPont and Corteva (another DuPont spin-off) dividing the remaining 50 percent (Stokes 2023). Most excitingly, in 2025, the Miteni case in Italy is the first time a national court has imposed criminal liability with substantial criminal sentences on corporate executives for PFAS-related environmental harm. The Court of Assizes of Vicenza convicted eleven former managers of the chemical company, imposing a total of 141 years in prison, while also awarding €56.8 million in damages to the Italian Ministry of the Environment (De Toni & Piva 2025).

What Does Corporate Behavior Reveal About Our System?

The documentation of concealed evidence, strategic liability shifting, and the substitution of one harmful chemical for another reveals foundational truths about the American regulatory system and the position of corporations within it.

Firstly, it speaks about the chemical regulatory framework. Specifically, the Toxic Substances Control Act (TSCA) has historically allowed chemicals to enter commerce without sufficient toxicity and exposure data, which creates what scholars term “undone science” or “structurally produced knowledge gaps” (Cordner et al. 2025; Richter et al. 2025). Secondly, for regulatory agencies, through statutory limitations or self-imposed reticence, they have failed to properly exercise their authority to mandate testing or restrict hazardous substances (Bartley & Fairbrother 2024).

Most fundamentally, the case of PFAS demonstrates that the current system does not just permit, but structurally incentivizes the kind of behavior DuPont and 3M exhibited. The profit motive to maximize shareholder returns creates a powerful pressure to minimize costs, including the costs of environmental protection, toxicity testing, and liability (Cordner et al. 2025). When companies can externalize such costs onto the families who lose loved ones to cancer, onto the communities of color and poverty whose water supplies are poisoned, and onto the taxpayers who ultimately fund the cleanup, they face no genuine penalty for doing so. Not only do they face no punishment, but the market incentive is so strong that corporations have purchased all of the seats at the table.

How Does Corporate Money Buy Political Power and Protection?

The primary manufacturers of PFAS, 3M and DuPont, have spent decades and hundreds of thousands to obtain political influence through campaign contributions, lobbying, and the placement of former industry executives in regulatory positions. The result of this is a system that rewards pollution. A system, as Pat Elder stated, that is “not an accident”⁸. The scale of corporate political spending is extravagant, as according to data compiled by the Environmental Working Group, the chemical industry spent more than \$110 million on federal lobbying in 2023. 3M and DuPont were top spenders, as spending translates into political access and legislative outcomes (EWG 2024).

When Congress considered the PFAS Action Act of 2021, the most comprehensive PFAS action bill to date, which would have designated PFOA and PFOs as hazardous substances under CERCLA, required the EPA to set drinking water standards, and allocated cleanup funding, the chemical industry ran an aggressive opposition campaign. The bill passed the House, but in the Senate, many politicians received substantial campaign contributions from PFAS manufacturers, thus preventing its advancement (Stokes 2023).

Moreover, former 3M and DuPont executives have held positions at the Environmental Protection Agency, the Department of Defense, and other agencies responsible for PFAS regulation and remediation. The current EPA administration has actively weakened PFAS protections while claiming progress, along with political appointees silencing career scientists and burying scientific findings (Southerland 2026). Regulatory capture, where the regulated entities control the regulators, explains why the EPA has delayed action for decades, suppressed toxicity assessments, and sought to rescind drinking water standards even after finalization. As Elder stated, “when you say EPA in the context of PFAS found at the Navy base, the EPA can hardly be considered as an independent agency” 9 .

The consequences of such political capture are human lives. When Congress passed the PACT Act in 2022, it was seen as a landmark expansion of benefits for veterans exposed to burn pits and other toxins. However, through Doris Brock’s observation, the final legislation fell short, as the government and VA “decided to only pick and choose which diseases or illnesses that they would cover,” which stripped out conditions that had been included, like male breast cancer 10 . The narrowing of a legislative scope after passing indicates the ongoing influence of lobbyists who work to limit the imposition of accountability. “Small pieces are going to cost the lives of tens of thousands of military members,” Brock noted, “I just think that that is a way the government can stall” 11 .

Delay saves money, and for as long as the system rewards delay through contributions and future industry employment, there is no fiscal reason to change. Despite the promise of representation, the seats where policy is made are occupied and purchased by the industry.

Preexisting Regulation and Precedent

Understanding the PFAS contamination crisis requires examining not only what the corporations did, but also how the system allowed them to do it. As environmental advocate and journalist Pat Elder put it, “The entire process is, I would go as far as to say criminal in nature,” referencing how the Navy serves “as the polluter but they’re also in charge of the cleanup and then they’re also in charge of deciding what to clean up and how to present that data to us” 12 . The arrangement of an entity being responsible for pollution simultaneously controlling the investigation, remediation, and communication of that pollution is a foundational governance failure.

Protector to Barrier

The Environmental Protection Agency (EPA) was established in 1970 during a time of bipartisan environmental consensus, created to serve as an independent entity capable of holding polluters accountable. As Elder reflected on his early career working with staffers connected to Senator Edmund Muskie of Maine (who was instrumental in the Clean Water Act and CERCLA), there was much excitement for the possibilities of progress available 13 . Although the formative years of the EPA were genuine, Elder also now states, “everything has kind of gone to hell” 14 . Dr. Betsy Southerland, who worked at the agency for over three decades, writes along similar lines as Elder. She discussed how the EPA under the Trump administration is gaslighting the public about PFAS contamination, claiming major progress while “actively weakening the very protections meant to reduce exposure to these toxic ‘forever chemicals’” (Southerland 2026).

The EPA’s most recent actions on PFAS show regulatory capture and intentional delay.

The Biden administration was giving water systems five years to comply with new PFAS drinking water standards for six chemicals, but the current EPA administrator, Lee Zeldin, is asking for four of those standards to be rescinded, along with plans to delay compliance for an additional two years. This request was rejected by federal courts, citing the lack of legal and scientific reasoning (Southerland 2026). Moreover, the EPA has delayed TSCA reporting requirements that required PFA manufacturers and importers to report uses, production volumes, exposures, hazards, and disposal from 2011 to 2022. Administrator Zeldin has proposed delaying that reporting by a year and exempting all importers of PFAS-containing articles (Southerland 2026). A main concern lies in the EPA suppressing scientific findings. The agency hid an updated PFAS toxicity assessment and eliminated the Office of Research and Development; “suppressing scientific findings does not make drinking water safer; it leaves people exposed” (Southerland 2026).

Perfect on Paper: CERCLA and Superfund Sites

CERCLA’s application to military installations has been largely compromised due to exceptions and exemptions. The Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA, also called Superfund), was enacted in 1980 to address the nation’s most hazardous waste sites. The statute establishes a framework for identifying contaminated sites, creating fiscal responsibility, and protecting communities from industrial contamination (Pollack et al. 2023). Substances designated as hazardous under CERCLA are subject to reporting requirements, specific spill rules, release tracking requirements, and additional reporting under different environmental statutes.

In theory, CERCLA represented a powerful tool for environmental justice and corporate accountability. As Pietkoski (2020) discusses, military facilities operate under exemptions, delayed compliance processes, or discretionary enforcement provisions, which allow contamination to continue without effective or appropriate remediation.

Instead, military sites operate through negotiated Federal Facility Agreements (FFAs) that prioritize smooth operations over quick remediation. These agreements are negotiated between the EPA, state environmental agencies, and the DOD, granting the military significant control over the pace and scope of investigation and/or remediation (Pietkoski 2020). As previously stated by Activist Pat Elder, a single entity having complete control over the consequences of its own actions is not effective.

State-Level Legislative Action

As federal action has stalled due to increasing political polarization, states have become a primary driver of PFAS regulation. As Elder noted, when evaluating PFAS health risks, the state of California created Proposition 65, where they designate PFOS and PFOA as human carcinogens ¹⁵. California's Proposition 65, enacted in 1986, mandates businesses to provide warnings about exposure to chemicals that cause cancer, birth defects, or any reproductive harm, with PFOA and PFOS on the long list of carcinogens.

Several other states have enacted comprehensive PFAS bans on consumer products. Colorado's SB 24-081 is the Perfluoroalkyl and Polyfluoroalkyl Chemicals Protection Act, banning the sale or distribution of cleaning products, cookware, dental floss, menstruation products, and ski wax containing intentionally added PFAS, as well as prohibiting the installation of artificial turf containing PFAS (Shook, Hardy & Bacon 2025). Connecticut's SB 292 prohibits the manufacture or sale of products with intentionally added PFAS unless manufacturers provide prior written notification to the state, and additional labeling requirements for outdoor apparel, turnout gear, and consumer products that have taken effect in 2026 (Shook, Hardy & Bacon 2025).

Maine's SP610 prohibits the sale or distribution of cleaning products, cookware, cosmetics, dental floss, juvenile products, menstruation products, textiles, ski wax, upholstered furniture, and HVAC equipment containing intentionally added PFAS, with one exception for what the

Department determines are “currently unavoidable.” This is a high bar that requires demonstrating the use is “essential for health, safety, or the functioning of society and for which alternatives are not reasonably available” (Shook, Hardy & Bacon 2025).

Minnesota’s Amara’s Law, named after a young girl affected by PFAS, requires manufacturers of all products containing intentionally added PFAS to submit detailed information to the state, with the inclusion of product descriptions, the purpose of PFAS use, and the amount of each PFAS present (Shook, Hardy & Bacon 2025). Those who fail to comply may be banned from selling the product in Minnesota entirely. The state of Maryland has also taken action with the Protecting State Waters from PFAS Pollution Act, which restricts PFAS discharges and increases reporting requirements. Federal facilities, including the NAS PAX base, remain subject to federal law rather than state law, creating a two-tiered regulatory system with less stringent standards for military installations (Handson 2017). As federal action has stalled, states across the country have enacted their own regulations. Although these various states are not geographically connected to PAX or Camp Lejeune, they represent a strong trend that advocates for the restriction and removal of PFAS. This signals that reform and change are possible, and that the two-tiered system offers hope.

SCOTUS Ruling: A Loss for State Jurisdiction

States nationwide in the U.S. have been seeking to remove PFAS from their markets. The two-tiered system of federalism grants states the power to do so, but federal preemption can overrule said power through Supreme Court rulings and federal decisions. On March 2, 2026, the U.S. Supreme Court denied certiorari to an appeal from Maryland and South Carolina, allowing the Fourth Circuit ruling to stand, which requires the states’ PFAS contamination cases against 3M to be heard in federal court (Taft PFAS Insights 2026; Maryland Daily Record 2026). There are four separate lawsuits filed by the states to recover damages for PFAS contamination.

The states strategically split their lawsuits by filing one suit for PFAS contamination generally, and a separate suit for AFFF contamination specifically. This was an effort to keep the general PFAS suits in state court (Maryland Daily Record 2026). The Fourth Circuit's ruling held that 3M produced some PFAS materials at the direction of the federal government for military use, thus allowing the company to invoke the "federal officer removal statute" (28 U.S.C. § 1442(a)(1)), pushing the cases to be heard in federal court (Tafts PFAS Insights 2026). 3M argued that the military used MilSpec AFFF across the country, including both states in the suit. The company also argued that the plaintiff's ability to identify whether the PFAS came from MilSpec AFFF or non-MilSpec AFFF is weak, drawing on the idea of scientific uncertainty. Their reasoning allows 3M to argue that the states' general PFAS suits were related to its federal contractor work, ultimately triggering federal jurisdiction. This ruling is significant because federal courts are less favorable to plaintiffs in environmental cases due to stricter pleading standards, more limited discovery, and a history of dismissing claims against federal contractors under government contractor immunity doctrines. For states like Maryland and South Carolina, the U.S.S.C decision means their cases proceed in federal court, where corporate defendants have a better track record.

Sovereign Immunity and the Limits of Litigation

The most powerful legal barrier to accountability for military contamination is the doctrine of sovereign immunity, holding that the federal government cannot be sued without its consent. Although the Federal Tort Claims Act (FTCA) waives sovereign immunity for certain claims, it has specific exceptions that are interpreted to prohibit most military environmental contamination lawsuits (Stokes 2023). As Elder recounted from his advocacy work with survivors of contamination at George Air Force Base: "They sued, and the Air Force claimed sovereign immunity in federal court... and then Wurtsmith Air Force Base in Michigan sued, and again, sovereign immunity, which means we have the right, in the name of national security, to poison you" 16 .

In addition to the doctrine of sovereign immunity, the Feres doctrine, established through a Supreme Court ruling in 1950, restricts military members' ability to sue the government for injuries stemming from military service. Under this doctrine, members cannot bring FTCA claims against the government for injuries "incident to service", which is a category that courts interpreted broadly to include environmental exposure at military installations (Stokes 2023). The Camp Lejeune Justice Act of 2022 created an exception to sovereign immunity for individuals exposed to contaminated water at Camp Lejeune between 1953 and 1987, allowing individuals to file claims for injuries resulting from exposure. However, it has a requirement of exhausting administrative remedies before filing suit, imposed strict deadlines, and capped attorneys' fees, reducing the accessibility and effectiveness of its supposed goal (Stokes 2023).

Illusion of Participation: The Restoration Advisory Board System

The Restoration Advisory Board (RAB) was established in the 1990s as a response to community activism demanding greater transparency (Pietkoski 2020). While RABs represent a theoretical partnership between the military, regulators, and affected communities, RABs are advisory only. There is no decision-making authority, and the military retains final control over all cleanup decisions. The PAX individual interviewed honed in on the base's transparency, and when asked about independent testing, they responded that they did not see the value of third-party verification 17 . M. Dibbles stated that this response is "anti-scientific" and that "the foundation of scientific review itself is to peer review and check and verify outside of your internal constituencies" 18 . Elder's experience with the MDE study shows how independent testing can be of use at times. When the state tested for PFAS in the St. Mary's River, they set detection thresholds at parts per billion rather than parts per trillion, which is 1000 times higher than appropriate. A result of no concern for additional investigation is drastically different from independent testing resulting in PFAS at levels as low as 1.7 parts per trillion.

The RAB system, designed to provide community input on military environmental cleanup, functions as co-opting community concerns rather than incorporating them. M. Dibbles noted that true accountability would require “a proactive disclosure system... as soon as a contaminant is detected, and even before a contaminant is detected, working with the community directly and the surrounding communities with very transparent communication” 19 . This would build upon community-led monitoring with independent verification, which is vastly different from the current system, where the polluter controls both the investigation and communication of the results.

In addition to this, any citizen can request information from/about federal agencies through the Freedom of Information Act. Although designed for the sake of transparency, there are nine exceptions to a FOIA, including national security, personal privacy, and trade secrets. If denied, an individual has the right to file an administrative appeal. It is a helpful initiative and useful resource to those governed, but there are several flaws, as those nine exceptions are quite broad, and plenty of information excluded from a FOIA contains information that, morally speaking, should be available to the public. Only 12.1 percent of FOI requests were fully granted in 2024, and there are over 200,000 requests backlogged for the third consecutive year (Massoglia 2026). A lack of response is to be treated as an automatic constructive denial. The process of requesting information is obscure or unknown to the average citizen, and the specificity of the contents in each agency can be nonexistent, meaning that individuals cannot properly utilize this resource if there is not an adequate understanding of an agency and its particular content. Moreover, FOIAs legally require agencies to respond to a request within 20 business days, but there is no reinforcement of this, resulting in all agencies violating this rule. Request wait times vary by agency, but many take two or more months to respond, and some will never respond (constructive denial). There is also a potential fee attached to FOIAs, which makes the process inaccessible to those of certain economic backgrounds.

Through personal experience, a FOIA was filed with the Department of the Navy in mid-December of 2026 and was answered in late January of 2026. The request was for the contracts between 3M and the Navy, as 3M has been the Navy's main supplier of AFFF and was the first company to develop a patented AFFF formula. The request could not be fulfilled due to the documents possibly being stored at the National Archives and Records Administration (NARA). After receiving this response, another FOIA was immediately filed (in January of 2026) to the specific subsection of NARA, the archives of the Navy, for the same material. By late March, there had yet to be a response, and this is most likely due to a silent denial. So, a Freedom of Information Act request is a brilliant idea on paper, but in reality, it is an empty promise. An initiative for transparency is absolutely futile when most information is not available to the public, the process is difficult, and it is placed behind a potential paywall. This is not a genuine promise of transparency, aligning with the pattern of “good in theory” structures that are ineffective in reality.

Environmental Justice: What Does PFAS Mean in the Context of Justice?

Scientific literature and empirical data demonstrate that PFAS contamination, similar to all forms of industrial pollution, disproportionately affects low-income communities, communities of color, and Indigenous populations. These communities have come to bear the health burdens, the economic costs of contamination, and the emotional toll of fighting for accountability. As M. Dibbles observed, the fundamental problem is that "the entire model moves the accountability from the contaminating party... onto the victims and the communities. The burden of proof for accountability should not be on the community members." 20 .

The concept of environmental justice came to be through grassroots movements in the 1980s when the documentation of the disproportionate siting of hazardous facilities in communities of color gained traction.

The 1987 report, *Toxic Wastes and Race in the United States*, found that race was the most significant predictor of hazardous waste facility locations, more significant than income, homeownership, or property values (Bullard 2000). Robert Bullard documented how African American communities in the South have been historically targeted for landfills, incinerators, and chemical plants because they lacked the political power to resist (Bullard 2000). Checker (2005) continued this analysis to understand the operation of “polluted promises,” which is the gap between government assurances of safety and the lived reality of contaminated communities. Checker documented how residents of a predominantly African American community in Georgia dealt with decades of contamination from an industrial facility while government agencies dismissed them, downplayed the health risks, and delayed action. She identified a pattern of community members bearing the burden of documenting contamination, fighting for acknowledgement, and suffering health consequences while institutions profit from delay. This pattern is recurring in PFAS-affected communities nationwide.

Peer-reviewed research has increasingly documented the burden of PFAS contamination on marginalized communities. A 2024 study analyzed PFAS concentrations in public water systems across the U.S. and found that communities with higher proportions of Black and Hispanic residents had significantly higher PFAS levels in their drinking water compared to predominantly white communities, even after controlling for income and other socioeconomic factors (Andrews et al. 2024). It was concluded that racial and ethnic disparities in PFAS exposure require specific intervention and attention (Andrews et al. 2024, p. 8). Moreover, a 2023 analysis found that communities of color are more likely to live near military installations and industrial facilities. It was documented that “Black Americans are 1.4 times more likely than white Americans to live within one mile of a PFAS-contaminated military base” (Environmental Working Group 2023). Such a disparity is a direct result of redlining, exclusionary zoning, and the intentional siting of hazardous facilities in communities without political power.

The PAX individual interviewed discussed that minority communities live in hazardous areas because the hazard was placed there before, lowering property values, coinciding with how lower economic classes are typically of color. M. Dibbles addressed that framing of "minority communities move into environmentally hazardous areas rather than turning minority communities into hazardous waste sites" 21 . Dibbles explained that this framing presents "a half-truth, as if it's the whole story." Even if market dynamics play a role, "the market itself was based on racist housing policies, redlining, discriminatory lending, and exclusionary zoning. So no matter how you look at it, I think it's a self-reinforcing cycle," rather than a which came first question 22 .

The discovery of PFAS contamination also imposes economic burdens on impacted communities, and these burdens fall disproportionately on those with fewer resources to combat it. Huang and Kumar (2021) document that PFAS discovery destabilizes municipal finances because local governments must bear the costs of testing, treatment infrastructure, and legal action. Such costs strain local budgets and can discourage aggressive investigation or disclosure, creating underreporting and inadequate recognition (Huang & Kumar 2021, p. 14). The issue of economic impacts extends to property values. A 2024 study found that homes located within one mile of a PFAS contamination site experienced property value declines of 5 to 15 percent, with the largest impacts in lower-income neighborhoods that have difficulty absorbing losses (Muehlenbachs et al. 2024). For families whose homes are their primary source of wealth, these declines can be detrimental.

An incredibly visible and relevant example of environmental justice in PFAS disposal is the landfill in Emelle, Alabama. Emelle is located in Sumter County, Alabama, which has one of the highest poverty rates and largest Black populations in the state. The landfill is operated by Waste Management and is the largest hazardous waste landfill in the United States (Tns 2023). As the PAX individual stated, describing where the base sends its PFAS-contaminated waste, "They [Waste

Management] said, [we're] already taking a bunch of hazardous waste stuff. We'll take yours'" 23 .

When asked about the factor of contamination hurting minority communities in Emelle, as PAX sent their own waste there, "As far as the impact upon minority community or any community around the landfill, I would be concerned if it was just a subtitle D that did not have a liner and did not have leachate treatment, I would absolutely be concerned." 24 . However, there should still be concern because lining and leachate treatment systems, especially those at Emelle, repeatedly violate regulations. In 2023, it was reported that over two dozen containers of hazardous waste, including PCBs, were rusting or damaged, with 16 reported as leaking, as well as a two-year-old uncleaned spill from a leachate tank (Tns 2023). M. Dibbles addressed this inadequacy of relying on liners for protection, stating it is a mistake to place full confidence in lining infrastructure, and that even the EPA has expressed concern about liners preventing all liquids from entering the ground, as there are many well-documented cases of chemical corrosion, temperature stress, and pipe collapse 25.

The Burden of Fighting

A defining feature of environmental justice is that the burden of fighting for accountability falls almost entirely on affected communities. As Pat Elder documented through his decades of environmental advocacy, it is community members, not government agencies or responsible corporations, who have to bear the costs of testing, the emotional weight and grief, and years of advocacy to achieve even minimal acknowledgement. Elder's experience at PAX aligns with this pattern as he has spent years collecting samples.

Elder's own advocacy at Patuxent River exemplifies this pattern: he has spent years collecting samples, filing Freedom of Information requests, publishing reports, and pushing state and federal agencies to acknowledge contamination that official studies systematically understate. The emotional and financial costs are immense. Elder

disclosed the expense of independent PFAS testing: a single test kit for a rockfish costs \$350, and testing for oyster, crab, and water samples costs multiple times that cost. Yet without independent testing, communities would solely rely on government studies that, as Elder documented in MDE's work, set detection thresholds 1,000 times higher than appropriate. The burden of producing the evidence to receive appropriate remediation falls on community members who are already bearing both the health and economic costs of contamination.

The personal toll of bearing the burden is evident in the stories and lives of those who have lost loved ones to military contamination. Doris Brock's experience of her husband's death from bladder cancer after 35 years of service, and experiencing 34 of their 66 friends' deaths, is not something to be taken lightly. As she described, "The disappointment was not being able to spend the rest of my life with him... after we retired, which is... horrible, and there's so many people that are in the same shoes" 26 . The reality that Brock had to document these deaths herself, to track the outcomes of her husband's colleagues when no government agency would do so, reveals the foundational injustice of a system that puts the burden of proof on victims instead of the polluters.

Reform or Transformation?

The literature revolving around PFAS contamination has documented and described a system where corporate polluters thrive, military installations are shielded from accountability, and the harm and burden will always fall on the average citizen. The evidence presented discusses PFAS as more than just a scientific or technical problem. The science and data behind the PFAS crisis have evolved rapidly and will continue to do so to the benefit of society, but its implications in policy, regulation, and justice are complex. The concept of reform in this existing framework is questioned, as the patterns documented at Camp Lejeune, PAX, and countless other bases show a system so fundamentally broken that it requires an entire transformation rather than just incremental adjustments.

DISCUSSION

The findings of this research show a crisis that goes further than the chemical properties of PFAS. Although the carbon-fluorine bond explains why these compounds are so persistent in the environment, the persistence of regulatory system failures requires a separate explanation. The results of the four interviews, in addition to the peer-reviewed literature and legal analysis, show that the understanding of PFAS toxicity has dramatically advanced, but governance and remediation efforts remain unaltered. Such a paralysis is not just a failure of the system; it may be a feature of it.

The Weaponization of Uncertainty

The weaponization of uncertainty in the PFAS crisis is best understood through the concept of “power/ignorance”, the idea that powerful institutions actively produce ignorance as a strategy of governance. In the context of PFAS, there is no lack of knowledge or presence of uncertainty, at least not to the extent that inaction is justified. As established by Michel Foucault, power and knowledge are interconnected. Power creates knowledge, and knowledge enables power (Foucault 1980). Andrew Mathews (2005) refined this through his study of forest fires in Mexico, where the state used fires as justification for complete control of forests, coining the term “power/ignorance”. The chemical industry and the DOD have produced scientific knowledge on toxicity and persistence, but they have also strategically released this knowledge to delay regulation, undermine its reliability, and maintain the status quo.

The official state testing conducted by MDE yielded results that were contradicted by two independent data sets. MDE’s no detection rate of concern is due to setting detection thresholds too high to capture contamination. Although technological limits existed in 2020, when the study was conducted, the agency has continued to stand by its initial results, disregarding scientifically valid evidence that shows detection of concern. There has also been suppression of scientific findings, as the EPA hid an updated PFAS toxicity assessment done by its own Office of

Research and Development in April of 2025 (Southerland 2026). There is also consistent invocation of inconclusive science as a justification for delay or denial, which is a pattern that individuals like Doris Brock and Jerry Ensimer personally identified across years of government-conducted studies and interactions.

Moreover, information is withheld through FOIA exceptions, with only 12.1 percent of FOIA requests fully granted in 2024 (Massoglia 2026). So, as Mathews stated, this deployment of ignorance is not a failure of the state or system; it is a strategy of governance. Through the claims that the science is uncertain, that more research is needed, and that causation has not been definitively proven, the DOD and EPA have delayed regulation, disability claims, and avoided overall liability while the average citizen pays the price. This dynamic was also identified by M. Dibbles through their observation that the EPA's use of needing more time and additional studies has been heard since the 1970s and 1980s²⁷, leading to the question of whether these hesitations hold genuine weight or are they just another form of regulatory capture and delay?

Bad Law and Supreme Inequality

The concept of “power/ignorance” functions within a systematically biased legal system. In *Bad Law*, Elie Mystal, a former litigator and legal analyst, builds a crucial foundation for understanding how this bias works. Many of the harmful laws in the U.S. were not created out of good intentions; they are statutes that were specifically designed to cause harm: “these laws do not come to us from on high; we write them, and we can and should unwrite them” (Mystal 2025). This has an important place in the PFAS crisis, as the laws that enable contamination to continue, like the Toxic Substances Control Act, the Federal Tort Claims Act's exceptions that bar military environmental lawsuits, and the federal officer removal statute that allows 3M to move state cases to federal courts, should not be mistaken for accidents or oversights. They are policies and statutes that were written, lobbied for, and passed by people who understood the harmful implications to the fullest extent.

While incredibly harmful, it is simply the result of a designed and desired function. These are not neutral procedures and processes. They are bad laws that “continue to perpetuate hate, racial bias, injustice, and inequality today” (Mystal 2025).

Moreover, these bad laws typically have popular and bipartisan support. For example, the 2005 Protection of Lawful Commerce in Arms Act (PLCAA), granting gun manufacturers near complete immunity from product liability lawsuits, was passed with bipartisan support. This matters in the context of PFAS because bad environmental laws expand beyond one party or ideology. They are the desired outcome of a political system where corporate power is systematically privileged over public health.

Laws have measurable consequences. In the context of PFAS, the bad laws include CERCLA’s Federal Facility Agreements that grant military control over cleanup, and the TSCA provisions that allowed PFAS to enter commerce without proper toxicity testing. The measurable consequences of these laws take form in a human cost, such as the 34 friends and husband Doris Brock lost far too soon, the 9-year-old girl, Janey, who died of leukemia from base contamination, and the communities that surround Emelle, Alabama, who live alongside the nation’s largest hazardous waste landfill.

In *Supreme Inequality* (Cohen 2020), it documents how the Supreme Court has ruled in favor of wealthy individuals and corporations over the past 50 years, creating a system that solidifies inequality. The judicial branch was supposedly designed to address and prohibit wrongdoings, but it appears to function beyond the idea of checks and balances, and instead wields extreme power to permanently reinforce improper decisions. The Court’s ruling in *Citizens United v. FEC* (2010), allowing unlimited corporate political spending, and its generously expansive interpretation of the Feres doctrine are not anomalies. They fit into a pattern of jurisprudence that prioritizes corporate power over individual rights and overall public health.

The Supreme Court's ruling on March 2, 2026, denying certiorari to Maryland and South Carolina, effectively allowing PFAS contamination cases against 3M to be heard in federal court over state court, is a solid example of this pattern. The Fourth Circuit's ruling that the Supreme Court declined to review held that, due to 3M producing some PFAS-containing materials at the direction of the federal government for military use, the company can invoke the "federal officer removal statute" (28 U.S.C. § 1442(a)(1)). However, in a dissent by the Fourth Circuit Judge Henry Floyd, he warned that the majority's ruling risked pushing "lawsuits properly before state courts" into the federal system, which moves "toward foreclosing state courtrooms to plaintiffs" (*State of Maryland V. 3M Company*, No. 24-1218 (4th Cir. 2025) 2025). By declining to review this ruling, the Supreme Court allowed a precedent that makes it substantially harder for states to hold PFAS manufacturers accountable, which is a decision that is consistent with the pattern described in *Supreme Inequality*.

The Accumulation of Disadvantage

Institutional and legal failures produce compounding harm for marginalized communities and can be characterized as cascading inequalities. A single environmental or societal disadvantage, like living near a military installation or hazardous waste site, being a person of color, or having a low income, does not exist in isolation. Rather, disadvantages accumulate and reinforce one another to produce outcomes that are worse than individual disadvantages. Through the lens of environmental racism, for a low-income family of color living near Naval Air Station Patuxent River, the cascading inequalities include proximity to contamination. Due to historical redlining and exclusionary zoning, a family of color lives in an area with higher PFAS levels in water, soil, and air (Andrews et al. 2024; Environmental Working Group 2023). In addition, the family would struggle to afford bottled water, home filtration systems, or the option of relocation. There would also be reduced property values, so their home, likely the family's primary source of generational wealth, would lose 5 to 15 percent of its value due to PFAS contamination (Muehlenbachs et al. 2024). State and

federal testing protocols understate contamination, denying the family information necessary to make informed decisions. The family would also face increased risks for cancer, thyroid disease, immune suppression, and developmental effects (ITRC 2025). The family's ability to sue for damages would be reduced due to sovereign immunity, the Feres doctrine, and the federal officer removal statute (Stokes 2023). Lastly, due to political marginalization, the chemical industry's campaign contributions and lobbying ensure that legislation like the PFAS Action Act is stalled and narrowed.

Each of the disadvantages compounds upon another. The family cannot sue, cannot afford to relocate, cannot trust the government's testing results, and has no political power to change the system. The landfill in Emelle, Alabama, where PAX sent PFAS-contaminated soils, is the largest hazardous waste facility in the U.S., while also being located in a county with one of the highest poverty rates and the largest Black populations in the state. This is a network of cascading inequalities that goes as far as from Maryland to Alabama, to the contaminated homes and beings of marginalized communities across the nation. Lastly, in a scientific study mentioned previously, PFAS exposure does increase the chances of PCOS (polycystic ovary syndrome), a diagnosis that is frequently dismissed due to medical racism. People of color are 20 to 30 percent more likely to experience medical misdiagnoses compared to white patients (Team 2024). Women are 66 percent more likely to receive an incorrect initial diagnosis than men; about 1 in 5 women report having their symptoms ignored, and 70 percent of women have experienced medical gaslighting (Peters & Peters 2025). This creates the issue of statistical accuracy in surveying contaminated communities due to misdiagnoses or lack of diagnoses, providing further evidence that inequalities continue to compound due to systematic biases and structural inequalities.

An Intentionally Ignorant System Creates Predictable Outcomes

The PFAS crisis begs the question: Is the system broken, or was it built this way? The evidence provided suggests that the patterns documented at Camp Lejeune, Patuxent River, and various other military

installations are not accidents or singular failures. They are predictable outcomes of a system that has been designed and reinforced through Supreme Court rulings, statutory frameworks, regulatory capture, and the strategic production of ignorance, all to prioritize corporate profit and military impunity over environmental justice and basic public health. As Pat Elder stated, the system is “not an accident” 28 . As M. Dibbles observed, “you can’t really have a functioning environmental regulatory body under authoritarianism” 29 . The issue of PFAS is a part of deeper structural problems in American governance and government. These problems will require new laws in addition to institutional reform, as well as the consideration of who bears the costs of industrial production.

FUTURE RESEARCH

The evidence provided demonstrates that PFAS contamination is more than a scientific and technological problem, as it is a systemic governance failure rooted in the structural inequalities of American environmental law. Although the extent of the issue can be paralyzing, parts of these findings point toward paths for continued research and advocacy. One avenue stemming from this work is the opportunity for additional and deeper engagement with the Naval Air Station Patuxent River through the Restoration Advisory Board. As the individual from PAX noted, the RAB also includes civilian community members and extended an invitation to attend future RAB meetings.

Moreover, the installation restoration project manager at PAX directly reached out to gauge interest in filling a vacant role on the board. Although temporary due to the time constraints, such an invitation represents a solid opportunity for St. Mary’s College of Maryland students and faculty to participate in the process of remediation. Attendance at RAB meetings would provide students with exposure to the Federal Facility Agreement process, the dynamics and importance of local community engagement and action, and it would allow the college to serve as an independent observer of the base’s claims about contamination levels and remediation progress. It would also begin to address the PAX interviewee’s concern about low public participation.

Due to the college's physical proximity to the base, the college community has a part in the outcome and in the responsibility to engage in change.

Q Methodology Continuation

The interviews conducted showed significant diversity in how different stakeholders understand the PFAS crisis. The PAX individual honed in on transparency, proactive action, and the validity of scientific uncertainty. Pat Elder discussed systematic corruption, intentional deception, and even criminal negligence. Doris Brock spoke on institutional denial, personal losses and burdens, and the limits of legislative routes. M. Dibbles gave a more structured analysis of environmental justice in relation to regulatory capture and the limits of the current system. These differing perspectives are fundamentally different ways of understanding the problem, its causes, and what the proper solution is. A Q methodology study would be useful in continuation of this work, as it is a research approach designed to study subjectivity by identifying and characterizing the different perspectives that exist within a group of people on a specific topic. Differing from the typical question that prompts agreement or disagreement, Q methodology lets individuals sort statements following their personal subjective importance, which ultimately reveals “subject positions” that might not have been previously apparent.

A Q methodology study on PFAS contamination at military sites could assist with identifying the range of perspectives held by different stakeholders, like military individuals, activists, impacted residents, regulators/policymakers, scientists, and industry representatives. It would also quantify the degree of consensus or disagreement on issues like the reliability and acceptability of current testing protocols, the credibility of government versus independent studies, the effectiveness of the RAB process, and the perceived fairness of current legal pathways of remediation. It could also put different perspectives into demographic variables like proximity to contamination, duration of exposure, institutional role or affiliation, and political orientation. Lastly, it would build an empirical basis to identify areas of potential agreement, which may be utilized to inform policy reform and community/overall engagement strategies.

Additional Contacts

The interviewees for this research provided contacts for additional experts and advocates whose work could strengthen any variation or continuation of this project. Beginning with scientific and medical experts, Nate Donley, the Environmental Health Science Director at the Center for Biological Diversity, has extensive experience on the health impacts of PFAS and various other contaminants. Donley's organization has been at the forefront of litigation and advocacy to push for EPA action on PFAS regulation. There is also Linda Birnbaum, the former Director of the National Institute of Environmental Health Sciences (NIEHS), who has been described as a pioneer in PFAS research. Birnbaum dealt with political pressure and opposition for her willingness to speak publicly on the risks of PFAS. Such a perspective from a scientist who led the federal agency responsible for monitoring environmental health would be useful for understanding the science of PFAS and the politics within federal research. For biomonitoring and exposure assessments, Sean Lynch of the Maryland Pesticide Education Network has done work that provided direct evidence of PFAS causing harm to the human body. Lynch's experience and access to biomonitoring data would be crucial for understanding and establishing the link between environmental contamination and human exposure. There is also Jared Hayes of the Environmental Working Group who works on PFAS policy and can access EWG's mapping of PFAS contamination sites across the country. The EWG has played a major role in bringing scientific research into public advocacy.

On the topic of scientific integrity, Kyla Bennett is a former EPA scientist who became a whistleblower by documenting agency failures on PFAS and other chemicals. Her perspective shows the pressures scientists deal with when attempting to speak the truth. Bennett's knowledge of EPA operations and processes, as well as her willingness to criticize a government agency, brings an important narrative to additional research. There is also Laura Dumais, who authored a commentary for the Public Employees for Environmental Responsibility (PEER), an organization that has been vital in supporting whistleblowers and

pushing back against agency failures. Dumais's work focused on PFAS contamination in the food supply, which is a path of exposure that Elder warned was more significant than drinking water. On a similar note, journalistic experts like Sharon Lerner are important, as she has broken many of the most significant PFAS contamination stories, which include the concealment of evidence and the military's role in contamination. Lerner's reporting work was directly mentioned in this research as the interviewee, Brock, noted that Lerner "was way ahead of everyone and still is" 30 . Her expertise in topics such as FOIA requests, understanding and accessing industry documents, and connecting scientific findings to policy outcomes would be extraordinarily helpful. Lastly, the inclusion of organizations like Earthwatch and the Natural Resources Defense Council would build institutional support and legal expertise.

To create a more inclusive participant cohort, Francis Gray, the Tribal Chair of the Piscataway Conoy Tribe, would be a crucial perspective. Gray has mentioned in a guest lecture at St. Mary's College of Maryland that PFAS is a frustrating issue that has changed cultural norms and typical practices. Native American communities face significant challenges to preserving their land and culture. The Piscataway Conoy Tribe has dealt with severe violence and oppression to the extent of forced migration. The tribe secured its first piece of land in 400 years in 2023, showing the significantly delayed reparations. The inclusion of Gary and the Piscataway Conoy Tribe is important, as Southern Maryland has a history that must be acknowledged, especially as historical injustices persist in the modern day. Understanding the place of indigenous communities within the PFAS crisis would provide invaluable knowledge for remediation efforts.

Final Remarks

The plan moving forward includes immediate action like attending RAB meetings, as well as long-term structural change like the repealment of bad laws, limiting corporate political spending, ending sovereign immunity for military pollution, and working toward the burden of proof

falling on polluters. The stories and perspectives included in this research, such as Elder's elevated serum levels, Brock's deceased husband and friends, and the Camp Lejeune families who fought for 40 years, are more than just evidence or markers of harm. These perspectives must be treated as a call to action. PFAS harms various types of communities and people, changing lives in drastic and unimaginable ways. This contamination requires efficient, expansive, and meaningful changes to our legislative bodies and choices; change that more accurately reflects the idea of representing the people instead of corporate or agency interest. Without such change, the toll of PFAS will continue to rise, a toll measured in human lives.

Notes

- [1] PAX Individual, interview by author, St. Mary's City, MD, February 6, 2026.
- [2] Individual, interview.
- [3] Individual, interview.
- [4] Individual, interview.
- [5] Doris Brock, interview by author, St. Mary's City, MD, February 10, 2026
- [6] *Id.* at 42
- [7] *Id.* at 42
- [8] Pat Elder, interviewed by author, St. Mary's City, MD, February 9, 2026.
- [9] *Id.* at 52.
- [10] *Id.* at 42.
- [11] *Id.* at 42.
- [12] *Id.* at 52.
- [13] *Id.* at 52.
- [14] *Id.* at 52.
- [15] *Id.* at 52.
- [16] *Id.* at 52.
- [17] Individual, interview.
- [18] M. Dibbles, interview by author, St. Mary's City, MD, February 23, 2026.
- [19] *Id.*

[20] *Id.* at 61

[21] *Id.* at 61

[22] *Id.* at 61

[23] Individual, interview.

[24] Individual, interview.

[25] *Id.* at 61

[26] *Id.* at 42

[27] *Id.* at 61

[28] *Id.* at 52

[29] *Id.* at 61

[30] *Id.* at 42

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Right to Repair in the United States: Definition and Defense Through Strategic Litigation

Afnaan Syed

Clark University

“THE BEST WAY FORWARD FOR THE RIGHT TO REPAIR MOVEMENT IN THE US IS TO FOCUS ON STRATEGIC LITIGATION.”

Abstract

In response to mounting anti-repair practices by major technology manufacturers, recent decades have seen the emergence of a global right to repair movement. It brings together tinkerers, hobbyists, repair professionals, policy advocates, environmental activists, and everyday consumers. This coalition operates both locally and globally to share parts, tools, information, and repair techniques (especially through online forums and Fixit Clinics) and, most importantly, to advocate for legislation protecting the right to repair in all jurisdictions in which it is found. This article examines the right to repair movement with an eye toward achieving three main goals. The first is to establish and defend the basis for the concept of a fundamental right to repair, in both normative and legal contexts – with the latter focusing primarily on the existence of a right to repair in the United States. The second is to highlight how tech corporations in recent decades have taken advantage of copyright law and technological advancement to restrict and monopolize repair markets, and more importantly, how the right to repair movement has fought back on two separate fronts – legislative advocacy and litigation. The third and most important goal of the article is to advance the argument that the best way forward for the right to repair movement in the US is to focus on strategic litigation rather than legislative advocacy by using existing precedent in intellectual property law, namely the longstanding doctrine of patent exhaustion.

Introduction

In response to mounting anti-repair practices by major technology manufacturers, recent decades have seen the emergence of a global right to repair movement. It brings together tinkerers, hobbyists, repair professionals, policy advocates, environmental activists, and everyday consumers. This coalition operates both locally and globally to share parts, tools, information, and repair techniques (especially through online forums and Fixit Clinics) and, most importantly, to advocate for legislation protecting the right to repair in all jurisdictions in which it is found. This article examines the right to repair movement with an eye toward achieving three main goals. The first is to establish and defend the basis for the concept of a fundamental right to repair, in both normative and legal contexts – with the latter focusing primarily on the existence of a right to repair in the United States. The second is to highlight how tech corporations in recent decades have taken advantage of copyright law and technological advancement to restrict and monopolize repair markets, and more importantly, how the right to repair movement has fought back on two separate fronts – legislative advocacy and litigation. The third and most important goal of the article is to advance the argument that the best way forward for the right to repair movement in the US is to focus on strategic litigation rather than legislative advocacy by using existing precedent in intellectual property law, namely the longstanding doctrine of patent exhaustion.

Part 1: Defining the Right to Repair

1.1 - A Normative Basis for the Right to Repair.

Before discussing the legal basis for a right to repair in the United States, the normative question must be addressed – *should* we have such a right? Is an equitable and sustainable society one in which there is a right to repair? On this question, scholars are largely in agreement that the answer is yes, for several reasons. First and foremost are the increasingly apparent environmental consequences of the “replace, don’t repair” culture brought about by repair restrictions. Electronic waste or “e-waste” accounts for roughly 70% of the toxic waste in US

landfills – a figure that continues to rise. That e-waste includes lithium, mercury, and lead—chemicals that endanger our water supplies and threaten human health. [1] Even before the end of the product lifecycle, manufacturing and global shipping produce significant pollution. And extracting the raw materials needed to produce our devices inflicts both environmental and human costs. Take cobalt, for example – a crucial component in the lithium-ion batteries found in phones, laptops, electric vehicles, and countless other consumer electronics. Global demand for cobalt exceeds 100,000 tons per year - a number that is expected to increase threefold over the next decade. [2] Most of the world's cobalt supply is found in the Democratic Republic of Congo. Roughly 20% of it is mined by hand by a quarter million local laborers who work in narrow, unstable tunnels, inhaling toxic cobalt dust, for less than a dollar a day - and an estimated 35,000 of whom are children as young as six years old. [3] By encouraging us to replace rather than repair our devices, manufacturers are increasing the already high demand for materials like cobalt. In the process, they are amplifying the negative environmental and human rights externalities created by global electronics production – and the problem gets worse upon realization that roughly 80% of e-waste globally is not collected for recycling [4], left instead to survive for thousands of years in landfills.

The societal consequences of restrictions on repair are made perhaps more dire when considered in the context of devices used in critical sectors of the globalized economy – for example, healthcare. At the height of the COVID-19 pandemic, as hospitals across the globe faced shortages of life-saving ventilators, and manufacturers scrambled to ramp up production, the ability to maintain and repair existing equipment emerged as a pressing problem. Authorized repair, which often requires shipping devices back to the manufacturer, can leave hospitals without critical equipment for days or weeks. In other instances, manufacturers failed to supply replacement parts, imperiling patients. When a hospital in Chiari, Italy, couldn't secure valves for its respirators from the manufacturer, local volunteers designed and 3D-printed 100 replacements that cost no more than a few dollars apiece. The volunteers managed this feat in just two days, with no help from the manufacturer, which refused to share design specifications. [5]

There is something to be said also about the implications of repair monopolies for exacerbating economic inequality. As Leah Chan Grinvald and Ofer Tur-Sinai argue: “A lack of access to repair information sustains classism because only the wealthy can afford to routinely purchase new products when theirs break. Combined with some manufacturers’ ‘planned obsolescence’ programs, some products are designed to break down sooner than others. For example, Apple has published the lifecycle expectancies of its iPhones, which is approximately three years.” [6] Put simply, making it more difficult to repair products has the effect of sustaining the economic disenfranchisement of those with fewer resources. [7]

Yet aside from staving off worsening economic inequality and the potentially disastrous consequences of repair restrictions on crucial electronic devices that our collective livelihood depends on, scholars in the field argue also that repair’s inherent benefit to an intellectually engaged society is also grounds for its protection. As Aaron Perzanowski has suggested, *The Right to Repair*: “Repair is a social practice that builds valuable skills. It demands analytic reasoning, strategic thinking, and creativity. And repair helps us develop a more complete understanding of how a device operates, enriching our awareness of the world around us. In that sense, it makes us freer, more autonomous, more in control of the world we occupy. Repair empowers us.” [8]

1.2 – The Right to Repair in the US

At the outset, there is an important distinction to be made regarding the term “right to repair.” In current popular discourse, “right to repair” is essentially a blanket term referring to actors on the legislative and legal fronts who advocate for increased access to repair for consumers. This includes fighting back against repair monopolies and pushing for manufacturers to design products that are inherently more repair friendly. This article, however, considers a more literal definition of the words: Just as we in the United States have a right to say whatever we please (with limited exceptions, of course) via the First Amendment, we also have a right to repair the products we own. Recognition of this right has limited relevance in terms of coercing tech manufacturers to make

their products more repair friendly, or to make repair data more accessible. It is, however, crucial for asserting our right to repair the things we own without fear of violating intellectual property law.

Although it is generally accepted among scholars that there is a strong pragmatic basis for the right to repair, there is some disagreement in terms of whether we have this right in the absence of explicit statutory protection.

Some scholars would argue that this right can already be found in the penumbras of existing copyright law. This may seem like judicial overreach at first glance, but the Supreme Court's watershed decision in *Quanta Computer v. LG Electronics* (2008) provides strong precedent suggesting that legitimate ownership of a patented technological device carries with it a degree of control over the use of that product, implicit in which is the right to modify or repair it as one sees fit. The decision does this by relying on a longstanding doctrine in intellectual property law known as the principle of exhaustion. We will proceed with an examination of the origins and theoretical underpinnings of exhaustion, followed by a discussion on the implications of the *Quanta* decision.

Article I, Section 8, Clause 8 of the United States Constitution grants Congress the power " [t]o promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries." These exclusive rights often take the form of patents for technological inventions and design processes, e.g. consumer electronics. [9] This provision of the Constitution recognizes that new ideas, once revealed, are easily appropriated by others, and prospective inventors might therefore decline to invest their time, energy, and resources in research without a mechanism to capture some benefit from their inventions. The patent system represents a bargain – society temporarily forgoes the benefits of market competition in return for the public good provided by the inventor's efforts. Thus, patents offer the right to exclude others from making, using, offering for sale, or selling the invention [10] – for a limited period. This period is limited by the exhaustion doctrine, which

declares that once the first sale of a patented item has occurred, the patent holder's rights over that item as set forth by the issued patent are "exhausted." [11] Put simply, exhaustion in its broadest form is what allows us to lend or resell physical copies of books, movies, and other forms of media, as well as electronic devices, at whatever price, if for a price at all, we choose. This might mean an individual lending or reselling to a friend, or it might mean an established bookstore or electronics retailer. However, a thorough examination of the relevant jurisprudence in the United States reveals that patent exhaustion does not merely apply to the reselling rights of the original purchaser. Rather, the courts have held that it applies to use of the patented product as well, as was affirmed in the *Quanta* decision (More on this below).

Exhaustion finds its roots in the common law's historic disfavor of the alienation of personal property. [12] Patent exhaustion as a legal principle in the United States has been expressly recognized by the Supreme Court as early as 1852 in its decision in *Bloomer v. McQuewan*. Here the Court held that when a patented machine passes to the hands of the purchaser from the holder of the patent right, it is no longer within the limits of the monopoly imposed by the patent. [13] 21 years later in *Adams v. Burke*, the Court provided a more clear and broadly encompassing articulation of what patent exhaustion entails: "...in the essential nature of things, when the patentee, or the person having his rights, sells a machine or instrument whose sole value is in its use, he receives the consideration for its use and he parts with the right to restrict that use." [14] Thereafter, the doctrine became fixed in US patent jurisprudence as a kind of axiomatic truth, "more slogan than policy," dutifully repeated but rarely explained. [15] Thus, exhaustion's theoretical underpinnings remain somewhat hazy. But an examination of the case law reveals two primary motivations for the exhaustion doctrine: (1) a desire to discourage unwarranted patentee compensation, and (2) deep misgivings about attaching permanent restrictions to personal property. [16]

As mentioned previously, *Quanta* is perhaps the most significant case for asserting patent exhaustion as the basis for the right to repair. At issue in *Quanta* was a post-sale use restriction that patent owner LG

Electronics imposed on Intel who used the patented processes to install Intel's microprocessors and other components in electronic products. Quanta was an electronic product manufacturer that installed Intel's microprocessors that included LG's patents. Rejecting LGE's claims of patent infringement against Quanta, the Court unanimously affirmed the application of the exhaustion to the use of a patented item after sale, holding: "The longstanding doctrine of patent exhaustion provides that the initial authorized sale of a patented item terminates all patent rights to that item." [17] The Court affirmed the earlier holding in *Bloomer* that any term extensions or other modifications to the patent in question did not affect the rights secured by purchasers who bought the item for use "in the ordinary pursuits of life." [18] Important to note is that the Court also affirmed that, despite LGE's arguments otherwise, exhaustion also applies to method patents – that is, patents granted for processes such as a manufacturing procedure or a recipe for an item, rather than the item itself. Relying on several earlier exhaustion decisions such as *Ethyl Gasoline Corp. v. United States* (holding that the sale of a motor fuel produced under patent also exhausted the patent for a method of using the fuel in combustion motors[19]), the *Quanta* Court held: "...this Court has repeatedly held that method patents were exhausted by the sale of an item that embodied the method...[e]liminating exhaustion for method patents would seriously undermine the exhaustion doctrine. Patentees seeking to avoid patent exhaustion could simply draft their patent claims to describe a method rather than an apparatus." [20] Patent exhaustion, in its most recently articulated form as embodied in the *Quanta* decision, provides as follows: when an individual purchases an item that embodies one or more patents within it, e.g. a desktop computer, the manufacturer of that desktop computer relinquishes its rights to any and all instances of the distribution and use of that product. The individual can face no legal repercussions for lending, reselling, or modifying that instance of the patented product, and that includes practicing the manufacturing methods essential to its function.

It should not come as a surprise that patent exhaustion, especially as embodied in *Quanta*, has been asserted to posit that there is, in fact, a right to repair. As Shubha Ghosh argues, the *Quanta* Court's definition of

“reuse” applies to repair as well: “The right to repair is an example of permitted reuse. At the outset, the right of repair is distinct from reconstructing a product from broken parts, which is an act of unpermitted making of a patented product, rather than permitted reuse.” [21] In doing so, he points to the distinction the Court drew between repair and reconstruction in its 1852 decision in *Wilson v. Simpson* – holding: “ [I]t does not follow, when one of the elements of the combination has become so much worn as to be inoperative, or has been broke, that the machine no longer exists, for restoration to its original use, by the owner who has bought its use. When the wearing or injury is partial, then repair is restoration, and not reconstruction.” [22]

However, it should be clarified that the Court’s distinction here is not infallible. Lower court decisions have affirmed the idea that the difference between repair and reconstruction is a difficult question that must be resolved case by case. [23] The line has been blurry going back thousands of years to the Greek philosopher Plutarch’s famous thought experiment surrounding the ship of Theseus. [24] Federal Circuit Judge Pauline Newman, presiding over the case of *Mallinckrodt, Inc. v. Medipart, Inc.* (1992), conceded that “it is not always clear where the boundary lies: How much ‘repair’ is fair before the device is deemed reconstructed.” [25] Mark Janis, in his seminal survey of the repair versus reconstruction jurisprudence in 1999, observed: “Courts long ago abandoned all efforts to cabin the repair-reconstruction dichotomy within a rigid framework of rules. Instead, they rest their decisions on ‘the exercise of sound common sense and an intelligent judgment.’” [26] According to Janis, the current test is based on “spentness”: Refurbishment is permissible as long as the product still retains some useful life. Janis argues that this test is indeterminate and better off replaced with a test based on the intention of the parties. [27]

In addition, some scholars, most notably Grinvald and Tur-Sinai, ground the right to repair in the utilitarian notion of intellectual property rights as a governmental tool to bargain for the development and disclosure of socially valuable information. [28]

Other scholars, however, seem to believe notions of a penumbral right

to repair within US patent exhaustion jurisprudence is insufficient by itself, and argue that codifying the right to repair in legislation is necessary to protect this right going forward, particularly in the face of new repair restrictions brought about by a clever combination of technological innovation and copyright law that tech corporations have leveraged. [29] In other words, the repair rights encompassed within patent exhaustion have been eroded by advancements in technology and instruments of copyright and contract law. In the words of Nicholas Mirr: "...the existing framework of copyright law and enforcement in the United States is insufficient to protect consumers' right to do what they will with their products. Through End User License Agreements and other contracts, companies are effectively able to limit the right to repair via contract even if there were to be a drastic overhaul of our current copyright system." [30] Mirr then goes on to argue that Congress should adopt legislation requiring companies to facilitate the repair process and should incorporate a "degree of repairability" requirement in the legislation to ensure the ability to repair remains an option for most consumers. [31] In other words, scholars such as Mirr argue that there at the very least should be a right to repair, but it must be codified through legislation, ideally at the federal level.

There are, therefore, at least two clear domains - litigation and lawmaking - in which the right to repair movement is playing out. The primary aim of this article is to evaluate which of these domains offers the more fertile ground for the newly emergent social movement surrounding the right to repair to mobilize. As the analysis below demonstrates, of these two policymaking venues, mobilization through strategic litigation is likely to be the more effective of the two. In other words, the issue of the right to repair should be "venue shopped" from the legislature to the courts. For this argument to work, it is a necessary assumption to make that there is, in fact, a right to repair that has been implicitly acknowledged by the courts, most notably the Supreme Court in its recent patent exhaustion jurisprudence as embodied in *Quanta* and other recent cases. This may be a tall order, considering the clever use of copyright law and technological innovation that tech giants have leveraged to monopolize repair in recent years. However, recall the previous discussion about the theoretical underpinnings of

patent/copyright exhaustion pertaining to the common law's disfavor of placing external restraints on one's personal property, along with the fact that the concept of exhaustion itself does not appear anywhere in the text of the US Constitution, and yet has been acknowledged and reiterated repeatedly by the Supreme Court. These factors, along with more recent lower court decisions in actions brought in the name of fighting back against repair restrictions (to be discussed in Part III) make clear that there is, in fact, a right to repair in the US that can be protected through litigation, even if not yet explicitly acknowledged by a court of law.

I proceed below with a discussion of how a social movement surrounding a right to repair came to emerge in the US. Then, I describe how corporations have launched a coordinated and sometimes quite effective campaign against the movement. This is followed by a discussion of how repair activists have fought back against corporations. These efforts have ranged from lobbying for legislation to antitrust litigation, each with somewhat mixed results. In Part 3, I evaluate the accomplishments of the movement in these two domains and suggest that strategic litigation is likely to be the most effective strategy for activists and others in the movement for a right to repair to succeed. Lawmaking depends on the vicissitudes of public attitudes and the political mood of the country. The legal case for a right to repair is much more stable because such a right can be independently justified drawing upon our common law tradition, doctrine, and case law.

Section II: Attack on Repair

The necessity of the growing right to repair movement across the globe is owed to the fact that technology corporations, in recent years, have adopted increasing measures to prevent consumers from repairing the products they own either by themselves or through an IRP (independent repair provider). This is done in the name of monopolizing repair markets and ultimately boosting profits by encouraging us, when our devices break, to replace our devices rather than repair them. [32] These measures take many forms – intellectual property law, hardware design, software restrictions, pricing strategies, and marketing messages – but for the purposes of this study, it is viable and somewhat

necessary to consolidate them into two main categories: anti-repair technological design and copyright law. Let's examine these one at a time.

2.1 - Anti-Repair Technological Design

There was a time when a clear line could be drawn between products and services. Once you purchased a TV, for example, the manufacturer had little say over how you used it. This is clearly different than purchasing a subscription to a streaming service to use on the TV you bought, such as Netflix or Disney+. In the latter case the service provider can add or subtract from the selection of the movies and shows your subscription grants you access to, change the price, or cancel the service altogether. Inconvenient as this may be, it is a reasonable thing to expect when purchasing access to an exclusively software-based streaming service that is accessed through the "cloud" – a distributed network of servers hosted in external data centers all over the world. [33] Today, however, the line between product and service is more of a blurry smudge. Our TVs, cars, appliances, smartphones, and countless other technological products we use every day are bundled with deeply intertwined software and external data services that are central to their functionality – a phenomenon that British technologist Kevin Ashton in 1999 aptly named "The Internet of Things." [34] The implications for repair are apparent and ominous.

2.1.1 - Apple

In 2016, several iPhone users reported that their devices had "bricked" - in other words, the phone had become completely unusable and thus turned into an expensive brick - after replacing a screen or Touch ID home button themselves or at another location not authorized by Apple. It was revealed that "Error 53" was a security measure implemented by Apple to prevent the devices from operating properly if they detected that third party components had been installed. Apple claimed this error was caused by a security check designed to prevent the Touch ID fingerprint reader from being exploited. Eventually, after complaints piled up, Apple released a software update which eliminated the error and restored the owner's phone to working condition. As it turned out, Australian courts declared Error 53 a violation of Australian consumer law, and Apple was eventually fined AUD \$9 million in June of 2018 as a result. [35]

2.1.2 - HP

“Security measures” like Error 53 are unfortunately not limited to cell phones. In March of 2025, tech giant Hewlett Packard (more commonly known as HP) settled a class action lawsuit in which it was accused of unlawfully blocking customers from using third-party toner cartridges. One of the named plaintiffs in the case, Mobile Emergency Housing Corp, an organization that works with emergency management organizations and government agencies to provide shelters for disaster victims and first responders across the US and Caribbean, bought an HP Color LaserJet Pro M254 in August 2019. In October 2020, the organization used toner cartridges from third-party supplier Greensky rather than pay for HP’s premium-priced toner. A month later, HP sent or activated a firmware update - part of its so-called “Dynamic Security” measures - rendering MEHC’s printer incompatible with third-party toner cartridges like those from Greensky. When MEHC CEO Joseph James tried to print out a document from the printer, he got a “Supply Problem” error message that rendered the printer completely unusable. [36]

2.1.3 – John Deere

If anyone could escape the repair restrictions brought about by the Internet of Things, you might think it would be farmers. We often imagine farmers living off land in rural communities, less dependent on modern technology and more rooted in traditions and practices that predate the digital era. But this image of farm life is out of step with reality. Farming today relies on a range of technologies, from moisture sensors and drones to genetically modified crops and patented biological pest controls. Even that classic symbol of rural America, the John Deere tractor, has been transformed into complex, software dependent piece of digital technology. [37]

Modern John Deere tractors can cost as much as \$800,000. No longer purely mechanical devices, they depend on multiple electronic control units (ECUs) to operate everything from the engine to the power seat. These embedded computers run software code essential to the operation and repair of the tractor. By controlling access to that code, John Deere can prevent independent diagnosis and repair. Without

enlisting a John Deere technician, the tractor’s software won’t even recognize replacement parts. That level of control forces farmers to rely on authorized John Deere dealers for service, rather than doing it themselves or turning to local IRPs. [38] So, a farmer who buys something as simple as a turn signal and installs it themselves or pays an independent repair shop to install it is still forced to pay the local John Deere dealer to send a technician to authenticate the part before it will work, for a fee of \$230, plus \$130 an hour for the technician. [39]

2.2 – Copyright Law

Aside from hostile design, one of the key legal tools firms have relied on to restrict repair is § 1201 of the Digital Millennium Copyright Act (“DMCA”). That statute makes it unlawful to circumvent - that is, to remove, bypass, or disable - technological protection measures (“TPMs”) that restrict access to copyrighted works. In addition, § 1201’s antitrafficking provisions prohibit the creation, sale, or distribution of tools that enable circumvention. [40] Device makers routinely use TPMs – such as the previously mentioned ECUs - to limit access to the software code that controls devices from smartphones to cars. That code is often necessary to diagnose and repair devices – see the previous discussion on the ECUs in John Deere tractors. These digital locks already pose practical hurdles for repair. But § 1201 compounds those difficulties by introducing potential criminal liability for repairers who remove or bypass such digital locks. [41]

The DMCA was enacted in 1998 to update and modernize US copyright law in response to the challenges presented by digital technologies and the internet. [42] At the time of the DMCA’s inception, that largely meant protecting the ownership rights of producers of digital media from unauthorized access to reproduce their works through decryption software. But as TPMs and Digital Rights Management (“DRM”) features have become increasingly intertwined into consumer technologies, the statute has been effectively hijacked by technology manufacturers to restrict consumers’ right to repair. The copyright protection afforded to the software code that is necessary to diagnose and repair our devices has, at least for now, overridden the patent exhaustion doctrine that theoretically protects this right. With that in mind, it should not come as

a surprise that the oppressive power afforded to technology giants by the DMCA has been challenged in courts of law, as will be discussed in further detail below.

Section III: The Right to Repair Movement

In response to these anti-repair practices, recent decades have seen the emergence of a global right to repair movement. It brings together tinkerers, hobbyists, repair professionals, policy advocates, environmental activists, and everyday consumers. [38] This coalition operates both locally and globally to share parts, tools, information, and repair techniques (especially through online forums and Fixit Clinics[39]) and, most importantly, to advocate for legislation protecting the right to repair in all jurisdictions in which it is found.

3.1 - Fight for Repair in the Legislature

Although there has been a strong push for right to repair legislation in the United States, the path has been made complicated by lobbying from tech companies. The rest of this section will briefly examine the history of attempts to codify the right to repair in various forms through legislation in the US.

In 2012, Massachusetts became the first state to take action preserving the right to repair when it passed the Act Protecting Motor Vehicle Owners and Small Businesses in Repairing Motor Vehicles. [45] Beyond its implications for vehicle repair, the Massachusetts automotive repair law has served as the template for a broader, nationwide effort to enshrine consumers' right to repair in state law [46] that has intensified in the past few years. In 2021, a total of twenty-seven states considered right to repair legislation, twelve of which were aimed specifically at agriculture equipment. In 2023, that number rose to thirty-three. Federal repair bills were also introduced in 2022 and 2023 in the US Senate and the House of Representatives respectively. [47] And as of February 2025, all 50 states have introduced right to repair legislation. [48] These bills have attempted to protect the repair rights of both consumers and IRPs in various ways. Some are narrowly tailored to protect access to repair information specifically for farming equipment,

in response to the more ubiquitous repair restriction practices by John Deere. [49] Others are more broad, such as Iowa's H.F. 556 introduced in March of 2017, which covers any and all "digital electronic products." [50] Finally, the Right to Repair Bill introduced in Washington in 2018 presents an approach that went a step further - it would prevent the sale of devices in that state which are not easily able to be repaired by independent repair providers. Washington's focus on legislating certain levels of reparability made it more comprehensive than any other bill yet proposed. [51]

Introducing legislation is one matter. Getting it passed in the face of the often prohibitively lengthy and contentious legislative process is another entirely – and in the case of repair, the problem has been exacerbated by tech corporations waging a fierce lobbying campaign against proposed laws protecting the right to repair. In Nebraska, for example, the 2017 Fair Repair Act was met with staunch opposition by tech companies like Apple and AT&T, as well as from John Deere and other agriculture-focused tech manufacturers and trade organizations. [52] This level of opposition is representative of the challenges facing the Right to Repair movement nationwide.

Despite this, the first half of the 2020s have seen some notable state-level successes. In 2020, Massachusetts voters were able to close a loophole in the 2012 statute protecting the right to repair for automobiles by voting "yes" on a ballot measure that would require car makers to share information they collect wirelessly about vehicles' mechanical health. Such information, known as telematics, can help anticipate problems before they arise. [53] Elsewhere, Washington was finally able to sign right to repair protections for consumers into law in the form of two bills: H.B. 1483 and S.B. 5680, both of which were signed into law in May of 2025. [54]

That said, the anti-repair coalition formed by technology giants remains a substantial barrier to legislative efforts to codify the right to repair for consumers and IRPs in both state and federal law, which brings us to the other major stage of the fight for repair: the courts.

3.2 – Fight for Repair in the Courts

Alongside the struggle to codify the right to repair in legislation, some consumer groups, in conjunction with the federal government, have taken it upon themselves to protect the right to repair through the judicial process. Before I illustrate some of these proceedings, here is a caveat: Much of the litigation undertaken in the name of protecting repair rights is still ongoing and, at the time of this writing, there is very little in the way of precedential court decisions from which inferences can be drawn. But what we can do is examine some of the most salient ongoing cases in this realm and what they might mean for the fight for repair in the courts going forward.

3.2.1 - Alliance for Automotive Innovation v. Campbell

The passage of the updated right to repair law for automobiles in Massachusetts in 2020 was followed by a slew of legal challenges by auto manufacturers, the last of which were dismissed in February of 2025, allowing the law to take effect. These challenges were brought through the Alliance for Automotive Innovation (“AAI”), a lobbying group representing more than forty major automakers and other tech manufacturers such as Ford, Mazda, Toyota, Mercedes, and many more. [55] AAI filed suit against Massachusetts Attorney General Andrea Joy Campbell, arguing that the law requiring the disclosure of telematic data was preempted by existing federal car safety laws – specifically, the Motor Vehicle Safety Act (“MVSA”) and several federal motor vehicle safety standards promulgated by the National Highway Traffic Safety Administration. This challenge failed, and the law was upheld. [56]

Despite being primarily concerned with questions of federal preemption and not directly addressing the implications of these questions on the existence or recognition of a “right to repair”, The District Court’s decision in *AAI v. Campbell* as the first major ruling in a slew of recent repair litigation should be interpreted as an instructive victory in the fight for repair in the courts. Regardless of what their actual motives may be for pushing back against the right to repair movement, tech corporations argue first and foremost that removing the repair restrictions they have levied on their devices through TPMs and embedded software poses risks for safety and data security. The AAI

decision sets the major precedent that courts disagree with this argument, and that state-level right to repair laws are workable in conjunction with existing federal regulations on the devices we own.

3.2.2 - Ongoing Deere Antitrust Litigation: Forest River Farms and the FTC

In January of 2022, multiple suits filed by farms across the country against John Deere were consolidated into a class action complaint in the US District Court for the Northern District of Illinois (Deere's home state) filed as *Forest River Farms v. Deere & Co.* The suit alleged, first and foremost, that Deere's repair restrictions on its farming equipment (as discussed in 3.1.1.3) violate the Sherman Antitrust Act "by monopolizing or attempting to monopolize the Deere Repair Services Market in a manner that harmed competition and injured the purchasers of such services by reducing choice and increasing prices in this market to supracompetitive levels." [57] The farms alleged also that Deere has leveraged its monopoly power over Deere software to tie sales of its tractors to sales of Deere Repair Services, in violation of Section 2 of the Sherman Act, and the farming giant has unjustly enriched itself by profiting from farmers' payment of supracompetitive prices for Deere Repair Services in violation of the antitrust laws. Alongside declaratory and injunctive relief, as well as treble and exemplary damages, the plaintiff farmers sought an order requiring Deere to make the necessary software available, at reasonable cost, to individuals and repair shops. [58] Three years later in January of 2025, the Federal Trade Commission filed its own suit, along with the states of Minnesota and Illinois, against Deere in the same district court, alleging much of the same. [59]

In both the 2022 class action suit led by Forest River Farms and the FTC action in 2025, Deere attempted to have the cases thrown out by the District Court. Motions to dismiss were filed by Deere's counsel in both cases and were each promptly denied by District Judge Iain Johnston. [55] Both cases are ongoing as of now, and no additional major ruling has been made in either. Regardless, this litigation still warrants mentioning because it shows how turning to litigation to fight back against the attack on repair, even if in the context of antitrust law rather

than protecting a conceptual “right to repair”, is a novel strategy by consumers and regulators that is rapidly gaining traction – and as seen from the AAI decision and the failed motions to dismiss by Deere, the results seem promising. Zooming out from its action against Deere in the District Court, the FTC along with the Department of Justice has also made ground in pushing back against the oppressive power of the DMCA in restricting repair.

3.3 – DMCA Repair Exemptions

As mentioned previously, copyright law – specifically, § 1201 of the DMCA – has been the critical legal instrument levied by tech manufacturers to restrict repair. This is because that provision normally prohibits bypassing technological protection measures (TPMs), even for lawful purposes like repair. However, the US Copyright Office’s triennial rulemaking process creates temporary exemptions (renewed every three years) to balance copyright protection with consumer rights. [61] Recent years have seen an increasing number of exemptions granted by the Copyright Office to § 1201 allowing circumvention of TPMs for repair purposes in the name of protecting repair. The first series of repair exemptions was granted in 2015 and allowed consumers and IRPs to circumvent the digital locks on motorized land vehicles. [62] Since then, the Copyright Office has expanded its repair exemptions to include smartphones and tablets, medical devices, industrial and some agricultural equipment, and commercial food preparation equipment as of the latest rulemaking process in 2024. [63] These exemptions have been advocated by the FTC and the DOJ, who in 2024 filed a joint comment with the Copyright Office supporting the renewal and expansion of repair exemptions. The joint comment argued that by limiting access to the data and software needed for independent repair and maintenance, TPMs can be used to squash competition for replacement parts, repair, and maintenance, which ultimately limits consumers’ and businesses’ choices and raises costs. [64]

Because the Copyright Office is technically a unit of the legislative branch, operating as a service unit of the Library of Congress, some may argue that the DMCA exemptions should be categorized under the legislative front of the fight for repair. I argue, however, that the

exemptions reflect not necessarily a change in the law but an updated interpretation of it, and moreover the two main government agencies argue that the DMCA exemptions should be categorized under the legislative front of the fight for repair. I argue, however, that the exemptions reflect not necessarily a change in the law but an updated interpretation of it, and moreover the two main government agencies advocating for the exemptions (the FTC and DOJ) operate as law enforcement agencies and often deal with litigation, therefore these exemptions fall more neatly under the litigation front. Regardless, the recent repair exemptions to § 1201 of the DMCA granted by the Copyright Office are a crucial victory for the right to repair movement, as advocates and regulators have been able to significantly weaken the primary legal bottleneck for protecting repair rights. But these exemptions are temporary and limited in nature, and thus ultimately represent only a temporary solution to the issue of the right to repair. With this in mind, let's move onto discussing the most effective strategy for protecting the right to repair going forward.

Section IV: Repair Going Forward: Strategic Litigation

Earlier I stated that it is a necessary assumption to make for purposes of writing this article that there does exist a “right to repair” embodied within US patent exhaustion jurisprudence. I would argue that this right exists, and the nomenclature of the “right to repair” movement in recent decades suggests that proponents of the movement would agree. But the only way to have this right explicitly acknowledged by the courts is through strategic litigation through a test case that incorporates the proper requirements for patent exhaustion but also represents a violation of the DMCA. Let's examine these factors one at a time.

4.1 - Strategic Litigation as a Force for Change

The power of strategic or “impact” litigation as a force for change in American society is too well-demonstrated historically to be denied. Some of our most critical civil rights that we enjoy today are the result of strategic litigation. These include the outlawing of racial discrimination through *Brown v. Board of Education* (1954), which was the culmination

of a strategic litigation campaign by the NAACP [65]; the constitutional right to privacy through *Griswold v. Connecticut* (1965), in which C. Lee Buxton of the Yale School of Medicine and Estelle Griswold of Planned Parenthood in Connecticut opened a birth control clinic in New Haven solely for the purpose of using the clinic to challenge the constitutionality of Connecticut's anti-contraceptive statute[66]; and the constitutional right to same-sex marriage through *Obergefell v. Hodges* (2015), in which groups of same-sex couples sued their relevant state agencies in Ohio, Michigan, Kentucky, and Tennessee to challenge the constitutionality of those states' bans on same-sex marriage or refusal to recognize legal same-sex marriages that occurred in jurisdictions that provided for such marriages. [67]

What these cases all have in common - and that the recent litigation efforts undertaken in the name of fighting back against repair restrictions thus far have lacked - is that they utilized grievances through test cases that sought to affect broad societal change. There is certainly something to be said about the risk of relying on strategic litigation in terms of short-circuiting the democratic process – that is, relying on a unilateral court order rather than elected representatives to protect our civil rights. But that is not a major concern of this article. The fact of the matter is that if the goal of the right to repair movement is indeed to protect a “right to repair”, then temporary exemptions to an oppressive statute can only go so far, and strategic litigation presents a promising opportunity to protect this right in a more meaningful and permanent way. The final section will discuss why and how this is the case.

4.2 – Patent Exhaustion vs. the DMCA

By all accounts, the principle of patent exhaustion on its face, and especially in its latest interpretation by the Supreme Court in the *Quanta* decision, protects the right to repair in most scenarios – see 1.2. The issue in recent years is that, as is often the case, technological innovation has outpaced the law, and tech manufacturers have made an end run around patent exhaustion by claiming copyright protection through the DMCA over the embedded software code that is increasingly essential to diagnose and repair the devices we use on a day-to-day basis. The patent on the device itself may be exhausted, but the

software essential to its function remains under the technological and legal control of the manufacturer. It is true that the copyright protection afforded to tech manufacturers by § 1201 of the DMCA has effectively overridden patent exhaustion, but this is the case only because no litigation has occurred yet that would force the courts to answer this question conclusively. When we consider the numerous exemptions that have been granted to § 1201 by the Copyright Office for repair purposes as covered in 2.4, as well as the general apparent sympathy of the courts to the right to repair movement thus far as covered in 2.3, it seems likely that if this question is eventually brought before a court, the ruling would bolster patent exhaustion and restore it to its former preeminence in dictating what manufacturers may or may not control over how we use their products. This is especially true when we consider what the Supreme Court said about method patents specifically in the *Quanta* decision. In the words of Justice Thomas, writing for a unanimous court: “It is true that a patented method may not be sold in the same way as an article or device, but methods nonetheless may be ‘embodied’ in a product, the sale of which exhausts patent rights.” [68] Applying this reasoning to TPMs, the questions arise: Would the software code essential to the function, diagnosis and repair of a car, tractor, or smartphone not exemplify a method for that device? And since that code is quite literally embedded into the device itself, would any patent or copyright protection on the code not also be exhausted by that device’s sale?

There is something to be said also about what the numerous repair exemptions granted to the DMCA say about the robustness of that statute. As stated previously, the DMCA was enacted almost thirty years ago, and the original intent of the statute – especially § 1201 – was mainly protecting the ownership rights of producers of digital media from unauthorized access to reproduce their works through decryption software. That tech firms have been able to levy the power of the statute to restrict third-party repair through TPMs is less an intentional goal and more an unforeseen implication. Perhaps this is why the Copyright Office has granted so many exemptions to it. But regardless, the question arises: if the Copyright Office is willing to grant so many exemptions to § 1201 of the DMCA, then what reason is there to not

rewrite or overturn the provision? My argument on this front is that this provision should be drastically reformed to eliminate the need for constantly having to grant and renew numerous exemptions for repair purposes for specific types of devices, or, if pursued through litigation, then it should be struck down entirely on grounds of violating the patent exhaustion principle and thus being unconstitutional as applied.

The right to repair movement thus far, both on the legislative and litigation fronts, has been rather decentralized. The initiative has attracted a diverse yet scattered coalition of sympathetic lawmakers, environmental activists, repair-industry organizations such as iFixit, and more general consumer-focused advocacy groups such as PIRG. [69] The drawback to this grassroots nature is that there is no single organization to consolidate the movement and pursue the most effective strategy going forward, such as what civil rights had in the NAACP. While lawmakers introduce state-level codified repair protections for certain categories of devices, other groups (see the previous discussion on *Forest River Farms v. Deere*) have taken it upon themselves to take legal action pursuant to existing antitrust law – both disjointed and temporary solutions in terms of defending repair as a fundamental right. Pursuing strategic litigation would solve this problem and, based on early rulings of the courts in recent repair restriction litigation as previously discussed, alongside the Supreme Court’s recent patent exhaustion jurisprudence, provides the best possible chance of providing a permanent protection for an actual “right” to repair. This of course would necessitate a much higher level of coordination between right to repair advocates for litigation purposes, especially in terms of utilizing a test case. This test case would have to satisfy the requirements for patent exhaustion but also involve a circumvention of TPMs in violation of § 1201 of the DMCA, with the primary goal being to argue that the DMCA provision is overridden by the doctrine of patent exhaustion.

As with any high-stakes litigation, there is, of course, a real risk of setting adverse precedent. However, in this case that risk is simply not large enough to preclude strategic litigation as a viable strategy. In terms of the previously discussed recent repair litigation, from AAI to

the Deere antitrust suits, all major rulings have come up in favor of repair. Perhaps it could be argued that the chances for strategic litigation pursuant to patent exhaustion are less certain, but this is only because this avenue has yet to be pursued. If recent jurisprudence regarding patent exhaustion in general (see *Quanta*) is anything to go by, then applying this principle to repair – an issue which courts thus far have apparently been sympathetic to – is a straightforward decision to make.

Moreover, as it stands, tech giants have been allowed free rein to use § 1201 of the DMCA as they please to lock down devices and inhibit repair or modification of any kind by the end user for the entirety of the product's life, making an end run around patent exhaustion entirely. As stated previously, this is the case only because the issue has not been brought to court. If strategic litigation of the kind advocated for in this article is undertaken and fails, thereby generating adverse precedent, then repair will be in the same position it is now, and the focus will have to be put solely on lobbying legislatures to update our outdated intellectual property laws and codify repair as a consumer protection. On the other hand, if such litigation were to prevail, either striking down the provision entirely or stripping its protection from TPMs, it would be perhaps the greatest victory for the right to repair movement in the US yet.

The avenue of strategic litigation for the right to repair movement is made more apparent by the nature of technological innovation on its face, and more specifically how it interacts with politics. It is hardly novel to say that technological innovation is outpacing the law. The appropriation of § 1201 of the DMCA by tech giants to claim copyright protection over software code embedded into physical devices is proof of this. While keeping in mind the risk of legislative appropriation, it is in situations like this where impact litigation may serve its most important purpose. The role of the courts, after all, is to interpret the law. A pristine example, although not explicitly strategic litigation, can be found in *In re Search of a Residence in Oakland*. Decided by the US District Court for the Northern District of California in January of 2019, this case dealt with a search and seizure warrant application by federal

authorities for a residence in Oakland, CA. As part of the application, federal authorities also sought the authority to compel any individuals present at the time of the warrant execution to unlock any digital device found via biometric means (i.e. a fingerprint scanner) for that device's contents to be searched. The District Court denied this warrant application on grounds of violating the Fifth Amendment privilege against self-incrimination. [70]

The Court reasoned that although the witness is not technically being compelled to speak, compelling the utilization of biometrics to unlock a device with potentially incriminating information on it is functionally equivalent to compelling a witness to reveal an alphanumeric passcode, thereby rendering it testimonial under the Fifth Amendment. [71] As part of her opinion, Magistrate Judge Kandis Westmore stated the following: "The challenge facing the courts is that technology is outpacing the law. In recognition of this reality, the United States Supreme Court recently instructed courts to adopt rules that 'take account of more sophisticated systems that are already in use or in development.'" [72] The same reasoning should be applied to TPMs and the DMCA.

4.3 – Patent Exhaustion vs. Contract Law (EULAs)

There is one major legal hurdle that has yet to be discussed, and which would present a serious challenge to the right to repair movement in the courts if strategic litigation is pursued – contract law. More specifically, End User License Agreements, commonly referred to as EULAs. An EULA is a type of contract between software publishers and end users, which governs the end user's right to use software, and is thus extremely important as it prescribes what consumers may and may not do with the product. Tech companies have long since adopted the strategy of weaponizing EULAs to explicitly preclude repair or modification of almost any kind by the end user. John Deere, for example, requires that all farmers that purchase their equipment sign an EULA that forbids nearly all repair and modification to farming equipment. [73]

Apple takes it a step further. Anyone who has purchased a MacBook is all too familiar with the experience of booting it up for the first time and

being greeted with a lengthy End User License Agreement. Most are also familiar with the routine of ignoring this EULA and dragging the slider on the screen to the bottom as fast as possible to activate the “Accept Terms and Conditions” button. But take the time to read it through, and you will find that although you have technically purchased the laptop itself, the software powering it has in fact only been licensed to you by Apple for reproduction, modification, publication and distribution of non-copyrighted materials, materials in which you own the copyright. The agreement also expressly forbids unauthorized modifications to the MacBook, such as by disabling hardware or software controls. [74]

The obstacle that EULAs present to the prospect of strategic litigation pursuant to patent exhaustion are apparent. Under this regime, it is not only TPMs but binding contract agreements that restrict the right to repair. How may we claim that a plaintiff’s property rights have been violated when they voluntarily agreed to not repair or modify the device? However, contract law is not the trump card to shut down any invocation of exhaustion that it may appear to be.

This article does not intend to present an exhaustive list of ways that repair advocates may overcome the contract obstacle in court. But upon preliminary review, there is a route to take that is quite simple: The principle of patent exhaustion is robust enough to override restrictive EULAs completely, rendering them null and void for purposes of repair and modification.

This suggestion may seem overly optimistic; perhaps even delusional. But consider the following: John Deere is far and away the pre-eminent manufacturer of commercial agricultural equipment, boasting a 70% market share within the industrial machinery and components industry during 2025. [75] Over in the world of personal computing, Apple may not boast the same market power on its own, but almost every major computer manufacturer, such as Lenovo and HP, now leverage similarly restrictive EULAs on all their products. [76]

In other words, the freedom of contract that may be invoked by tech giants in court to justify repair restrictions is not actually free. While a much more pragmatic interpretation of the law by a court would be

required, the argument can and should be made that end users are not “voluntarily” signing away their right to repair and modify what they own as they please in any real sense of the word.

But even ignoring these circumstances, what is even more crucial to note is that the *Quanta* decision (cited heavily throughout this article and which strategic litigation pursuant to patent exhaustion in the name of repair must also cite) also involved a breach of contract. Stipulated to by all parties to that case is the fact that LGE’s licensing agreement with Intel explicitly stated that the license does not extend to a product made by combining an Intel product with a non-Intel product, and that Intel must give its customers written notice of this. Stipulated to also is the fact that one of Intel’s customers – Quanta Computer, Inc. – disregarded this part of the agreement and manufactured computers using the licensed Intel parts with non-Intel parts. [77] A clear breach of contract was present, and yet the breaching party prevailed under patent exhaustion. The goal of the strategic litigation advocated for in this article is already to prove in a court of law that the exhaustion principle under the common law tradition has enough force to override TPMs and federal copyright law. To prove also that it overrides EULAs is little more than a minor adjustment, and the precedent is there.

Conclusion

Patent exhaustion jurisprudence supports the idea that the concept of an inherent right to repair does indeed exist. But because the grassroots political movement for protecting this right is – relatively speaking – very young, it shouldn’t come as a surprise that in its current form it is rather decentralized and hasn’t yet been able to decide on the most effective strategy to focus all its efforts on, despite the victories it has achieved in the legislature and the courts. That being said, because the ideal outcome for the recently emerged right to repair movement is to in fact gain recognition for an inherent “right” to repair the things we own, and because the courts thus far in proceedings pertaining to repair restrictions have been sympathetic to the movement, coupled with the fact that technological innovation and the DMCA have left open several questions as they pertain to patent exhaustion, strategic litigation is the most effective strategy going forward for protecting the right to repair in the United States.

Notes

- [1] Aaron Perzanowski, "Consumer Perceptions of the Right to Repair," *Indiana Law Journal* 96, no. 2 (May 1, 2020): 363.
- [2] *Id.* at 364.
- [3] *Id.*
- [4] Leah Chan Grinvald and Ofer Tur-Sinai, "Defending the Right to Repair," in *Feminist Cyberlaw*, ed. Meg Leta Jones and Amanda Levendowski (Berkeley: University of California Press, 2024), 30.
- [5] Perzanowski, "Consumer perceptions of the right to repair," 364.
- [6] Grinvald and Tur-Sinai, "Defending the Right to Repair," 29.
- [7] *Id.* at 30.
- [8] Aaron Perzanowski, *The Right to Repair: Reclaiming the Things We Own* (Cambridge: Cambridge University Press, 2022), 10.
- [9] "Patent Essentials," United States Patent and Trademark Office, February 18, 2025, www.uspto.gov/patents/basics/essentials.
- [10] Andrew T. Dufresne, "The Exhaustion Doctrine Revived? Assessing the Scope and Possible Effects of the Supreme Court's *Quanta* Decision," *Berkeley Technology Law Journal* 24, no. 1 (2009): 13.
- [11] Note that this is a separate term from, but closely related to, the "first sale" doctrine in copyright law, which limits a copyright owner's distribution right such that their right to control the sale of a particular copy of a work is terminated once the owner releases that copy to the public through a sale, gift, or loan. Kimberley Byer, "The Death of the First Sale Doctrine," *Journal on Telecommunications & High Technology Law* 11, no. 2 (Fall 2013): 392.
- [12] *Id.* at 392.
- [13] *Bloomer v. McQuewan*, 55 U.S. 539, 549 (1852).
- [14] *Adams v. Burke*, 84 U.S. 453, 456 (1873).

- [15] Dufresne, "The Exhaustion Doctrine Revived?" 12–13.
- [16] *Id.* at 12-13.
- [17] *Quanta Computer, Inc. et al. v. LG Electronics, Inc.*, 553 US 617, 625 (2008).
- [18] *Id.* at 625.
- [19] *Ethyl Gasoline Corp. v. United States*, 309 U.S. 436, 457 (1940).
- [20] *Quanta Computer, Inc.*, 553 U.S. 617, 629 (2008).
- [21] Shubha Ghosh, "The Continuing Right to Repair," *Berkeley Technology Law Journal* 37, no. 3 (2022): 1103.
- [22] *Wilson v. Simpson*, 50 U.S. (9 How.) 109, 123 (1850).
- [23] Liu, Benjamin. "Toward a Patent Exhaustion Regime for Sustainable Toward a Patent Exhaustion Regime for Sustainable Development." *Berkeley Journal of International Law* 32, no. 2 (December 2014).
- [24] Perzanowski, The Right to Repair, 129. "The sequential replacement of parts hints at a deeper puzzle that dates back at least 2,000 years. Plutarch described how the Athenians preserved the ship that carried Theseus home from Crete. 'They took away the old planks as they decayed, putting in new and stronger timber in their places.' Among philosophers, the ship of Theseus embodied the problem of identity over time. Was the preserved ship the same vessel Theseus captained, or was it a new ship altogether?"
- [25] *Mallinckrodt, Inc. v. Medipart, Inc.*, 976 F.2d 700 (Fed. Cir. 1992).
- [26] Janis, Mark D. "A Tale of the Apocryphal Axe: Repair, Reconstruction, and the Implied License in Intellectual Property Law." *Maryland Law Review* 58, no. 2 (1999)
- [27] *Id.*
- [28] Grinvald and Tur-Sinai, "Defending the Right to Repair," 29.
- [29] Perzanowski, *The Right to Repair*, 72-110.

[30] Nicholas A. Mirr, "Defending the Right to Repair: An Argument for Federal Legislation Guaranteeing the Right to Repair," *Iowa Law Review* 105, no. 5 (July 2020): 2393.

[31] *Id.* at 2393-2428.

[32] *Id.* at 10.

[33] "What Is the Cloud?," Cloudflare, accessed November 4, 2025, www.cloudflare.com/learning/cloud/what-is-the-cloud/.

[34] Samuel Greengard, "Internet of Things," *Britannica*, November 14, 2022, www.britannica.com/science/Internet-of-Things.

[35] Mirr, "Defending the Right to Repair", 2397.

[36] Iain Thomson, "HP Settles Lawsuit after Killing First Responder's Printers," *The Register*, March 19, 2025, www.theregister.com/2025/03/19/hp_printer_lawsuit_settled/; see also "Notice of Proposed Class Action Settlement," HP, December 24, 2024, support.hp.com/ie-en/document/ish_11775540-11775594-16.

[37] Perzanowski, *The Right to Repair*, 9.

[38] *Id.*

[39] Perzanowski, "Consumer Perceptions of the Right to Repair", 370.

[40] 17 U.S.C. § 1201 (2018).

[41] Perzanowski, "Consumer Conceptions of the Right to Repair", 370-371.

[42] Grinvald and Tur-Sinai, "Defending the Right to Repair.", 26-27. See also Jane Ginsburg, "Copyright Legislation for the 'Digital Millennium,'" *Columbia VLA Journal of Law & the Arts* 23 (1999): 137-179.

[43] Perzanowski, *The Right to Repair*, 10-11.

[39] *Id.* at 11.

[45] The Massachusetts 2012 bill required manufacturers of motor vehicles to “make available for purchase by owners . . . and by independent repair facilities the same diagnostic and repair information” as well as “all diagnostic repair tools” made available to dealers, on “fair and reasonable terms.” *Mirr*, “Defending the Right to Repair”, 2399. See also H. 4362, 187th Gen. Ct. (Mass. 2012).

[46] Perzanowski, “Consumer Perceptions of the Right to Repair”, 376.

[47] Gabriel Dominic Gomez, "Have Some Heart for the Heartland: A Call for a Federal Right to Repair Law," *Journal of Legislation* 50, no. 1 (January 2024): 149–178.

[48] Elizabeth Chamberlain, "Right to Repair Laws Have Now Been Introduced in All 50 US States," iFixit, February 24, 2025, www.ifixit.com/News/108371/right-to-repair-laws-have-now-been-introduced-in-all-50-us-states.

[49] Wyoming H.B. 0199, 2017 Legislature, General Session (2017).

[50] These bills define "digital electronic products" as "a part or product containing a microprocessor originally manufactured for distribution and sale in the United States," which would ostensibly include everything from a smartphone or computer to household appliances and cars, trucks, or tractors. Iowa H.F. 556, 87th General Assembly, Regular Session, § 1(3) (2017).

[51] *Mirr*, “Defending the Right to Repair”, 2402-2403.

[52] *Id.*, 2405. According to Senator Brasch, who introduced Nebraska's bill, Apple sent its representative Steve Kester to speak with her. Kester reportedly told Senator Brasch that if Nebraska enacted a Right to Repair law, it would become a “mecca” for bad actors and hackers. These technology and equipment companies were joined by groups like the Association of Equipment Manufacturers (“AEM”) and Equipment Dealers Association (“EDA”) who jointly authored a “Statement of Principles” which frames the “right to repair” movement narrowly and not as a “right to modify” movement. Their website claimed that the right to repair laws proposed in state legislatures “would jeopardize the safety and sustainability regulations governing

modern farm equipment." The website jointly operated by AEM and EDA claimed that, in their current form, the right to repair laws jeopardize: (1) safety, as users would be able to override safety features; (2) sustainability, as emissions standards could be circumvented; and (3) innovation, as permitting "access to source code would jeopardize manufacturers' intellectual property and stifle innovation." This website, at one point accessible under the domain R2RSolutions.org, has since been deactivated. The Fair Repair Act was indefinitely postponed in 2018.

[53] Callum Borchers, "Mass. Voters Say 'Yes' on Question 1, Expanding Access to Car Repair Data," WBUR News, November 3, 2020, www.wbur.org/news/2020/11/03/ballot-question-1-right-to-repair-pas-ses. With the passage of Question 1, automakers would be required to make mechanical data available to drivers through a mobile app and, with drivers' consent, to independent shops through a new database. This ballot measure was passed despite fierce opposition from automakers, who spent upwards of \$26 million in lobbying in ad campaigns alleging the measure as a cybersecurity risk.

[54] Washington H.B. 1483, Regular Session (2025); see also Washington S.B. 5680, Regular Session (2025). H.B. 1483 supports the servicing and repair of certain products with digital electronics in a secure and reliable manner to increase access and affordability for Washingtonians, while S.B. 5680 established a right to repair for mobility equipment (i.e., scooters and wheelchairs) for people with physical disabilities.

[55] "Our Members," Alliance for Automotive Innovation, accessed November 14, 2025, www.autosinnovate.org/about/our-members.

[56] *Memorandum of Decision*, Alliance for Automotive Innovation v. Andrea Joy Campbell, Attorney General of the Commonwealth of Massachusetts, United States District Court for the District of Massachusetts, February 11, 2025. AAI contended that the requirements of the Massachusetts law conflicted with the "make inoperative" provision of the MVSA, which prohibits OEMs from "knowingly [making] inoperative any part of a device or element of

design installed on or in a motor vehicle or motor vehicle equipment in compliance with an applicable motor vehicle safety standard." It further asserted that the Massachusetts law was preempted by various federal safety standards that concern acceleration, braking, steering and air bag systems because OEMs "have installed a variety of cybersecurity protections around regulated vehicle functions" and "[y]et the [Massachusetts law] requires motor vehicle manufacturers to remove cybersecurity protections." District Court Judge Denise Casper was unconvinced, holding: "AAI has not met its burden to prove that the [Massachusetts law] stands as an obstacle to the accomplishment and execution of the full purposes and objectives of Congress with respect to the MVSA... No provision in the MVSA directly addresses motor vehicle cybersecurity or access to motor vehicle diagnostic data, nor do any of the [safety standards] that AAI cites discuss cybersecurity protections, much less adopt a federal cybersecurity objective that conflicts with the [Massachusetts law]."

[57] "Class Action Complaint," *Forest River Farms v. Deere & Co.*, United States District Court, Northern District of Illinois, Eastern Division, January 12, 2022.

[58] *Id.*

[59] Tom Koch and Stephanie T. Nguyen, "FTC, States Sue Deere & Company to Protect Farmers from Unfair Corporate Tactics, High Repair Costs," Federal Trade Commission, January 15, 2025, www.ftc.gov/news-events/news/press-releases/2025/01/ftc-states-sue-deere-company-protect-farmers-unfair-corporate-tactics-high-repair-costs. See also "Complaint," *Federal Trade Commission, State of Illinois, and State of Minnesota v. Deere & Co.*, United States District Court for the Northern District of Illinois, Western Division, Case No. 3:25-cv-50017. The FTC alleged that, for decades, Deere's unlawful practices have limited the ability of farmers and independent repair providers to repair Deere equipment, forcing farmers to instead rely on Deere's network of authorized dealers for necessary repairs, and that this unfair steering practice has boosted Deere's multi-billion dollar profits on agricultural equipment and parts, growing its repair parts

business while burdening farmers with higher repair costs. In the words of then-FTC chair, Lina Khan: "The FTC's action today seeks to ensure that farmers across America are free to repair their own equipment or use repair shops of their choice—lowering costs, preventing ruinous delays, and promoting fair competition for independent repair shops."

[65] Rod Sutton, "Deere Motion to Drop Right to Repair Lawsuit Denied by Court," *Construction Equipment*, December 1, 2023, www.constructionequipment.com/equipment-maintenance-and-repair/news/33015818/deere-motion-to-drop-right-to-repair-lawsuit-denied-by-court. See also Todd Neeley, "Federal Court Rejects Deere's Motion to Dismiss FTC Right-to-Repair Lawsuit," *DTN Progressive Farmer*, June 11, 2025, www.dtnpf.com/agriculture/web/ag/equipment/article/2025/06/11/federal-court-rejects-deeres-motion. In the case of the class action, Deere argued that the lawsuit should be dismissed because farmer groups bringing the suit lacked antitrust standing and failed to "plausibly allege relevant markets." District Court Judge Iain Johnston disagreed, holding in a November 2023 ruling that standing does exist and that the plaintiffs had indeed alleged constitutional and antitrust standing, relevant markets, and the necessary requirements in the complaint. The story seems to repeat itself verbatim in 2025 with the FTC action. Deere filed a motion to dismiss, arguing that the statute of limitations had run out on the states' claims and that the states lacked standing. The company also again argued that the plaintiffs failed to plead a cognizable aftermarket, and that farmers had full knowledge of Deere's decades-long repair practices when they bought equipment. Judge Johnston again disagreed and denied the motion to dismiss, this time stating, "Sequels so rarely beat their originals that even the acclaimed Steve Martin couldn't do it on three tries."

[61] "Rulemaking Proceedings under Section 1201 of Title 17," U.S. Copyright Office, accessed November 14, 2025, www.copyright.gov/1201/.

[62] Mitch Stoltz et al., "2015 DMCA Rulemaking," Electronic Frontier Foundation, December 14, 2020, www.eff.org/tl/cases/2015-dmca-rulemaking.

[63] U.S. Copyright Office, "Rules and Regulations," *Federal Register* 89, no. 208 (October 28, 2024), www.govinfo.gov/content/pkg/FR-2024-10-28/pdf/2024-24563.pdf.

[64] Federal Trade Commission, "FTC and DOJ File Comment with the U.S. Copyright Office Supporting Renewal and Expansion of Exemptions Facilitating Consumers' and Businesses' Right to Repair Their Own Products," March 14, 2024, www.ftc.gov/news-events/news/press-releases/2024/03/ftc-doj-file-comment-us-copyright-office-supporting-renewal-expansion-exemptions-facilitating.

[65] *Brown v. Board of Education of Topeka* (1), 347 U.S. 483 (1954)

[66] *Griswold v. Connecticut*, 381 U.S. 479 (1965)

[67] *Obergefell v. Hodges*, 576 U.S. __ (2015)

[68] *Quanta Computer, Inc.*, 553 U.S. 617 (2008), 628.

[69] "About." *PIRG*, 19 Aug. 2025, www.pirg.org/about/. "Being a consumer advocate means being a champion for a marketplace that's more honest and transparent, and a society that's healthier, safer and more sustainable. It means advocating less waste in favor of reducing, reusing and recycling more. It means preventing toxic threats to our health and safety. And it means considering the impact of new products and technologies on the next generation, not just the next quarterly earnings report."

[70] *In re Search of a Residence in Oakland*, 354 F. Supp. 3d 1010 (N.D. Cal. 2019), 1. "The Government, however, also seeks the authority to compel any individual present at the time of the search to press a finger (including a thumb) or utilize other biometric features, such as facial or iris recognition, for the purposes of unlocking the digital devices found in order to permit a search of the contents as authorized by the search warrant. For the reasons set forth below, the Court finds that the Government's request runs afoul of the Fourth and Fifth Amendments, and the search warrant application must be DENIED."

[71] *Id.* at 3-4. "The Court finds that utilizing a biometric feature to unlock an electronic device is not akin to submitting to fingerprinting or

a DNA swab, because it differs in two fundamental ways. First, the Government concedes that a finger, thumb, or other biometric feature may be used to unlock a device in lieu of a passcode. In this context, biometric features serve the same purpose of a passcode, which is to secure the owner's content, pragmatically rendering them functionally equivalent.”

[72] *Id.* at 3.

[73] Mirr, “Defending the Right to Repair”, 2397.

[74] “Software License agreement for MacOS Sequoia”, Apple Inc., July 11, 2024. <https://www.apple.com/legal/sla/docs/macOSSequoia.pdf>.

[75] “DE’s Market Share Relative to Its Competitors, as of Q1 2026.” CSIMarket, June 16, 2026. <https://csimarket.com/stocks/competition/SEG2.php?code=DE>.

[76] “Lenovo License Agreement (COE-30002-01)”, Lenovo Support US, July 2022. <https://support.lenovo.com/us/en/solutions/ht100141-lenovo-license-agreement-l505-0009-06>. See also “End-User License Agreement.” HP, June 2021. https://support.hp.com/us-en/document/ish_4416646-4390016-16.

[77] *Quanta Computer, Inc.*, 553 U.S. 617, 629 (2008), 1. “A separate agreement (Master Agreement) required Intel to give its customers written notice that the license does not extend to a product made by combining an Intel Product with a non-Intel product, and provided that a breach of the agreement would not affect the License Agreement. Petitioner computer manufacturers (Quanta) purchased microprocessors and chipsets from Intel. Quanta then manufactured computers using Intel parts in combination with non-Intel parts, but did not modify the Intel components. LGE sued, asserting that this combination infringed the LGE Patents.”

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The Inference is the Search: AI-Driven Predictive Surveillance and the Fourth Amendment

Sheeliza Deoram

City University of New York - York College

**“ THE TRANSFORMATION OF
AGGREGATED PERSONAL DATA INTO A
PERSON-SPECIFIC RISK SCORE OR
IDENTITY MATCH REQUIRES NO LESS
THAN A WARRANT. ”**

Abstract

This article argues that when law enforcement uses an AI system to generate a person specific probabilistic inference and acts on that inference, it has conducted a search within the meaning of the Fourth Amendment requiring a warrant or established exception. Existing doctrine, from *Katz v. United States* through *United States v. Jones* to *Carpenter v. United States*, has moved progressively closer to this problem without resolving it. *Carpenter* held that the comprehensiveness of digital surveillance can trigger Fourth Amendment protection regardless of the third-party doctrine, but addressed only data collection, not the AI synthesis of collected data into actionable person-specific outputs. This article fills that gap. It develops four arguments: that the constitutional harm occurs when a person-specific inference is used as the basis for law enforcement action, not when the underlying data is collected; that AI inference is categorically different from human analysis in ways that make constitutional protection more urgent; that probabilistic outputs are structurally incompatible with the probable cause standard; and that AI systems' documented capacity to hallucinate with confidence makes judicial oversight more urgent, not less. It then proposes a narrow, judicially administrable standard requiring a warrant before law enforcement takes investigative action based on a person-specific AI output. The argument does not require courts to break new ground. It requires them to apply the logic of *Carpenter* to the next question *Carpenter* did not answer.

I. Introduction

In February 2019, Nijeer Parks was arrested for a crime he did not commit. Police identified him not through a witness or physical evidence, but through a facial recognition algorithm that matched his photo to a blurry surveillance image from a hotel gift shop in Woodbridge, New Jersey. The match was wrong. Parks had never owned a car and had never held a driver's license, yet he stood accused of shoplifting and fleeing the scene in a vehicle. He turned himself in to clear up the mistake and was arrested on the spot. He spent ten days in jail and paid approximately five thousand dollars to defend himself before the case was dismissed for lack of evidence. Parks is not alone. [1] Robert Williams in Detroit, Michael Oliver in New Orleans, and Porcha Woodruff, also in Detroit, were each wrongfully arrested after police acted on a facial recognition match without independent verification. In each case, the same constitutional question went unanswered: when law enforcement uses an AI-generated output to target a specific person and acts on that output, has it conducted a search under the Fourth Amendment?

The legal framework that should govern cases like Parks' remains undeveloped. The Fourth Amendment guarantees the right of people to be secure against unreasonable searches and seizures, and for decades courts have asked a single question when evaluating digital surveillance: was there a search when the data was collected? In *Carpenter v. United States*, the Supreme Court took a significant step forward, holding that police must obtain a warrant before accessing long-term cell phone location data and acknowledging that seismic shifts in digital technology may require rethinking foundational Fourth Amendment doctrine. [2] But *Carpenter* was deliberately narrow. The Court addressed data collection by third parties and explicitly declined to disturb conventional surveillance techniques or address other collection technologies. It said nothing about what happens after collection, when an AI system ingests that data and synthesizes it into a probabilistic inference targeting a specific individual. That inference, the output the government acts on, the score or match that puts a name on a warrant, falls entirely outside what *Carpenter* addressed. Every wrongful arrest driven by a facial

recognition match, every knock on a door justified by a risk score, occurs in a constitutional space the Court has not yet entered. This article enters it.

This article argues that when law enforcement uses an AI system to generate a person specific probabilistic inference and acts on that inference, it has conducted a search within the meaning of the Fourth Amendment, and that search requires a warrant or an established exception. The constitutional moment is not when the underlying data was collected. It is when the government relies on a person-specific output, an inference targeting a named individual, as the basis for action against that person. A raw dataset is inert. A list of location pings, purchase records, or facial images sitting in a database does not accuse anyone, flag anyone, or authorize action against anyone. An AI output attaching a risk score or identity match to a specific person is a different thing entirely. It is actionable state knowledge. The government can arrest on it, surveil on it, and prosecute on it. The person it targets is harmed by it. That harm, not the collection of data that preceded it, is where the Fourth Amendment's protection must attach.

This article proceeds in six parts. Part II explains how predictive AI surveillance systems operate, distinguishing place-based tools from person-based tools that generate individual risk scores and identity matches. Part III traces the Fourth Amendment's evolution from *Katz* through *Jones* to *Carpenter*, showing how each case moved doctrine closer to the problem AI inference presents without resolving it. Part IV develops the core argument: that the constitutional harm occurs when a person-specific inference is used as the basis for action, that AI inference is categorically different from human analysis, and that probabilistic outputs are structurally incompatible with the probable cause standard. Part V addresses the two most significant counter arguments. Part VI proposes a narrow, judicially administrable standard and considers whether that standard carries a practical remedy. Part VII concludes.

II. How Predictive AI Surveillance Works

Law enforcement agencies across the United States have integrated artificial intelligence into their investigative and patrol operations at a pace that has outrun the legal frameworks designed to govern them. This section does not attempt a comprehensive survey of AI surveillance technology. It establishes one factual premise on which the legal argument that follows depends: AI does not merely collect or organize data. It produces something new. That new thing, a risk score, a probability estimate, an identity match attached to a specific person's name, is what the government acts on, and it is what the law must reckon with.

Predictive AI surveillance tools fall into two broad categories. The first, place-based systems, analyze historical crime data to generate forecasts about where and when criminal activity is likely to occur. Platforms like PredPol, now rebranded as Geolitica, and ShotSpotter produce hotspot maps and acoustic gunshot alerts that police departments use to allocate patrol resources. These tools target locations and times, not individuals. The constitutional questions they raise, while real, are not the focus of this article.

The second category, person-based systems, is what this article addresses. These tools analyze data about specific individuals and produce outputs that target those individuals for police attention. The Chicago Police Department's Strategic Subject List, widely known as the Heat List, is the most documented example. The system assigned every person in its database a numeric risk score between zero and five hundred, calculated from factors including prior arrests, age, victimization history, and connections to others on the list. [3] A person with a high score could be visited at home by officers, placed in intervention programs, or subjected to intensified surveillance, all on the basis of a number generated by an algorithm. The score was not a finding that the person had done anything. It was a prediction about what they might do. By 2019, when the program was discontinued, the list contained approximately four hundred thousand names. [4] Facial.

recognition systems operate on a similar logic. These tools scan an image, typically from surveillance footage, and compare it against a database of photographs to generate a ranked list of potential matches. The system does not identify a suspect. It produces a probability: this image resembles this person in the database at this level of confidence. Detroit's Project Green Light program and similar systems in use across the country feed facial recognition outputs directly into investigative workflows. When a detective receives a match, what they receive is not evidence. It is an inference, a statistical judgment produced by a machine that the person in the surveillance image is the person in the database. In the cases of Robert Williams, Nijeer Parks, Michael Oliver, and Porcha Woodruff, that inference was wrong. In each case, police acted on it anyway. [5]

What unites these systems is the nature of their output. Before the algorithm runs, there is data: images, records, location pings, arrest histories. That data does not point at anyone. After the algorithm runs, there is something qualitatively different: a number attached to a name, a match attached to a face, a judgment that this specific person warrants police attention. That transformation, from raw data to person-specific inference, is the constitutional event this article argues the Fourth Amendment must address. The data that fed the algorithm may have been collected lawfully. The inference the algorithm produced is a different thing entirely, and it is the inference, not the data, that put Nijeer Parks in a jail cell for ten days. Understanding why that distinction matters constitutionally requires first tracing how Fourth Amendment doctrine has evolved to meet earlier generations of surveillance technology.

III. Fourth Amendment Doctrine and Its Digital Evolution

The Fourth Amendment provides that the right of the people to be secure in their persons, houses, papers, and effects against unreasonable searches and seizures shall not be violated, and that no warrants shall issue but upon probable cause. For most of American history, courts applied this guarantee to physical intrusions: an officer

entering a home, searching a car, or seizing papers. The digital age forced a reckoning. Three cases define the arc of that evolution. Each moved doctrine closer to the problem AI inference presents. None solved it.

A. Katz v. United States, 389 U.S. 347 (1967)

The modern Fourth Amendment framework begins with *Katz v. United States*, in which the Supreme Court held that the Amendment protects people, not places. Federal agents had attached a listening device to the outside of a public phone booth and recorded Charles Katz's end of a conversation without physically entering the booth. The Court held this constituted a search. Writing separately, Justice Harlan II articulated the test that has governed Fourth Amendment analysis ever since: a person enjoys constitutional protection when they have a subjective expectation of privacy that society is prepared to recognize as reasonable. [6]

Katz was a landmark, but its framework was built for a different kind of *surveillance* problem. The reasonable expectation of privacy test asks whether an individual had a reasonable expectation of privacy with respect to a specific act of government observation. It was designed for cases involving a defined moment of intrusion. It was not designed to address what happens when thousands of individually unremarkable data points, none of which would trigger constitutional protection on its own, are aggregated over months and synthesized by an algorithm into a probabilistic judgment about a specific person. *Katz* gives us the baseline. It does not give us the answer.

B. United States v. Jones, 565 U.S. 400 (2012)

United States v. Jones advanced the doctrine significantly, though the most important reasoning came not from the majority but from the concurrences. The majority held that attaching a GPS tracking device to Antoine Jones's vehicle and monitoring its movements for twenty-eight days constituted a trespassory search under the Fourth Amendment. [7] Five justices pressed the harder question in separate writings: what about surveillance that involves no physical intrusion at all? Justice Sotomayor wrote that GPS monitoring generates a precise,

comprehensive record of a person's movements that reflects a wealth of detail about her familial, political, professional, religious, and sexual associations. [8] Justice Alito, joined by three others, agreed that long-term surveillance of this kind was constitutionally different in kind from any single act of observation. [9]

Together these concurrences articulated what scholars have called the mosaic theory: that aggregating individually innocuous data points over time creates a constitutionally significant picture of a person's life, even if each individual point would not trigger Fourth Amendment protection standing alone. [10] When an AI system aggregates months of location data, purchase records, arrest histories, and social network connections to generate a risk score, it is doing exactly what the *Jones* concurrences warned about, and doing it faster, at greater scale, and with less human judgment than any detective could replicate. *Jones* brought doctrine to the edge of this problem. It did not cross it.

C. Carpenter v. United States, 585 U.S. 296 (2018)

Carpenter v. United States is the essential foundation for any Fourth Amendment argument about AI surveillance. The Court held that the government's acquisition of seven or more days of cell-site location information constitutes a search requiring a warrant, rejecting the argument that the third-party doctrine eliminated any constitutional protection for records voluntarily shared with a service provider. [11] Chief Justice Roberts acknowledged that seismic shifts in digital technology may require a different application of the Fourth Amendment's principles and located the constitutional interest not in the abstract fact that data existed somewhere but in the government's ability to chronicle a person's past movements and access what he called the privacies of life. [12]

But the *Carpenter* Court was emphatic about the limits of its holding. The opinion does not disturb conventional surveillance techniques and does not address other collection technologies. [13] Critically, it said nothing about what happens after collection, when an AI system takes that data and synthesizes it into a probabilistic inference targeting a

specific person. *Carpenter* addressed the moment data enters government hands. It did not address the moment an algorithm transforms that data into an actionable judgment about who you are, what you might do, and whether you should be the subject of law enforcement attention. Every person-based AI surveillance system in operation today produces outputs in that gap. The argument that follows explains why the Constitution must reach them.

IV. The Inference Is the Search: Why AI-Generated Outputs Constitute a Fourth Amendment Search

The preceding sections established two things: that person-based AI surveillance systems produce new, person-specific outputs that serve as the operative basis for law enforcement action, and that existing Fourth Amendment doctrine has moved progressively toward this problem without resolving it. This section makes the affirmative case. The argument proceeds in four steps, each building on the last.

A. The Constitutional Harm Occurs When the Inference Is Used, Not When the Data Is Collected

The first and most fundamental argument is that the Fourth Amendment's protection must attach when law enforcement relies on a person-specific AI output to act against a named individual, not at the earlier moment when the underlying data was collected. This argument follows directly from the logic of *Carpenter*, properly understood. It also requires being precise about where, within that later window, the constitutional line falls. It does not fall at the moment an algorithm generates an inference. It falls at the moment that inference becomes the operative basis for government action against a specific person. Generation without use is constitutionally inert in the same way collection without inference is inert: a database of matches no officer has acted on accuses no one. It is the act of relying on a person-specific output, to detain, arrest, surveil, or otherwise target an individual, that converts an algorithmic byproduct into a Fourth Amendment event.

A raw dataset is constitutionally inert. A database containing millions of facial images, location records, arrest histories, and social network

connections does not, by virtue of its existence, accuse anyone, flag anyone, or authorize action against anyone. The data points that fed the algorithm targeting Nijeer Parks had been collected and stored without constitutional incident. No Fourth Amendment violation occurred when that data was gathered. The violation occurred when the facial recognition system synthesized those inputs into a single output: this person is the suspect. That output was what the police acted on. That output was what put Parks in a cell.

This distinction between collection and use maps directly onto the analytical move the Supreme Court made in *Carpenter*. The Court located the constitutional interest not in the existence of data but in the government's capacity to transform that data into actionable knowledge about a specific person. [14] An AI inference is precisely that transformation, executed at a scale, speed, and level of comprehensiveness that no human analyst could replicate. If the transformation of cell-site data into a movement map required a warrant in *Carpenter*, the transformation of aggregated personal data into a person-specific risk score or identity match requires no less.

Paul Ohm has argued that *Carpenter* creates a rule of technological equivalence: if a technology gives police the power to gather information equivalent to a recognized Fourth Amendment search, using that technology is also a search. [15] The equivalence here is direct. The output of a facial recognition system is functionally equivalent to a detective pointing at a photograph and saying that is the person who committed this crime. The output of a risk-scoring algorithm is functionally equivalent to a detective saying this person is likely to commit a crime. Both outputs are actionable. Both authorize law enforcement intervention against a named individual. The Constitution does not permit the detective to make that judgment without probable cause. It should not permit the algorithm to do so either.

It is worth being explicit about what this argument does and does not claim, since the two can be conflated. It does not follow that every act of algorithmic inference is itself a search requiring prior judicial authorization. An AI system may generate outputs continuously, and the

government does not conduct a search each time an algorithm quietly returns a probability that no one acts on. What triggers the Fourth Amendment is the government's reliance on that output as the operative basis for action against a specific person, an arrest, a stop, a search, a knock at the door. This is not a retreat from the article's central claim; it is a specification of it. The inference is the search only once it is put to use. The title reflects the point at which the constitutional harm crystallizes, not a claim that generation and use are constitutionally identical moments requiring identical judicial process. Part VI develops this operative-basis principle into a standard courts can apply.

B. AI Inference Is Categorically Different From Human Analysis

The most intuitive objection to the argument advanced in Part A is that AI inference is simply a faster, more efficient version of what detectives have always done. A detective who examines a surveillance photograph and identifies a suspect has not conducted a new search. Neither, the argument goes, has a facial recognition system that does the same thing more quickly. This objection fails for three independent reasons.

The first reason is scale. A human detective reviewing evidence operates within the constraints of human cognition: limited working memory, finite time, and the practical inability to cross-reference more than a handful of data sources simultaneously. An AI system faces none of these constraints. The Chicago Strategic Subject List processed data on approximately four hundred thousand individuals simultaneously, cross-referencing arrest histories, victimization records, age, and social network connections to generate individualized risk scores. [16] No human analyst could replicate this process. When the Supreme Court held in *Carpenter* that the third party doctrine could not be mechanically applied to digital data because the nature of the intrusion was qualitatively different from anything the doctrine's authors contemplated, it was making exactly this point. [17]

The second reason is speed. Human analysis involves deliberation. A detective who reviews evidence, forms a hypothesis, and acts on it has engaged in a process that involves human judgment at each step, creating at least the possibility of pause and error correction before

action is taken. An AI system produces its output instantly and delivers it in a form that invites immediate action. There is no deliberative interval between the algorithm's conclusion and the officer's response. When Detroit police received a facial recognition match in the Robert Williams case, they acted on it the same day, without independent verification, and without any mechanism for Williams to challenge the basis for the identification before his arrest. [18]

The third reason is opacity. A human detective who forms a conclusion about a suspect can, at least in principle, explain the basis for that conclusion. That reasoning can be tested and subjected to adversarial scrutiny. The reasoning behind an AI output frequently cannot. The COMPAS recidivism algorithm used in *State v. Loomis* was proprietary. The defendant could not access the weights assigned to the factors that produced his risk score, could not challenge the data inputs, and could not test the model's validity as applied to his circumstances. [19] The Wisconsin Supreme Court upheld the use of the score at sentencing despite acknowledging these concerns. [20] The opacity problem is not incidental to AI surveillance systems. It is structural. When the government acts on a black box output to deprive a person of liberty, the Constitution's requirement that searches be reasonable demands more than the assurance that the box exists and produced a number.

Taken together, scale, speed, and opacity distinguish AI inference from human analysis in ways that make it more constitutionally dangerous, not equivalent. The *Carpenter* Court recognized that when a technology enables surveillance of a comprehensiveness that would have been impossible in an earlier era, the old rules do not automatically apply. AI inference is the technology for the next generation of Fourth Amendment cases.

C. Probabilistic Outputs Are Structurally Incompatible With Probable Cause

The third and most original argument is that AI-generated probabilistic outputs cannot satisfy the probable cause standard, and that acting on them as if they could represent a structural violation of the Fourth Amendment's core requirement of individualized suspicion.

Probable cause requires a fair probability, based on specific articulable facts, that a particular person has committed or will commit a particular crime. [21] The standard is individualized: it attaches to a specific person based on specific facts about that person's conduct or circumstances. It is not satisfied by a showing that a person belongs to a group that tends toward criminal activity, or that a person's characteristics correlate statistically with past offending. [22]

An AI risk score does not produce probable cause. It produces what Bernard Harcourt has called actuarial suspicion: a statistical judgment that a person resembles, in some algorithmically determined way, a population that has historically offended. [23] The COMPAS algorithm, ProPublica found, correctly predicted violent recidivism in only twenty percent of cases, meaning that for every person accurately flagged as likely to reoffend violently, four people were flagged who would not. [24] A fifty-three percent overall accuracy rate is only marginally better than a coin flip. [25] These are not the statistics of probable cause. They are the statistics of a guess dressed in algorithmic clothing.

This is not an objection to probability as such. Probable cause is itself a probabilistic standard, and courts already treat other probabilistic inputs, a trained narcotics dog's alert, an informant's tip, a drug-courier profile, as capable of contributing to it. What distinguishes those accepted inputs from an algorithmic risk score is not that they involve probability but how that probability is generated, tested, and corroborated. A canine's alert is anchored to a specific search of a specific vehicle or bag, and its reliability is tested against the dog's certification and field performance record. An informant's tip is weighed under the totality-of-the-circumstances test this Court adopted in *Gates* itself, which asks not whether the tip is probabilistic but whether the informant's basis of knowledge and veracity can be independently corroborated. [26] Both inputs are event-specific: they say something about this stop, this vehicle, this encounter. An algorithmic risk score is not. It says something about a population the subject statistically resembles, generated in advance of and independent from the specific encounter, and its reliability is typically unknown to the officer relying on

it and, as *Loomis* illustrates, sometimes unknown even to the court reviewing it. That is the distinction this article's probable cause argument turns on, not that the output is a probability, but that it is a categorical probability substituting for an individualized one, generated by a process that neither the officer nor the person it targets can test.

This distinction tracks, and extends, existing scholarship on predictive policing and reasonable suspicion. Andrew Ferguson has argued that predictive policing outputs function much like anonymous tips and should be subject to comparable corroboration requirements before they support a stop. [27] Elizabeth Joh has described the discretion embedded in algorithmic suspicion systems as a new and largely unregulated form of surveillance discretion. [28] Both bodies of work address reasonable suspicion, the lower threshold governing investigative stops, and both are principally concerned with place-based and area-based predictive tools. The move this article makes is twofold: from reasonable suspicion to probable cause, where the Fourth Amendment's particularity requirement is at its most demanding and has the least tolerance for a categorical substitute, and from area-based outputs to the person-based, name-attaching outputs, facial recognition matches and individual risk scores, that are this article's focus. An algorithmic output can inform probable cause the way a tip can, as one corroborated data point among several. It cannot substitute for probable cause the way this article's opening cases show it currently does.

The problem runs deeper than accuracy. Even a perfectly accurate probabilistic output would not satisfy probable cause, because probable cause is not a statistical concept. It does not ask whether a person is more likely than not to have committed a crime. It asks whether there are specific articulable facts about this person that justify the inference that they did. [29] A risk score answers a different question entirely: given what this person looks like statistically, how similar are they to people who have offended? The person is being evaluated as an instance of a category, not as an individual. The Fourth Amendment's particularity requirement is structurally incompatible with this mode of reasoning. [30]

This incompatibility is not merely theoretical. In the cases of Robert Williams, Nijeer Parks, Michael Oliver, and Porcha Woodruff, police obtained or executed arrest warrants on the basis of facial recognition matches without independent verification of the match's reliability. [31] In each case, the algorithmic output substituted for the individualized probable cause determination the Fourth Amendment requires. The result was not probable cause. It was actuarial suspicion translated into an arrest warrant, with no judicial check on the translation.

D. The Hallucination Problem: When AI Fabricates With Confidence

The structural incompatibility between AI outputs and constitutional standards is compounded by a well-documented failure mode unique to AI systems: hallucination. AI systems, particularly large language models and pattern-matching tools, do not merely make mistakes of degree. They generate outputs that are internally confident, authoritative in presentation, and factually wrong in ways that a human analyst would not be. The system does not flag its own uncertainty. It produces a result and presents it as a conclusion.

The legal system has already confronted this problem directly. In *Mata v. Avianca, Inc.*, attorneys submitted a brief to a federal court containing six case citations generated by ChatGPT. The cases did not exist. The AI had fabricated the case names, the reporters, the courts, and the holdings, and presented them in the standard format of real legal authority. [32] When confronted, counsel admitted they had not verified the citations against any legal database. The court sanctioned the attorneys for submitting AI-generated work product as reliable without independent verification, holding that the duty of candor to the tribunal and the obligation of professional competence require more than deference to an algorithm's output. [33]

The *Mata* case is not merely an anecdote about careless lawyers. It is a window into the specific danger AI outputs pose when treated as authoritative conclusions rather than starting points for human verification. A facial recognition system does not produce a match and then note its own error rate. A risk-scoring algorithm does not attach a

confidence interval that the reviewing officer is trained to interpret. The output arrives as a number or a name, and the institutional incentives of law enforcement, the pressure to clear cases, to act on available intelligence, to trust the technology procurement process that acquired the tool, all push toward treating that output as reliable. *Mata* illustrates what happens when that trust is misplaced in a context where the stakes are high, and the error is visible. In the wrongful arrest cases, the error was equally dramatic and the institutional pressures equally real, but the person who paid the price was not a lawyer facing sanctions. It was Nijeer Parks spending ten days in a cell.

The hallucination problem strengthens the constitutional case for a warrant requirement. A warrant does not merely require that police have evidence. It requires that a neutral magistrate evaluate that evidence before action is taken. In the context of AI-generated outputs, that requirement serves a function it has always served, but with heightened urgency: it inserts a human check between an algorithmically produced conclusion and the deprivation of a person's liberty. If the legal profession's response to AI hallucination is that lawyers must verify AI outputs before submitting them to courts, the Constitution's response to AI surveillance must be at least as demanding: that law enforcement must present AI outputs to a magistrate before acting on them against a named individual.

V. Counterarguments and Responses

No serious argument goes unanswered, and the argument advanced in this article faces two significant objections. The first challenges the premise that AI inference is a constitutionally distinct act. The second challenges the remedy. Both deserve honest engagement, and neither survives scrutiny.

A. *Lawful Collection Immunizes Downstream Analysis*

The most doctrinally grounded objection is that the Fourth Amendment violation, if any exists, occurred when the data was collected, not when it was analyzed. On this view, AI inference is simply investigative work

product: the government obtained data lawfully, a system processed it, and an officer drew a conclusion. A detective who reviews a lawfully obtained surveillance photograph and identifies a suspect has not conducted a new search. Neither, the argument goes, has a facial recognition system that performs the same function more efficiently.

This objection has surface plausibility, but it proves too much and is inconsistent with the reasoning of *Carpenter*. Taken to its logical conclusion, it would permit the government to compile every lawfully obtained data point about every person in the country into a single comprehensive profile and act on the resulting inferences without constitutional constraint, so long as each individual piece of data was collected lawfully. That cannot be the law. The *Carpenter* Court explicitly rejected the argument that lawful collection immunizes government use of the resulting data, holding that the comprehensiveness and intimacy of the information obtained, not the technical legality of its collection, determines whether a Fourth Amendment search has occurred. [34] The mosaic assembled from individually innocuous data points is qualitatively different from any single point, and the Constitution must reach the mosaic even when it cannot reach the points.

Applied to AI inference, the logic is identical and, if anything, stronger. The output of a person-based AI surveillance system is not merely a mosaic. It is a mosaic that has been interpreted, weighted, and translated into an actionable judgment about a specific individual. The raw data did not accuse Nijeer Parks. The algorithm did. The constitutional harm is located precisely where this article locates it: at the moment the inference is produced and acted upon.

Orin Kerr has argued that the sequential approach to Fourth Amendment analysis, evaluating each investigative step individually rather than treating the aggregate as a constitutional event, is essential to doctrinal coherence. [35] Kerr's concern is real: a rule that treats aggregation itself as constitutionally significant could sweep too broadly, capturing ordinary investigative practices that pose no serious

threat to privacy. [36] But this article's proposed standard does not adopt the mosaic theory in its broadest form. It does not argue that aggregation alone triggers Fourth Amendment protection. It argues that the production of a person-specific AI output that forms the operative basis for investigative action is the constitutional event. That is a narrower and more administrable rule than the mosaic theory, and it directly addresses Kerr's administrability concern without abandoning the insight that motivates it.

B. Warrant Requirements Would Cripple Law Enforcement

The second objection is practical rather than doctrinal. Requiring a warrant before law enforcement acts on an AI inference, the argument goes, would impose an impossible burden on agencies that rely on these tools for time-sensitive operations. Predictive policing systems generate thousands of outputs continuously. Facial recognition systems return matches in seconds. Requiring judicial approval before acting on each output would eliminate the operational utility of these technologies.

This objection mischaracterizes the rule this article proposes. The proposed standard does not require a warrant every time an AI system produces an output. It requires a warrant, or an established Fourth Amendment exception, before law enforcement takes person-specific investigative action based on that output. A police department may use a place-based predictive policing system to allocate patrol resources without constitutional constraint. A department may run a facial recognition search without a warrant, because running the search is not the constitutional event. The warrant requirement attaches at the moment the department acts on the person-specific result, using it as the operative basis to detain, arrest, or surveil a named individual.

Nor is this requirement operationally unworkable. The Fourth Amendment already requires probable cause and, in most circumstances, a warrant before police arrest a named individual. The rule this article proposes adds one requirement to that existing

framework: that the AI output on which the arrest is based cannot substitute for the probable cause determination. Police may still use AI tools. They may still act quickly in exigent circumstances under existing exceptions to the warrant requirement. What they may not do is treat an algorithmic output as the equivalent of probable cause and act on it without independent verification or judicial oversight. That is not a new burden. It is the existing constitutional framework applied to a new kind of evidence.

The Supreme Court rejected a nearly identical efficiency argument in *Riley v. California*, when it held that police must obtain a warrant before searching a cell phone incident to arrest, despite law enforcement's argument that immediate access to phone data was operationally essential. [37] The Court acknowledged the burden but held that the privacy interests at stake outweighed the efficiency concern. [38] The privacy interests implicated by person-based AI surveillance are at least as significant as those implicated by a cell phone search, and the potential for harm, as demonstrated by the wrongful arrests documented in this article, is correspondingly severe. Efficiency is a legitimate consideration in Fourth Amendment balancing. It is not a trump card.

VI. A Proposed Standard

Having established that person-based AI inference constitutes a Fourth Amendment search, and having answered the principal objections to that conclusion, this article turns to the practical question: what rule should govern law enforcement use of AI-generated outputs? The standard proposed here is intentionally narrow. It identifies the minimum that existing constitutional doctrine requires, states that minimum as a clear and administrable rule, explains why courts can adopt it without breaking new ground, and confronts directly whether the rule would carry any practical remedy once existing suppression doctrine is taken into account.

A. The Proposed Rule

When law enforcement uses an AI system to generate a person-specific

inference that forms the operative basis for investigative action against that individual, that inference constitutes a search within the meaning of the Fourth Amendment and requires either a warrant supported by probable cause or an established Fourth Amendment exception. Three elements of this rule require definition. The first is person-specific inference. This element distinguishes the outputs this rule covers from those it does not. A place-based predictive policing system that identifies a neighborhood or a time window as high-risk does not generate a person-specific inference. A hotspot map does not name anyone. A facial recognition match that attaches an identity to a named individual does. A risk score that assigns a number to a specific person in a database does. The rule applies when the AI output identifies and targets an individual, not when it identifies and targets a location or a population.

The second element is the operative basis for investigative action. This element limits the rule to cases where the AI output is the reason police act, not merely one consideration among many in a broader investigation. A detective who has developed probable cause through traditional investigative work and who uses a facial recognition match as one confirming data point among several is not acting on an AI inference as an operative basis. The rule attaches when the AI output is what moves the investigation from suspicion to action against a specific named individual, when it is, as it was in the Parks and Williams cases, the thing that generates the warrant.

The third element is that the rule uses existing Fourth Amendment framework rather than creating new doctrine. It does not require courts to adopt the mosaic theory, to abandon the third party doctrine entirely, or to impose technology-specific regulations that would require ongoing judicial management of evolving AI systems. It asks only that courts apply the warrant requirement and the probable cause standard, tools already in their possession, to a new category of evidence.

B. What the Standard Does Not Require

It is equally important to be precise about what this standard does not

require. The standard does not require that AI surveillance tools be accurate. Accuracy is an important policy concern, but it is not a Fourth Amendment category. A warrant issued on the basis of an AI output that has been subjected to proper judicial scrutiny and evaluated against independent corroborating evidence would satisfy this standard even if the underlying AI system has a meaningful error rate, provided the totality of the evidence established probable cause.

The standard does not require that AI tools be auditable or that their source code be disclosed. It requires judicial oversight of the decision to act on an AI output, not judicial inspection of the algorithm that produced it. And the standard does not prohibit the use of AI surveillance tools in law enforcement. It requires that before a person's liberty is constrained on the basis of an AI-generated output, the government satisfy the same constitutional standard it must satisfy before constraining liberty on any other basis: probable cause, evaluated by a neutral magistrate, with the specific individual identified with particularity. That is not a new requirement. It is the Fourth Amendment.

C. Why Courts Can Adopt This Standard

The proposed standard is judicially administrable because it requires no technological expertise to apply. A court asked to evaluate whether police acted on a person-specific AI inference as the operative basis for investigative action need not understand how the algorithm works, what data it used, or how it weighted its inputs. It need only ask two questions: did the AI output identify a specific individual, and did police use that output as the reason to act against that individual? If the answer to both questions is yes, the Fourth Amendment requires a warrant or an exception. If the answer to either question is no, the rule does not apply.

The standard is also consistent with the trajectory of Supreme Court doctrine. *Katz* established that the Fourth Amendment follows the person, not the place. *Jones* established that long-term aggregation of data creates constitutionally significant information that no single data

point would. *Carpenter* established that the comprehensiveness of digital surveillance can trigger Fourth Amendment protection regardless of the third-party doctrine. Each of these holdings reflects the same underlying principle: that the Constitution must reach new technologies when those technologies enable the government to intrude on the privacies of life in ways the Amendment was designed to prevent. [39] The proposed standard is the next application of that principle. It does not ask courts to change direction. It asks them to keep going in the direction they have already chosen.

D. Remedy: Why the Good-Faith Exception Does Not Swallow the Rule

A rule without a remedy is an invitation, not a requirement. The standard proposed above only has practical bite if violating it produces consequences, and the most immediate objection is that it will not. Under *United States v. Leon* and *Herring v. United States*, evidence obtained through a Fourth Amendment violation is not excluded where police acted in objectively reasonable reliance on a warrant, or in reasonable reliance on a system or record they did not know was flawed. [40] Wisconsin's highest court sidestepped a strikingly similar concern in *Loomis* by treating COMPAS's proprietary weighting as a limitation the sentencing court could work around through a written advisement of the tool's limitations, rather than a defect requiring exclusion of the score itself. [41] An officer who acts on a facial recognition match or a risk score without knowing its underlying error rate has a substantial argument for good-faith reliance, and if that argument prevails as a matter of course, the warrant requirement proposed here would rarely produce suppression, whatever its doctrinal force on paper.

This concern is real, but it does not defeat the proposed standard, for two reasons. First, the good-faith exception protects reasonable reliance on existing authorization. It does not protect the absence of authorization altogether. *Leon* excused police who executed a warrant a neutral magistrate had already approved. *Herring* excused a recordkeeping error inside an otherwise routine, lawfully authorized arrest. [42] Neither excuses proceeding without a warrant at all where

the Fourth Amendment required one. Once this article's standard is adopted, an arrest built on an unwarranted AI-driven inference is not a good-faith error committed inside an authorized search. It is the absence of the authorization the Fourth Amendment demands, and that absence is closer to the conduct Leon and Herring leave outside their protection than to the conduct those cases were designed to excuse.

Second, to the extent courts remain inclined to extend good-faith protection to officers who reasonably believed an AI output was accurate, that inclination is itself a reason to require the warrant step this article proposes, not a reason to abandon it. A magistrate's independent review before action is taken is the mechanism that keeps good-faith reliance from becoming a standing excuse: it moves the reasonableness inquiry from after the arrest, where the exclusionary rule operates and good faith can forgive an officer's error, to before the arrest, where a neutral judicial officer, not the arresting officer, bears the burden of evaluating the AI output's reliability in the first instance. A rule that front-loads that inquiry does more to prevent the underlying harm than a rule that leaves the inquiry to be litigated after the fact ever could. The proposed standard's remedy is prevention rather than suppression, and it is not defeated by a doctrine designed to excuse good-faith conduct within an authorized search when the point of the standard is to require the authorization in the first place.

VII. Conclusion

The Fourth Amendment was written for a world of physical intrusions: officers entering homes, seizing papers, searching persons. It has survived into the digital age because the Supreme Court has been willing, imperfectly and incrementally, to extend its principles to new technologies when those technologies enable the government to reach into the privacies of life in ways the Amendment was designed to prevent. *Katz* extended it to electronic eavesdropping. *Jones* brought it to GPS tracking. *Carpenter* applied it to the comprehensive digital record of a person's movements. Each extension was contested. Each was necessary. Each followed from the same underlying recognition that the

Constitution's guarantee of security against unreasonable government intrusion cannot be rendered meaningless by technological change.

AI-driven predictive surveillance is the next technology that demands this recognition. When a facial recognition system scans a database and attaches a name to a crime, it has done something the Fourth Amendment has always cared about: it has pointed the government at a specific person and authorized action against them. When a risk-scoring algorithm assigns a number to a named individual, and that number becomes the reason officers appear at their door, the same thing has happened. The government has made a judgment about a person, acted on it, and constrained their liberty. The fact that an algorithm made the judgment rather than a detective does not diminish its constitutional significance. It amplifies it. Algorithms operate at a scale no detective can match, at a speed that forecloses deliberation, through reasoning that neither the officer acting on the output nor the person targeted by it can examine or challenge, and with a documented capacity to fabricate conclusions with the same confident presentation as accurate ones.

Nijeer Parks spent ten days in jail because an algorithm was wrong and nothing in the constitutional framework required anyone to check. Robert Williams was arrested in front of his daughters because a facial recognition match substituted for the individualized probable cause determination the Fourth Amendment requires, and no court had yet said it could not. The lawyers in *Mata v. Avianca* faced sanctions because they submitted an AI's fabricated conclusions to a federal court without verification. In each case, the common thread is the same: an AI system produced an output, a human actor treated it as authoritative, and the consequences fell on someone else. The Constitution exists precisely to interrupt that chain before it ends in a cell.

The Supreme Court said in *Carpenter* that it would not allow the Constitution to be defeated by seismic shifts in digital technology. [43] AI-driven predictive surveillance is that shift, arrived and accelerating. The inference is the search. The warrant is required.

Notes

[1] Kashmir Hill, “Another Arrest, and Jail Time, Due to a Bad Facial Recognition Match,” *New York Times*, December 29, 2020, <https://www.nytimes.com/2020/12/29/technology/facial-recognition-misidentify-jail.html>.

[2] *Carpenter v. United States*, 585 U.S. 296 (2018).

[3] Andrew Guthrie Ferguson, *The Rise of Big Data Policing: Surveillance, Race, and the Future of Law Enforcement* 34–61 (New York University Press 2017).

[4] *Id.*

[5] American Civil Liberties Union, “More Than a Dozen Wrongful Arrests Due to Police Reliance on Facial Recognition Technology,” ACLU (2023), <https://www.aclu.org/news/privacy-technology/more-than-a-dozen-wrongful-arrests-due-to-police-reliance-on-facial-recognition-technology>.

[6] *Katz v. United States*, 389 U.S. 347, 360–61 (1967).

[7] *United States v. Jones*, 565 U.S. 400, 404–05 (2012).

[8] *Id.* at 415 (Sotomayor, J., concurring).

[9] *Id.* at 430 (Alito, J., concurring in the judgment).

[10] Orin S. Kerr, *The Mosaic Theory of the Fourth Amendment*, 111 *Mich. L. Rev.* 311, 313 (2012), <https://doi.org/10.36644/mlr.111.3.mosaic>.

[11] *Carpenter*, 585 U.S. at 310.

[12] *Id.* at 306, 310.

[13] *Id.* at 316.

[14] *Id.* at 310.

[15] Paul Ohm, *The Many Revolutions of Carpenter*, 32 *Harv. J.L. & Tech.* 357, 380 (2019).

- [16] Ferguson, *supra* note 3, at 34–61.
- [17] Carpenter, 585 U.S. at 310.
- [18] ACLU of Michigan, “*Williams v. City of Detroit*” (2021), <https://www.aclumich.org/cases/facial-recognition/>.
- [19] *State v. Loomis*, 881 N.W.2d 749, 767 (Wis. 2016).
- [20] *Id.* at 769.
- [21] *Illinois v. Gates*, 462 U.S. 213, 238 (1983).
- [22] *Ybarra v. Illinois*, 444 U.S. 85, 91 (1979).
- [23] Bernard E. Harcourt, *Against Prediction: Profiling, Policing, and Punishing in an Actuarial Age 4–5* (University of Chicago Press 2007).
- [24] Julia Angwin et al., “Machine Bias,” ProPublica (May 23, 2016), <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>.
- [25] *Id.*
- [26] *Gates*, 462 U.S. at 230–31.
- [27] Andrew Guthrie Ferguson, *Predictive Policing and Reasonable Suspicion*, 62 Emory L.J. 259, 305 (2012).
- [28] Elizabeth E. Joh, *The New Surveillance Discretion: Automated Suspicion, Big Data, and Policing*, 10 Harv. L. & Pol’y Rev. 15 (2016).
- [29] *Gates*, 462 U.S. at 238.
- [30] *Groh v. Ramirez*, 540 U.S. 551, 557 (2004).
- [31] ACLU, *supra* note 5.
- [32] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443, 448 (S.D.N.Y. 2023).
- [33] *Id.* at 461.
- [34] Carpenter, 585 U.S. at 310–12.

[35] Kerr, *supra* note 10, at 314–25.

[36] *Id.* at 350–68.

[37] *Riley v. California*, 573 U.S. 373 (2014).

[38] *Id.* at 403.

[39] *Carpenter*, 585 U.S. at 306.

[40] *United States v. Leon*, 468 U.S. 897 (1984); *Herring v. United States*, 555 U.S. 135 (2009).

[41] See *State v. Loomis*, 881 N.W.2d 749 (Wis. 2016) (permitting continued use of the score at sentencing so long as the presentence report includes a written advisement of the tool’s limitations, rather than requiring disclosure of its methodology or exclusion of the score itself).

[42] *Leon*, 468 U.S. 897; *Herring*, 555 U.S. 135.

[43] *Carpenter*, 585 U.S. at 304.

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